



ASIA PACIFIC COLLEGE
School of Computing and Information Technologies

LitPath AI: Smart Pathfinder for Theses and Dissertations

Final Documentation Submitted to the Faculty of the
School of Computing and Information Technologies

In Partial Fulfillment of the Requirements for the Course
Project Management (PROJMAN)
3rd Term A.Y. 2025-2026

By:

Charijoy Cempron

Jenine Elaine Dulay

Joshua Sta Ana

Tracie Tomon

Mr. Jose Eugenio L. Quesada

June 2026



Table of Contents

01 Initiation Phase	8
01 Project Charter	8
Project Charter	8
03 Business Case	10
Executive Summary	10
Business Case Analysis Team	12
Problem Definition	12
Project Overview	14
Strategic Alignment	17
Cost Benefit Analysis	18
Alternatives Analysis	19
04 Business Model Canvas	21
05 Project Brief	24
06 Feasibility Study	26
Executive Summary	26
Description of Products and Services	27
Technology Considerations	28
Product/Service Marketplace	29
Marketing Strategy	30
Organization and Staffing	31
Schedule	32
Financial Projections	33
Findings and Recommendations	34
07 Stakeholder Management Strategy	36
Introduction	36
Identify Stakeholders	37
Key Stakeholders	38
Stakeholder Analysis	38
07B Stakeholder Analysis	41
08 Statement of Work (SOW)	41
Introduction	41
Scope of Work	42
Period of Performance	42
Place of Performance	42
Work Requirements	43
Schedule/Milestones	44



Acceptance Criteria	44
Other Requirements.....	45
02 Planning Phase	46
02 Communication Management Plan.....	46
Introduction.....	46
Communications Management Approach	46
Communications Management Constraints.....	47
Stakeholder Communication Requirements	47
Roles	47
Project Team Directory	49
Communication Methods and Technologies.....	49
Communications Matrix	50
Communication Flowchart	51
Guidelines for Meetings	52
Communication Standards.....	53
Communication Escalation Process	54
Glossary of Communication Terminology	55
03 Risk Management Plan	56
Introduction.....	56
Top Three Risks	56
Risk Management Approach	57
Risk Identification	57
Risk Qualification and Prioritization	58
Risk Monitoring.....	59
Risk Mitigation and Avoidance.....	59
Risk Register	60
04 Scope Management Plan.....	62
Introduction.....	62
Scope Management Approach	62
Roles and Responsibilities.....	62
Scope Definition.....	63
Project Scope Statement.....	64
Work Breakdown Structure	64
Scope Verification.....	67
Scope Control	67
05 Work Breakdown Structure (WBS)	68



Introduction	68
Outline View	68
Hierarchical Structure	69
Tabular View	71
Tree Structure View	72
WBS Dictionary	73
Glossary of Terms	74
06 Schedule Management Plan	75
Introduction	75
Schedule Management Approach	75
Schedule Control	77
Schedule Changes and Thresholds	78
Scope Change	79
07 Requirements Management Plan	80
Introduction	80
Requirements Management Approach	80
Configuration Management	81
Requirements Prioritization Process	81
Product Metrics	82
Requirements Traceability Matrix	82
08 Risk Register	86
Risk matrix	87
RISK SUMMARY	87
Risk assessment Voting Table	88
Risk assessment Voting Table	88
Risk Score Color Legend	89
10 Project Quality Management Plan	89
Introduction	89
Quality Management Approach	89
Quality Requirements / Standards	90
Quality Assurance	91
Quality Control	92
Quality Control Measurements	93
13 Cost Management Plan	94
Introduction	94
Cost Management Approach	94



Project Assumptions & Sprint Configuration	94
Sprint Budget Details.....	95
Measuring Project Costs	97
Cost Performance Thresholds	99
Reporting Format.....	99
Cost Variance Response Process.....	99
Cost Change Control Process	100
Project Budget Summary	100
14 Configuration Management Plan	101
Introduction.....	101
Roles and Responsibilities.....	102
Configuration Control.....	103
Configuration Management Database (CMDB)	104
Configuration Status Accounting	105
Configuration Audits	106
15 Project Change Management Plan.....	107
Introduction.....	107
Change Management Approach.....	108
Definitions of Change	108
Change Control Board.....	109
Roles and Responsibilities.....	110
Change Control Process	111
16 Human Resource Plan	114
Introduction.....	114
Roles and Responsibilities.....	114
Staffing Management.....	116
03 Execution Phase	118
Sprint 1 Ceremonies	118
Sprint 2 Ceremonies	121
Sprint 3 Ceremonies.....	123
04 Monitoring and Controlling Phase	126
02 Project Status Report	126
Project Status Summary.....	126
Work Planned for Last Month	127
Work Completed Last Week	127
Work Planned For Next Week.....	128
Open Issues	128



Open Risks	129
Deliverables and Milestones	129
Open Change Requests	130
05 Closing Phase	131
01 Transition Out Plan	131
Executive Summary	131
Transition Approach	131
Transition Team Organization	131
Workforce Transition	132
Work Execution During Transition	132
Subcontracts	132
Property Transition	133
Knowledge Transfer	134
Schedule	134
Handover and Acceptance	135
02 Project Acceptance	135
Project Deliverables	135
Acceptance Criteria	136
Transition to Operations	136
Project Closeout Authorization	137
03 Post Project Review	138
Project Summary	138
Project Team and Staffing	138
Project Deliverables (Planned vs. Actual)	139
Transition to Operations	139
Project Costs	140
Project Schedule	141
Recommendations	142
04 Lessons Learned	143
Introduction	143
Lessons Learned Approach	143
Lessons Learned from this Project	144
Lessons Learned Knowledge Base / Database	145
Lessons Learned Applied from Previous Projects	145
Process Improvement Recommendations	146
06 Agile	147



03 Agile Team Charter	147
Project Vision	147
Mission and Objectives	147
Team Members and Roles	147
Team Communication	148
Tools	148
Rules of Behavior	148
03B Agile Team Values	149
07 Project Documents	156
04 System Design Document	156
Introduction	156
Purpose	156
System Overview	156
Design Constraints	157
Roles and Responsibilities	157
Project References	157
System Architecture	157
Database Design	159
Hardware and Software Detailed Design	159
System Security and Integrity Controls	160



01 Initiation Phase

01 Project Charter

Project Charter

This Charter formally authorizes the LitPath AI Project to design and develop a standalone AI-powered web application that provides intelligent academic thesis and dissertation search capabilities for the DOST-STII Library. A project plan will be developed and submitted to the Project Sponsor for approval. All resources will be assigned by the Project Sponsor, Khasian Eunice Romulo, Science Research Specialist II / Unit Head of Digital Information and Processing Delivery Unit.

The purpose of the LitPath AI project is to improve the quality, efficiency, and relevance of research experiences in academic libraries, particularly within the DOST'S OPAC system. This project meets DOST-STII Library's need for a standalone AI-powered web application that addresses vague results and limited keyword matching through advanced AI techniques including vector embeddings and large language models. The system delivers semantic search results, intelligent summaries, and citation generation with faster query processing and result display compared to traditional library OPAC systems. The project deliverables shall include an AI-powered query engine, an in-text citation and bookmark feature, research history, follow-up questions, and an admin dashboard. The objectives of the LitPath AI project are to improve search result relevance and research efficiency for student researchers and library staff. High level risks include completing all features before the June–July 2026 deployment deadline. Success will be determined by the Project Sponsor once the system is deployed and usability testing confirms improved research outcomes.

The Project Manager, Tracie Tomon, is hereby authorized to interface with management as required, negotiate for resources, delegate responsibilities within the framework of the project, and communicate with all contractors and management to ensure successful and timely completion. The Project Manager is responsible for developing the project plan, monitoring schedule and scope, assigning tasks, overseeing system integration, testing, and documentation, and organizing final outputs for defense and presentation.

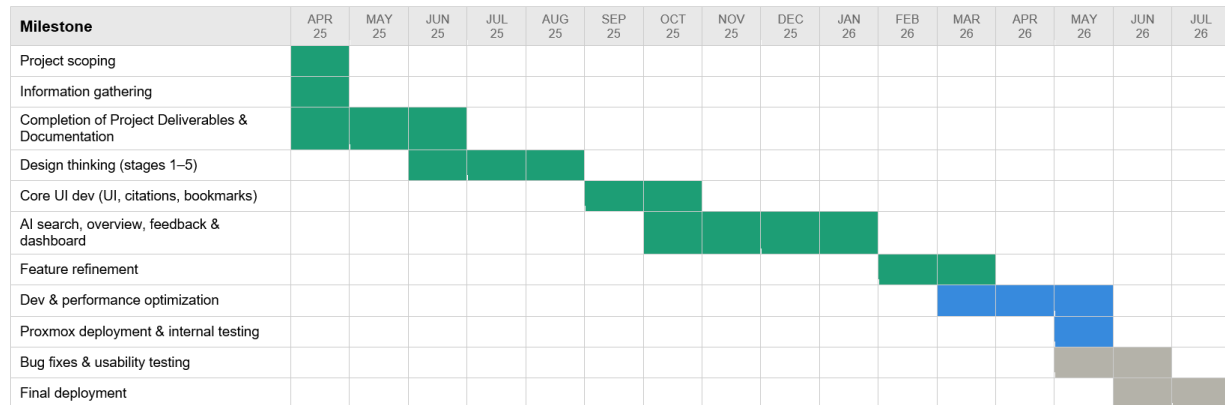
The project plan will be submitted and approved in accordance with the milestone schedule outlined below. Upon approval, resources will be assigned and development work will proceed in phases from planning and design through backend development, system integration, user testing, and final deployment. The Project Sponsor must approve any changes that may impact milestone dates. A detailed schedule will be maintained throughout the project lifecycle. The high-level milestone schedule is:



LitPath AI: Smart Pathfinder for Theses and Dissertations

April 2025 – July 2026

Completed In progress Not started



Timeline runs Apr 2025 → Jul 2026

Figure 1. LitPath AI Milestone (Gantt Chart)

Milestone	Target Date	Status
Project Scoping	April 2025	Completed
Information Gathering	April 2025	Completed
Completion of Project Deliverables & Documentation	June 2025	Completed
Design Thinking Process (Stages 1–5)	Aug 2025	Completed
Core UI Development (System UI, Citation Generator, Bookmark)	Sep – Oct 2025	Completed
AI Search Engine, Generate Overview, Feedback Mechanism & Admin Dashboard	Oct 2025 – Jan 2026	Completed
Feature Refinement	Feb – Mar 2026	Completed
Continued Development & Performance Optimization (incl. codebase restructuring)	Mar – May 2026	In Progress
Proxmox Server Deployment & Internal Testing	May 2026	In Progress
Bug Fixes & Usability Testing	May – Jun 2026	Not Started
Final Deployment	Jun – Jul 2026	Not Started

Table 1. High Level Milestone

LitPath AI initially did not require a formal budget, however, with the planned cloud deployment and the integration of AI/LLM services, operational costs are now anticipated. Expected costs include cloud hosting fees for the final production deployment in June–July 2026, as well as AI/LLM API usage costs incurred during development, testing, and live operation of the system. These costs are to be shouldered by DOST-STII as the primary client and sponsor of the project. A detailed cost breakdown will be finalized once the specific cloud infrastructure and AI/LLM services are confirmed with the client.



03 Business Case

Executive Summary

This business case outlines how the LitPath AI project will address the research discovery limitations of the DOST-STII Library's current Online Public Access Catalog (OPAC) system, the anticipated benefits of the project, and the recommendation and justification for moving forward. The business case also discusses detailed project goals, performance measures, assumptions, constraints, and alternative options considered.

The DOST-STII Library maintains a valuable collection of graduate theses and dissertations, yet this collection remains significantly underutilized due to the limitations of its current OPAC, specifically its reliance on exact keyword matching, lack of AI-powered search, slow performance, and missing citation and bookmarking tools. These limitations create barriers for student researchers, DOST employees, and academic professionals who depend on this collection for their work.

LitPath AI is proposed as a standalone AI-powered web application designed to complement the existing OPAC. Rather than modifying or replacing the current system, it provides an independent platform that enhances research discovery through advanced AI capabilities. By leveraging semantic search, automated citation generation, bookmarking features, and analytics, LitPath AI significantly improves research efficiency and accessibility for users.

1.1 Issue

The DOST-STII Library's current Online Public Access Catalog (OPAC) is limited in its ability to support modern research needs. The system relies solely on exact keyword matching, making it difficult for users who are exploring new topics or are unsure of precise terminology to discover relevant materials. Internal assessments show that fewer than 7 out of 10 search results align with the user's actual research interests, and users must spend approximately 15 minutes manually sifting through results to find useful materials. Search queries also take over a minute to return results which is far exceeding the acceptable standard, and the system lacks essential features such as built-in citation and bookmarking tools. These deficiencies result in research inefficiencies and the underutilization of the library's key academic resources.

1.2 Anticipated Outcomes

Implementing LitPath AI as a standalone web application for the DOST-STII Library will enhance the research experience, particularly for users searching for theses and dissertations. By augmenting the existing keyword-dependent system with an AI-powered search capability, users will be able to discover relevant academic works even when they are unsure of the exact terminology. This improvement will reduce reliance on precise keyword matching and allow for more intuitive and efficient research.

A built-in automated citation generator supporting APA, MLA, Chicago, and IEEE formats will eliminate the need for users to manually format references, improving accuracy and saving



time. In addition, a bookmarking feature will allow users to save and easily revisit selected theses and dissertations, supporting better organization of research materials.

For DOST-STII as an institution, LitPath AI will improve service delivery by enhancing access to its academic collections. Librarians will benefit from an admin dashboard that provides analytics on user search trends related to theses and dissertations, enabling more informed decisions in collection development and management.

1.3 Recommendation

Various options and alternatives were analyzed to determine the best way to address the research discovery limitations of the DOST-STII OPAC. The recommended approach is developing LitPath AI as a standalone AI-powered web application for theses and dissertations built by HecTech which was selected because it directly addresses the identified problems, is technically feasible using currently available tools, and does not require a formal budget from the HecTech side, as cloud and AI/LLM operational costs will be shouldered by DOST-STII.

LitPath AI will achieve its desired results through the following:

- AI-powered query analysis that enables the system to understand user research intent beyond exact keyword matches, returning semantically relevant theses and dissertations.
- A built-in citation generator that eliminates the manual effort required to format references in APA, MLA, Chicago, and IEEE styles.
- A bookmarking feature that allows users to save, organize, and quickly access selected research materials for future use.
- An admin dashboard that provides DOST-STII librarians with visibility into search trends and usage patterns related to academic works.

1.4 Justification

LitPath AI was selected as the recommended solution because it directly addresses the critical limitations of the DOST-STII Library's current OPAC while remaining technically and financially feasible for all parties involved. The system was chosen over all other alternatives as it delivers the greatest improvement in research efficiency without requiring a formal development budget from HecTech, since cloud hosting and AI/LLM API operational costs will be covered by DOST-STII.

Key stakeholders including Science Research Specialists Khasian Eunice Romulo and Jonathan Abalon have confirmed the need for modernized search capabilities and improved user satisfaction with the library's thesis and dissertation collection. Their direct involvement in requirements definition and deliverable approval further validates the alignment of LitPath AI with DOST-STII's institutional objectives.

If LitPath AI is not implemented, the DOST-STII Library will continue to operate with a search system that fails to meet modern research expectations. Users will continue experiencing poor search relevance, slow query performance of over 2–3 minutes, and the absence of essential tools such as citation generation and bookmarking, directly impacting



the productivity of students, academic researchers, and DOST employees who rely on the library's collection.

Initial estimates for the LitPath AI project are:

- At least 70% of search relevance accuracy in the top 10 results, based on user feedback, compared to the current rate of fewer than 7 out of 10 relevant results
- Search queries are processed and returned significantly faster compared to the existing system, resulting in more responsive and efficient user experience.
- Elimination of manual citation formatting through an automated citation generator supporting APA 7th, MLA 9th, Chicago, and IEEE formats
- Reduction in time spent manually filtering through irrelevant results from approximately 15 minutes per session to just a few minutes through improved semantic search relevance

Business Case Analysis Team

The following individuals comprise the business case analysis team. They are responsible for the analysis and creation of the LitPath AI project business case.

Role	Description	Name/Title
Project Manager	Manages business case development and leads the project team	Tracie Tomon, Group Leader, HecTech
Project Adviser	Provides project guidance	Gonzalo Guumogda, SoCIT Faculty
Development Team	Contributes to business case analysis through system development context and technical inputs	HecTech Team, Project Developers
Client	Provides project requirements and approval	Khasian Romulo, Science Research Specialist II

Problem Definition

3.1 Problem Statement

The DOST-STII Library houses a valuable and growing collection of graduate theses and dissertations from DOST Scholars, cataloged through its Online Public Access Catalog (OPAC). Despite the value of this collection, it remains underutilized due to critical limitations in the OPAC's search functionality, including exact keyword matching, slow processing times, and a lack of Artificial Intelligence. These limitations create a significant barrier to knowledge discovery for students, academic researchers, and DOST employees. Users who are exploring new research areas or are unsure of precise terminology find it difficult to discover



relevant materials, leading to research inefficiencies and preventing the library from maximizing the impact of its key academic resources.

Specifically, the key challenges with the current system are:

1. **Limited Search Mechanism:** The current system relies heavily on exact keyword and title-based searches, lacking AI-powered search capabilities to interpret a user's research interests or intent. This makes the system ineffective for users exploring broader topics or those unfamiliar with precise terminology, often yielding results that are irrelevant or incomplete.
2. **Poor Relevance and Efficiency:** The OPAC struggles to consistently generate accurate results due to the absence of advanced, intent-driven search algorithms. Internal assessments reveal that fewer than 7 out of 10 results match a user's actual research interests. Consequently, users are forced to spend approximately 15 minutes per session manually reviewing and sifting through irrelevant results to find useful materials.
3. **Lengthy Searching Performance:** The search function exhibits severe performance issues, with search queries taking over a minute to return results. This processing time is significantly slower than the acceptable industry standard, leading to a poor user experience and discouraging further exploration of the platform.
4. **Lack of Essential Research Support Features:** The current system lacks key built-in tools essential for modern academic research, specifically automated citation generation and bookmarking capabilities. As a result, users must manually compile citations and manage their research materials independently, which increases the time spent on administrative tasks and raises the risk of formatting errors or disorganized references.

3.2 Organizational Impact

LitPath AI will affect DOST-STII and the HecTech development team in the following ways:

Tools:

The existing OPAC will remain in place as a separate system, while LitPath AI operates as a standalone web application focused on theses and dissertations. Users can access LitPath AI independently to take advantage of its advanced AI-powered search capabilities without modifying the current OPAC. This approach ensures no disruption to existing operations while providing a significantly enhanced research experience. In addition, DOST-STII staff will be provided with an admin dashboard to monitor usage analytics and gain insights into user behavior.

Processes:

Research workflows for students, researchers, and DOST employees will be streamlined. Users will be able to quickly discover relevant theses and dissertations, generate citations automatically, and bookmark selected materials for future reference. This reduces the need to manually sift through irrelevant results or manage references separately.

Roles and Responsibilities:



DOST-STII library staff will take on a new role in monitoring and interpreting analytics from the admin dashboard. HecTech will continue to support the system during testing and deployment phases, with full technical documentation and a user manual provided upon project completion.

Hardware/Software:

The system is currently deployed on a Proxmox server instance hosted on a team member's local device for development and testing purposes. For final production deployment, the system will be transitioned to a hosting environment to be confirmed with DOST-STII, with associated costs shouldered by the client. AI/LLM API services will also be covered by DOST-STII. No additional hardware purchases are required from HecTech.

3.3 Technology Migration

LitPath AI will be implemented using a phased approach to ensure continuity of the existing OPAC system and minimize disruption to DOST-STII operations.

- Phase I — Planning & Design (Apr–Aug 2025, Completed):
 - Project scoping, requirements gathering, stakeholder interviews, and UI/UX design using Figma.
- Phase II — Core Development (Sep 2025–Jan 2026, Completed):
 - Development of the AI search engine, core UI, citation generator, bookmarking feature, feedback mechanism, and admin dashboard.
- Phase III — Feature Refinement (Feb–Mar 2026, Completed):
 - Continued feature development (research history, follow-up questions), and performance optimization.
- Phase IV — Internal Testing & Temporary Deployment (May 2026):
 - Deployment on Proxmox server for internal testing, unit testing, and comprehensive testing of the standalone system prior to final production handover.
- Phase V — Usability Testing & Bug Fixes (May–Jun 2026):
 - Usability testing with student researchers and DOST employees, performance optimization, and bug fixes.
- Phase VI — Final Deployment (Jun–Jul 2026):
 - Full production deployment as a standalone system, operating independently from the existing DOST-STII OPAC, followed by the formal handover of the system and all documentation.

Project Overview

The LitPath AI Project Overview provides detail for how this project will address the DOST-STII Library's research discovery problem. The overview covers the project description, goals and objectives, performance criteria, assumptions, constraints, and major milestones.



4.1 Project Description

LitPath AI is a standalone AI-powered web application developed by HecTech for the DOST-STII Library. Operating independently from the existing OPAC, the system enhances the discovery of graduate theses and dissertations by moving beyond simple keyword matching. It understands user research intent through AI-powered query analysis, returning semantically relevant results.

In addition to improved search functionality, the system includes a built-in citation generator and a bookmarking feature, enabling users to efficiently manage and organize their research materials.

The project will be completed through six development and deployment phases, leveraging the four-member HecTech team's expertise in software development, AI integration, and UI/UX design. Upon final deployment in June–July 2026, DOST-STII will receive a fully operational system, along with complete technical documentation and a user manual. All cloud hosting and AI/LLM API costs will be covered by DOST-STII as the project's primary client and sponsor.

4.2 Goals and Objectives

The LitPath AI project directly supports the research and information management goals of DOST-STII. The following table lists the business goals and objectives that the LitPath AI project supports and how it addresses them:

Business Goal/Objective	Description
Improve research discovery and search relevance	LitPath AI replaces exact keyword matching with AI-powered semantic search using RAG and Gemini AI, enabling users to find relevant theses and dissertations even without precise terminology, targeting at least 70% relevance accuracy in top 10 results
Enhance user efficiency and reduce manual workload	Automated citation generation in APA 7th, MLA 9th, Chicago, and IEEE formats, along with bookmarking features, eliminates manual formatting tasks and reduces time spent managing research materials from approximately 15 minutes to just a few minutes per session
Improve system performance and user experience	Search queries are processed and returned significantly faster compared to the existing system, resulting in more responsive and efficient user experience.
Support informed collection management for DOST-STII	An admin analytics dashboard will provide DOST-STII librarians with visibility into search trends and usage patterns to guide collection development and institutional decision-making



4.3 Project Performance

The following table lists the key resources, processes, or services and their anticipated business outcomes in measuring the performance of the project. These performance measures will be quantified and further defined in the detailed project plan.

Key Resource/Process/Service	Performance Measure
Search Relevance	Top 10 search results achieve a minimum of 70% relevance accuracy based on user feedback and expert validation
User Satisfaction	Average satisfaction score of at least 3.5 out of 5 from usability testing participants
System Response Time	Search queries processed and results displayed faster compared to the existing system
Admin Dashboard	DOST-STII librarians able to access and interpret search trend data to inform collection management

4.4 Project Assumptions

This section outlines the key conditions considered true for the planning and scoping of the LitPath AI project. If any of these assumptions prove false, it may introduce risks that impact on the timeline, budget, or scope, requiring a formal change request:

1. The current library servers or the planned cloud hosting environment has the capacity to support the AI engine
2. The project team will have continuous and stable access to the DOST-STII database and necessary technical systems throughout the project timeline.
3. The project scope will remain stable throughout the development phase.
4. The Project Sponsors, Khasian Eunice Romulo and Jonathan Abalon, will remain available for milestone approvals, client feedback, and formal acceptance of deliverables.
5. Cloud hosting and AI/LLM API costs will be fully covered by DOST-STII as agreed.
6. Users attempting to log in to LitPath AI will be able to register directly in the system, independent of the parent OPAC.

4.5 Project Constraints

This section outlines the specific limitations and boundaries within which the LitPath AI project must be executed. This strict schedule, resource, financial, and scope parameters dictate the team's operations and must be adhered to throughout the project lifecycle:

1. The project must be fully deployed by June–July 2026 in alignment with the academic and institutional timeline.
2. Development is limited to the four-member HecTech team with no external outsourcing anticipated.
3. No formal development budget is allocated to HecTech, and the team's primary resource is time and effort.



4. Full-text access to undigitized physical theses/dissertations is out of scope; the system focuses on abstracts and metadata only.

4.6 Major Project Milestones

The following are the major project milestones identified at this time. As the project planning moves forward and the schedule is developed, the milestones and their target completion dates will be modified, adjusted, and finalized as necessary to establish the baseline schedule.

Milestones/Deliverables	Target Month
Project Scoping	April 2025
Information Gathering	April 2025
Completion of Project Deliverables & Documentation	June 2025
Design Thinking Process (Stages 1–5)	August 2025
Core UI Development	September – October 2025
AI Search Engine, Feedback Mechanism & Admin Dashboard Development	October 2025 – January 2026
Feature Refinement	February – March 2026
Continued Development & Performance Optimization	March – May 2026
Proxmox Deployment & Internal Testing	May 2026
Bug Fixes & Usability Testing	May – June 2026
Final Deployment & Project Handover	June – July 2026

Strategic Alignment

The LitPath AI project is in direct support of DOST-STII's institutional goals and HecTech's academic development objectives. By supporting these strategic plans, this project advances knowledge accessibility and research innovation within the institution.

Plan	Goals/Objectives	Relationship to Project
DOST-STII Digital Information Access Plan	Improve accessibility and discoverability of academic resources,	LitPath AI directly enhances the discoverability of the DOST-STII Library's thesis and dissertation collection through AI-



	particularly graduate-level theses and dissertations, for researchers, students, and DOST employees	powered semantic search using RAG and Gemini AI, making the collection significantly more accessible and useful to its target users
DOST-STII Technology Modernization Plan	Leverage modern technology to improve institutional service delivery without disrupting existing systems	LitPath AI operates as an independent, standalone application alongside the existing OPAC system. It introduces AI-powered search, automated citation generation in APA, MLA, Chicago, and IEEE formats, and an admin analytics dashboard, serving as a dedicated research tool without replacing or altering the legacy catalog.
DOST-STII Research Support and Scholar Empowerment Plan	Support DOST scholars and employees in conducting efficient, accurate, and well-documented research	LitPath AI reduces research friction by cutting response time up to industry standard, delivering semantically relevant results with at least 70% accuracy, and providing tools that help users organize, bookmark, and cite their materials with ease

Cost Benefit Analysis

The following table captures the cost and benefit actions associated with the LitPath AI project, descriptions of these actions, and their estimated costs or savings.

Action	Action Type	Description	First year costs (- indicate anticipated savings)
System Development by HecTech	Cost	Labor and time investment by the 4-member HecTech team across six development phases spanning the 2025–2026 academic year	₱0 (in-kind contribution — no formal budget allocated to HecTech)
Cloud Hosting & Infrastructure	Cost	Currently hosted on a Proxmox server instance on a team member's local device for development and testing. Final production hosting environment and costs to be confirmed with DOST-STII upon deployment.	Shouldered by DOST-STII



AI / LLM API Integration (Gemini AI)	Cost	Operational costs for AI-powered query analysis using Retrieval-Augmented Generation (RAG) and Gemini AI during development, testing, and live operation	Shouldered by DOST-STII
Reduction in manual research time	Benefit	Users currently spend approximately 15 minutes per session manually sifting through irrelevant results. Improved semantic search is estimated to reduce this to just a few minutes per session	Significant time savings per user session upon deployment
Elimination of manual citation formatting	Benefit	Automated citation generation in APA 7th, MLA 9th, Chicago, and IEEE formats removes the need for users to manually format references, reducing errors and saving time per research session	Estimated hours saved per researcher per project
Improved system response time	Benefit	The system gives search results up to industry standard which drastically reduces researcher wait time and improves overall productivity	Immediate upon deployment

Since HecTech contributes its full development effort at no formal cost and DOST-STII shouldered all cloud and AI/LLM operational expenses, the LitPath AI project represents a high-value, low-financial-risk initiative for both parties. The primary return on investment is measured in improved research efficiency, faster search performance, and greater utilization of the DOST-STII Library's thesis and dissertation collection.

Alternatives Analysis

The following alternative options have been considered to address the business problem. These alternatives were not selected for several reasons which are also explained below.

No Project (Status Quo)	Reasons For Not Selecting Alternative
Keep the existing DOST-STII OPAC system without any enhancements or modifications	- Users continue to experience poor search relevance with fewer than 7 out of 10 results matching their actual research needs - Search queries continue to take over 10 minutes to return results, far exceeding the acceptable industry standard



	• No citation generation or bookmarking tools available, forcing users to manually compile and format references • The library's thesis and dissertation collection continues to be significantly underutilized due to the limitations of the current search system
Alternative Option	Reasons For Not Selecting Alternative
Purchase and deploy a third-party AI-powered library or academic search system	• Significantly higher licensing, subscription, and implementation costs that are outside the project's budget constraints • Third-party solutions are not tailored to the specific structure and content of the DOST-STII OPAC and its thesis and dissertation collection • May require full replacement of the existing OPAC, causing major disruption to ongoing library operations • Vendor solutions may not comply with the Data Privacy Act of 2012 (RA 10173) or DOST-STII's internal security requirements
Alternative Option	Reasons For Not Selecting Alternative
Full replacement and redevelopment of the OPAC system from scratch	• Requires a substantially larger budget and a significantly longer development timeline beyond what HecTech can deliver within the 2025–2026 academic year • Poses high operational risk by completely replacing the existing catalog system that users and library staff currently depend on daily • A full redevelopment goes far beyond the defined project scope and the available resources of the five-member HecTech team • Does not align with DOST-STII's preference for a non-disruptive supplementary solution that works alongside the existing OPAC



04 Business Model Canvas



KEY PARTNERS DOST-STII (Department of Science and Technology – Science and Technology Information Institute) — primary client and data provider	KEY ACTIVITIES - Semantic search indexing of thesis and dissertation metadata using vector embeddings - AI-powered overview and summary generation via Gemini API - Development and maintenance of the web interface - User registration and account management and session handling - Bookmark, citation generation, and search history features - Admin dashboard development for usage analytics and feedback management	VALUE PROPOSITIONS LitPath AI is a research tool that enhances access to theses and dissertations in the DOST-STII library catalog. It uses semantic search to deliver fast, context-aware results that are up to the industry standard, and automatically generates citations in APA, MLA, Chicago, and IEEE formats. It also has a bookmarking feature to easily access what theses or dissertations the user needs. By reducing manual filtering from minutes to just a few clicks, LitPath AI improves research efficiency and supports SDG 4 by making academic resources more accessible.	CUSTOMER RELATIONSHIPS - Self-service web platform accessible via standard browsers (e.g., Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge). - AI-generated overviews and inline citations guide users independently - User feedback mechanism (thumbs up/down ratings + comments and Client Satisfaction Measurement Form) enables continuous improvement	CUSTOMER SEGMENTS Students & Academic Researchers — need efficient, relevant access to graduate-level theses and are frustrated by slow, inaccurate keyword searches DOST Professionals & Employees — require quick access to science and technology research materials for policy and project work
	KEY RESOURCES - Development team (HecTech) - Cloud hosting infrastructure (servers, storage, and networking) - AI integration via Google Gemini API - Database for library catalog and metadata - Access to DOST-STII library data		CHANNELS - Direct access via the LitPath AI website (web browser) - Institutional deployment through DOST-STII Library - Word-of-mouth and institutional endorsement within the DOST research community	

COST STRUCTURE Fixed Costs: • Development labor (student developer team – HecTech) • Initial system setup and deployment • Basic web cloud hosting (baseline server allocation) Variable Costs (scale with system usage): • API usage (Google Gemini API), which depends on request volume and user activity • Additional cloud resources (storage, bandwidth, and compute scaling) • System maintenance and updates (depending on system usage and enhancements) Scalability Requirements: As system adoption increases, costs are expected to scale primarily due to API usage and cloud resource demand. Post-handover operational costs include API consumption, hosting, and ongoing maintenance, with monthly expenses varying based on user activity levels.	REVENUE STREAMS LitPath AI is an academic/institutional platform developed as a value-added service for the DOST-STII Library. Its primary "return" is institutional value: improved research efficiency, increased utilization of the library's collection, and support for SDG 4, 9, and 17. Long-term sustainability may involve institutional funding from DOST-STII for maintenance, or potential expansion to other government library systems.
--	--



05 Project Brief

PROJECT NAME	LitPath AI: Smart Pathfinder for Theses and Dissertations		
PROJECT MANAGER	Tracie Tomon	EMAIL	tctomon@student.apc.edu.ph
START DATE	March 2025	BUDGET	
END DATE	June-July 2026		
PROJECT OVERVIEW	The DOST-STII Library houses a valuable and growing collection of graduate theses and dissertations from DOST Scholars, cataloged through its Online Public Access Catalog (OPAC). Despite the value of this collection, it remains underutilized due to critical limitations in the OPAC's search functionality, including exact keyword matching, slow processing times, and a lack of Artificial Intelligence. These limitations create a significant barrier to knowledge discovery for students, academic researchers, and DOST employees. Users who are exploring new research areas or are unsure of precise terminology find it difficult to discover relevant materials, leading to research inefficiencies and preventing the library from maximizing the impact of its key academic resources. Specifically, the key challenges with the current system are: 1. Limited Search Mechanism: The current system relies heavily on exact keyword and title-based searches, lacking AI-powered search capabilities to interpret a user's research interests or intent. This makes the system ineffective for users exploring broader topics or those unfamiliar with precise terminology, often yielding results that are irrelevant or incomplete. 2. Poor Relevance and Efficiency: The OPAC struggles to consistently generate accurate results due to the absence of advanced, intent-driven search algorithms. Internal assessments reveal that fewer than 7 out of 10 results match a user's actual research interests. Consequently, users are forced to spend approximately 15 minutes per session manually reviewing and sifting through irrelevant results to find useful materials. 3. Lengthy Searching Performance: The search function exhibits severe performance issues, with search queries taking over a minute to return results. This processing time is significantly slower than the acceptable industry standard of 30 seconds, leading to poor user experience and discouraging further exploration of the platform. 4. Lack of Essential Research Support Features: The current system lacks key built-in tools essential for modern academic research, specifically automated citation generation and bookmarking capabilities. As a result, users must manually compile citations and manage their research materials independently, which increases the time spent on administrative tasks and raises the risk of formatting errors or disorganized references.		
OBJECTIVES	The primary goal of this project is to develop and implement an AI-powered search application to increase the discovery and use of the DOST-STII Library's collection of theses and dissertations. The project will achieve this through the following specific objectives: 1. To improve search efficiency and precision, the system will directly solve the limitations of the current exact-match search mechanism. Success will be measured by achieving a user satisfaction score of at least 3.5 out of 5 from testers on the relevance and ease of use of the new search features. 2. To help users discover more relevant studies, the system will leverage AI-powered algorithms to provide semantically relevant search results. These algorithms will analyze user queries and content to ensure that the top 10		



	search suggestions have a minimum relevance accuracy of 70%, based on user feedback and expert validation. 3. To create a better and faster user experience, the interface will be designed to be intuitive and responsive, with the goal of returning results significantly faster compared to the existing system, resulting in a more efficient user experience. 4. To make citing sources easier for researchers, the system will feature a built-in automated citation generator that supports multiple citation styles, including APA 7th Edition, MLA 9th Edition, Chicago Manual of Style, and IEEE. Accuracy will be verified through expert review and user testing to ensure correct citation formatting.
SCOPE	The scope of the LitPath AI project includes the delivery of the following tangible outputs: 1. Standalone Web Application: A fully functional, independent user interface that operates separately from the existing DOST-STII OPAC. 2. AI-Powered Semantic Search Engine: A backend search algorithm designed to interpret the meaning, context, and research intent behind user queries, moving beyond traditional exact-match keyword searching. 3. Client-Provided Database: An independent backend database populated exclusively with the sample thesis and dissertation papers provided directly by DOST-STII for the system's search repository. 4. Automated Citation Generator: A built-in tool that automatically formats citations for selected research materials in APA, MLA, Chicago Manual of Style, and IEEE formats. 5. Research Management & Feedback Tools: A bookmarking feature allowing users to save and organize their discovered research materials, alongside an integrated feedback tool to capture user ratings and comments. 6. Admin Analytics Dashboard: A dedicated dashboard providing DOST-STII library administrators with visibility into search trends, usage patterns, and user interactions.
DELIVERABLES	1. Working Prototype Ready for Deployment: A fully functional, standalone web application featuring an AI-powered semantic search engine capable of analyzing user research intent. This complete system includes a responsive user interface, a built-in automated citation generator (supporting APA 7th, MLA 9th, Chicago, and IEEE styles), a bookmarking feature for saving and organizing research materials, an admin dashboard for tracking search trends and usage patterns, and an integrated user feedback tool to measure satisfaction and validate search relevance. 2. Comprehensive Project Documentation: A complete package containing all project documents required for formal handover, future use, and maintenance. This includes the technical manual, a user guide for researchers and library staff, and a final report verifying that all project objectives have been successfully met.
SUCCESS CRITERIA	a. The project delivers a 100% working application with all defined features fully complete and operational.



	b. System achieves an average user satisfaction score of at least 3.5 out of 5 from the test group of students and DOST employees. Feedback must also confirm that the system makes their research process feel faster and easier. c. The new search system is successfully deployed as a standalone web application via Proxmox and consistently processes queries and displays results faster than the traditional library OPAC system. d. All key deliverables - the AI engine, the user interface including the citation tool, user feedback system, and all documentation, have been formally signed off on and accepted by the project stakeholders.
TARGET AUDIENCE	The project is intended for multiple audiences: 1. Students / Researchers – Individuals conducting research for academic papers or projects. 2. System Admin – DOST-STII administrators and facilitators who use the system to assist library users during reference interviews and understand usage analytics.

06 Feasibility Study

Executive Summary

The DOST-Science and Technology Information Institute (DOST-STII) Library maintains a significant collection of graduate theses and dissertations produced by DOST Scholars, cataloged through its Online Public Access Catalog (OPAC). Despite the academic value of this collection, it remains consistently underutilized. The existing OPAC relies on exact keyword-based searches, suffers from processing times exceeding one minute per query, and lacks any artificial intelligence capability. These shortcomings create a critical barrier to knowledge discovery, preventing students, researchers, and DOST personnel from efficiently accessing and building upon existing research.

LitPath AI is a proposed standalone web application developed by HecTech, a four-member team from Asia Pacific College, to address these challenges directly. The system introduces an AI-powered semantic search engine built on a Retrieval-Augmented Generation (RAG) pipeline using SentenceTransformers for embedding generation, ChromaDB for vector storage, and Gemini AI for query understanding and result summarization. Additional features include an automated multi-format citation generator, research bookmarking, a user feedback mechanism, and an administrative analytics dashboard for DOST-STII library staff. The platform will be deployed as a publicly accessible web application via Cloudflare, developed using a React.js frontend and a Django backend.



This feasibility study assesses the viability of the LitPath AI project across technical, organizational, market, and financial dimensions. Findings indicate that the project is technically achievable using the defined open-source and cloud technology stack, operationally manageable within the existing DOST-STII structure, and financially sound within the scope of an academic project engagement. The primary open cost items, AI model API access and Cloudflare deployment tier, are currently under evaluation by the team and are noted in the financial projections as pending finalization. The study recommends proceeding with project approval and initiation of formal development.

Description of Products and Services

LitPath AI is an AI-powered search engine platform developed exclusively for the DOST-STII Library. The system is designed to dramatically improve how students, academic researchers, and DOST employees search for and interact with the library's collection of graduate theses and dissertations. Unlike the current OPAC, LitPath AI employs a semantic search engine backed by a Retrieval-Augmented Generation (RAG) pipeline to interpret the meaning and research intent behind user queries, surfacing relevant results even when users are unfamiliar with precise academic terminology.

The platform is delivered as a fully functional standalone web application, accessible via cloud deployment on Cloudflare so that users may access it from any location without software installation. The system operates on a backend database populated exclusively with thesis and dissertation metadata and abstracts provided directly by DOST-STII. No changes to the existing OPAC infrastructure are required. The core features of LitPath AI are as follows:

1. **AI-Powered Semantic Search Engine:** A search module that accepts natural language queries, converts them into vector embeddings via SentenceTransformers, and retrieves semantically relevant results from a ChromaDB vector database. Results are reranked using AI to maximize relevance. The system targets a minimum relevance accuracy of 70% based on user feedback.
2. **AI-Generated Overview Summaries:** Gemini AI generates contextual overview summaries of search results with embedded citations, enabling users to quickly assess relevance without reading full documents.
3. **Automated Citation Generator:** A built-in tool that automatically formats citations for selected materials in APA, MLA, Chicago Manual of Style, and IEEE formats, eliminating manual citation compilation.
4. **Research Bookmarking and History Tracking:** Users can save and organize discovered research materials and revisit their search history within their session.
5. **User Feedback Tool:** An integrated mechanism allowing users to rate search results and submit comments, providing continuous data for system validation and improvement.
6. **Admin Analytics Dashboard:** A dedicated interface for DOST-STII library administrators to monitor search trends, usage patterns, and user engagement, supporting data-driven library management.



Technology Considerations

LitPath AI is built on a well-defined, modular technology stack selected to balance capability, cost, and implementation risk within the constraints of a four-member academic development team. The architecture follows a three-tier model: a Presentation Layer (frontend), an Application Layer (backend and AI processing), and a Data Layer (relational and vector databases). Each component is described below.

Frontend

- React.js for a dynamic, component-based user interface.
- Tailwind CSS for responsive, utility-first styling.
- Vite as the build tool for fast development and optimized production builds.

Backend

- Django (Python) as the web framework, exposing a RESTful API architecture to the frontend.
- Handles user authentication, session management, bookmarking, citation generation, feedback collection, and admin analytics logic.

AI and Semantic Search Components

- SentenceTransformers: Generates vector embeddings from user queries and thesis metadata, enabling semantic similarity matching beyond keyword lookup.
- ChromaDB: A vector database that stores document embeddings and supports fast, high-dimensional similarity search for ranked result retrieval.
- Gemini AI API: Powers query understanding, result reranking, and AI-generated overview summaries with citations. The team is currently evaluating which tier of the Gemini API (free vs. paid) is sustainable for production use. This cost item is pending finalization.
- Retrieval-Augmented Generation (RAG) Pipeline: Combines embedding-based retrieval with Gemini AI generation to produce contextually grounded summaries and search results.

Database Systems

- Supabase (PostgreSQL): Relational database storing user accounts, sessions, bookmarks, search history, feedback data, and activity logs.
- ChromaDB: Vector database storing semantic embeddings of thesis documents. The hybrid architecture ensures both structured data queries and fast semantic retrieval are supported.

Deployment and Infrastructure



- **Development and Testing:** Currently performed on a local Proxmox-based server (HP ProDesk 600 G2 MT, Intel Core i5-6500, 32 GB RAM, 512 GB SSD), which is sufficient for development, testing, and beta deployment phases.
- **Production Deployment:** The team is planning cloud deployment via Cloudflare, which provides DDoS protection, and SSL/TLS termination. The appropriate Cloudflare service tier (free vs. paid Workers/Pages plan) is currently under evaluation. This cost item is pending finalization.
- **Security:** The system implements HTTPS communication, password hashing and encryption, role-based access control (User and Admin roles), secure token-based session management, firewall and network restrictions, and logging of user activities for audit and monitoring purposes.

The use of open-source frameworks (React, Django, SentenceTransformers, ChromaDB) eliminates licensing costs for most of the stack, substantially reducing the project's cost base. The two remaining open cost items, Gemini AI API access and Cloudflare deployment tier, are the only external expenses requiring resolution before final budget confirmation. Both are commercially available services with free tiers that may be sufficient for an initial deployment phase, with upgrade paths available should usage scale beyond free-tier limits.

Product/Service Marketplace

The market for AI-powered academic research tools is experiencing rapid growth globally, driven by increasing demand for efficient knowledge discovery in higher education and research institutions. Within the Philippine government science sector, this trend is especially relevant as DOST actively invests in digital transformation initiatives that improve access to its knowledge assets.

LitPath AI serves two distinct user roles. The primary users are students and academic researchers engaged in thesis and dissertation research, as well as DOST employees conducting literature reviews and knowledge discovery activities. The secondary users are system administrators, divided into two roles: IT Administrators, who are responsible for managing user accounts, configuring AI model parameters and APIs, maintaining database structure and backups, overseeing system security, integrating LitPath AI with other institutional systems, monitoring performance and uptime, and deploying updates and new features; and Librarian Administrators, who review and validate user feedback, generate usage reports, and review search results for accuracy.

The current landscape of competing tools and alternatives that users rely on includes:

- **Google Scholar:** The dominant free academic search tool globally. While highly capable of general academic literature, it does not index the DOST-STII library's internal thesis collection, meaning DOST Scholars' work remains undiscoverable through this platform.
- **University and Institutional OPAC Systems:** Philippine universities and research institutions operate OPAC systems for internal catalog management. These systems, including DOST-STII's own OPAC, share the same fundamental limitation of exact keyword matching and lack semantic or AI-powered search capability.



- ResearchGate and Scopus/Web of Science: These platforms serve the broader global research community but require institutional subscriptions for full access and do not include unpublished theses or grey literature from DOST-STII's collection.

LitPath AI's key competitive differentiator is its exclusive, purpose-built focus on the DOST-STII thesis and dissertation collection combined with a RAG-based AI search layer. No existing commercial or open-access platform provides semantic search specifically over this dataset. This positions LitPath AI not as a competitor to Google Scholar or Scopus, but as a complementary, institution-specific discovery tool that unlocks value from a previously underutilized collection. Distribution of the platform occurs through cloud-based web access via Cloudflare, requiring no software installation from users.

Marketing Strategy

As a government-funded academic tool, LitPath AI will not engage in commercial advertising. Its adoption strategy focuses on institutional promotion, structured user onboarding, and demonstrating value through measurable performance improvements. The goal is to maximize awareness and active usage among the target user groups: DOST Scholars, students, academic researchers, and DOST personnel.

- Institutional Endorsement and Communication: DOST-STII will promote LitPath AI through official internal communications, including email circulars, the DOST-STII website, and announcements distributed to partner universities and DOST scholarship recipients. The institutional credibility of DOST-STII serves as a strong endorsement for the platform.
- Differentiation Through Relevance and Speed: LitPath AI's primary differentiating value proposition is its ability to display relevant research faster and more accurately than any existing tool available to DOST Scholars. Performance benchmarks comparing LitPath AI directly against the current OPAC, in terms of query response time and result relevance, will be used in onboarding materials and institutional communications to demonstrate concrete improvement.
- User Onboarding and Documentation: A comprehensive user guide will be made available within the platform and through DOST-STII library staff. Administrators will be trained to assist users during reference interviews, ensuring smooth adoption.
- Feedback-Driven Improvement and Credibility: The integrated user feedback tool will collect ratings and comments from day one of launch. Aggregated satisfaction data will be used to validate system performance against the target score of at least 3.5 out of 5 and will be shared with stakeholders to build confidence in the system.
- Beta Testing as Social Proof: A structured beta testing phase involving DOST employees and students will precede full launch. Positive feedback gathered during this phase will be used to refine the system and support broader institutional promotion.



Because the platform is free to all authorized users, there are no subscription or pricing barriers to adoption. Marketing costs are minimal, focused primarily on documentation preparation and internal communications. The primary success metric for reach will be active user volume and average satisfaction scores.

Organization and Staffing

The LitPath AI project is developed under the supervision of DOST-STII and executed by HecTech, a four-member development team from Asia Pacific College. The organizational structure is lean and requires no new permanent positions or departmental changes within DOST-STII. The project operates within an existing academic and institutional framework, with clear role description between the client organization and the development team.

Client Organization: DOST-STII

DOST-STII serves as the project owner and primary stakeholder. The institute's library staff and designated project supervisors are responsible for providing the thesis and dissertation dataset (metadata and abstracts), validating system requirements, participating in user acceptance testing, and formally signing off on all deliverables. The admin analytics dashboard is specifically designed for DOST-STII library staff and requires no technical expertise to operate.

Development Team: HecTech

The HecTech team is collectively responsible for all design, development, testing, and documentation work packages. Communication is conducted through daily interactions, weekly team meetings, and sprint board updates managed in OpenProject. Each member maintains current task statuses and escalates blockers immediately to the Project Manager.

Name	Role	Technical Focus	Contact
Tracie Tomon	Project Manager	Project oversight, sprint planning, stakeholder communication	tctomon@student.apc.edu.ph
Charijoy Cempron	System Architect	AI/backend architecture, RAG pipeline, ChromaDB integration	ctcempron@student.apc.edu.ph
Jenine Elaine Dulay	UI/UX Designer	React.js frontend, Tailwind CSS, user experience flow	jcdulay@student.apc.edu.ph
Joshua Sta Ana	Security Lead	Django backend, authentication, admin dashboard, QA	jtstaana2@student.apc.edu.ph

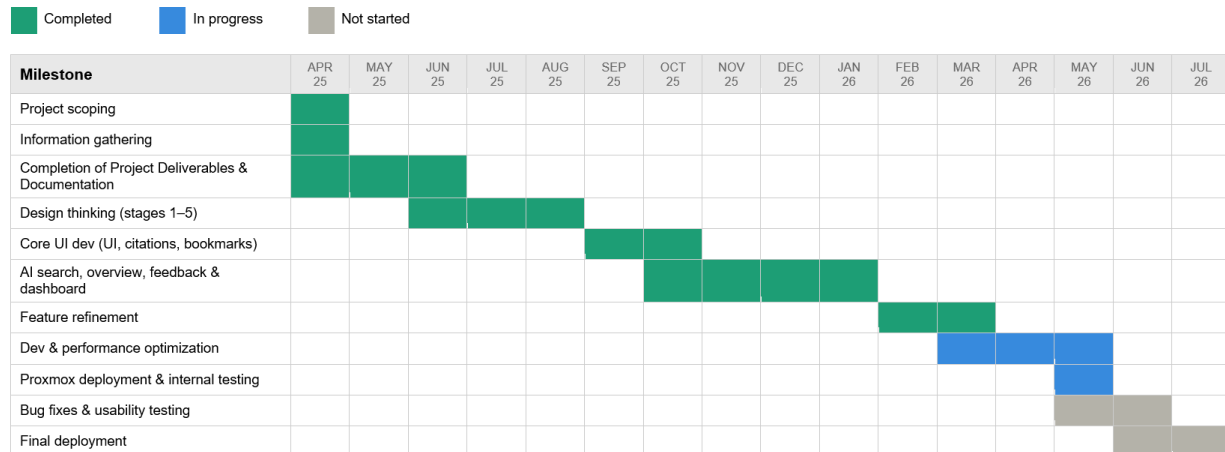
No additional permanent hires are anticipated. Post-deployment, system operation and maintenance will be handled by DOST-STII's existing technical staff using the technical manual and documentation produced as project deliverables.

Schedule

The LitPath AI project targets completion by June-July 2026, following an agile sprint methodology with progress tracked through OpenProject. The following high-level milestones govern the project schedule:

LitPath AI: Smart Pathfinder for Theses and Dissertations

April 2025 – July 2026



Timeline runs Apr 2025 → Jul 2026

Milestone	Target Date	Status
Project Scoping	April 2025	Completed
Information Gathering	April 2025	Completed
Completion of Project Deliverables & Documentation	June 2025	Completed
Design Thinking Process (Stages 1–5)	Aug 2025	Completed
Core UI Development (System UI, Citation Generator, Bookmark)	Sep – Oct 2025	Completed
AI Search Engine, Generate Overview, Feedback Mechanism & Admin Dashboard	Oct 2025 – Jan 2026	Completed
Feature Refinement	Feb – Mar 2026	Completed
Continued Development & Performance Optimization (incl. codebase restructuring)	Mar – May 2026	In Progress
Proxmox Server Deployment & Internal Testing	May 2026	In Progress
Bug Fixes & Usability Testing	May – Jun 2026	Not Started
Final Deployment	Jun – Jul 2026	Not Started

A detailed work breakdown structure and sprint schedule will be established by the HecTech team in OpenProject upon formal project approval. All milestones are subject to stakeholder review and sign-off before proceeding to subsequent phases.



Financial Projections

The financial projections for LitPath AI reflect the cost structure of an academic technology project developed entirely through free, open-source, and zero-cost resources. As a non-commercial platform built for a government institution, no development budget was required from either DOST-STII or the HecTech team. The financial analysis distinguishes between the development phase, which incurred zero cost, and the post-handover operational phase, which will be assumed by DOST-STII as the platform owner.

The following assumptions underline these projections:

- The entire development phase was completed at zero direct cost. All tools, frameworks, libraries, and infrastructure used during development are open-source or available under free-tier plans.
- No honoraria, allowances, or personnel costs were incurred. The HecTech team completed all design, development, testing, and documentation as part of an academic project engagement.
- Development and testing were conducted using existing hardware (Proxmox-based local server), requiring no hardware acquisition.
- Free-tier plans for the Gemini AI API and Cloudflare were used throughout the development and testing phases without incurring charges.
- Upon formal project handover and deployment to production, DOST-STII will assume ownership and management of all operational costs, specifically the Gemini AI API subscription and Cloudflare hosting plan, scaled to actual production usage.
- Post-launch system maintenance will be handled by DOST-STII's existing IT Administrator and Librarian Administrator roles, requiring no additional staffing costs.

Cost Item	Dev Phase (HecTech)	Post-Handover (DOST-STII)
DEVELOPMENT COSTS (HecTech)		
Personnel, Honoraria & Allowances	P0	—
Software Licenses & Tools (React, Django, SentenceTransformers, ChromaDB, Supabase)	P0	—
Hardware (Existing Proxmox server — no new acquisition)	P0	—
Testing, QA, & Documentation	P0	—
Gemini AI API (free tier used during development)	P0	—
Cloudflare Deployment (free tier used during development)	P0	—



TOTAL DEVELOPMENT COST (HecTech)	₱0	—
POST-HANDOVER OPERATIONAL COSTS (DOST-STII)		
Gemini AI API (production tier — scaled to usage volume)	—	See recommendation below*
Cloudflare Hosting Plan (production tier — scaled to traffic)	—	See recommendation below*
System Maintenance (existing IT Admin & Librarian Admin roles)	—	₱0 (existing staff)

Note:

* *Gemini AI API — The team recommends Gemini 2.5 Flash as the most suitable model for an institutional library platform, offering strong AI performance at a cost-efficient rate. The free tier supports up to 1,500 requests per day at no cost, which may be sufficient for initial deployment. Should usage exceed free-tier limits, the paid tier is estimated at approximately \$0.30 per million input tokens and \$2.50 per million output tokens (roughly ₱350 – ₱1,700/month depending on query volume). DOST-STII's IT Administrator may evaluate and select the appropriate tier upon taking ownership of the system.*

** *Cloudflare Hosting Plan — A standard cloud hosting plan is estimated at approximately ₱0 – ₱570/month depending on the selected service tier. The free tier may remain sufficient for initial institutional usage. DOST-STII may upgrade to a paid plan as traffic scales.*

The project delivers a fully operational AI-powered research platform at zero development cost to the institution. The only post-handover expenses are the Gemini AI API and Cloudflare hosting, both of which offer scalable pricing tiers, including free tiers that may remain sufficient depending on production traffic volume. These estimates are provided as a reference recommendation for DOST-STII's planning purposes. The IT Administrator may evaluate actual usage patterns post-deployment and select the most appropriate service tiers accordingly. The projected monthly operational cost ranges from ₱0 (if free tiers remain sufficient) to approximately ₱2,270 at full paid-tier usage, both well within the budget expectations of a government institutional technology initiative. The institutional value delivered, a measurably faster and more efficient research experience compared to the existing OPAC, improved search relevance to a minimum of 70%, and a modern AI-powered discovery platform for DOST Scholars and researchers, significantly outweighs any ongoing operational costs.

Findings and Recommendations

Based on the analysis presented across all sections of this feasibility study, it is recommended that DOST-STII formally approve the LitPath AI project and authorize the



HecTech team to proceed with development. The study findings consistently support the technical viability, organizational readiness, market relevance, and financial soundness of the proposed system. The only open items, Gemini AI API tier and Cloudflare deployment plan, are minor decisions with low-cost or free options available, and do not represent blockers to project initiation.

Key findings by dimension are summarized below:

Technology

- The full technology stack (React.js, Django, SentenceTransformers, ChromaDB, Supabase, RAG pipeline) is defined, proven, and in use in comparable academic systems globally, substantially reducing technical risk.
- Open-source frameworks eliminate licensing costs for most of the stack.
- Existing Proxmox-based hardware is sufficient for development and testing phases, avoiding hardware capital expenditure.
- The system will be hosted on a cloud-based web hosting platform providing standard security (HTTPS/SSL) and DDoS protection, accessible to any user with an internet connection at potentially zero cost using available free tiers.
- The HecTech team has technical competencies, covering system architecture, AI/ML, frontend design, backend development, and security, to deliver the full system within the June 2026 deadline.

Market and User Need

- A clear and documented gap exists between current OPAC capability and user needs: fewer than 70% of results match user intent, queries take over a minute, and essential features like citation generation are absent.
- No existing platform provides semantic search specifically over the DOST-STII thesis collection, confirming that LitPath AI fills a genuine, unmet institutional need.
- The target user base is well-defined and accessible through DOST-STII's existing communication channels.

Organization

- No new permanent staffing or structural changes are required within DOST-STII.
- Post-deployment maintenance can be handled by DOST-STII's existing technical staff using the project's documentation.
- The HecTech team's structured use of OpenProject for sprint tracking and daily communication ensures accountability and on-time delivery.

Financial

- The entire development phase was completed at zero cost to both HecTech and DOST-STII. All tools, frameworks, hardware, and cloud services used during development were sourced from free and open-source solutions.



- Post-handover, the only operational expenses are the Gemini AI API (recommended: Gemini 2.5 Flash, estimated at ₱0–₱1,700/month depending on usage tier) and web hosting (estimated at ₱0–₱570/month). Both are manageable within a government institutional technology budget.
- The absence of licensing fees, personnel costs, and hardware acquisition makes LitPath AI an exceptionally cost-efficient solution for DOST-STII.
- The project delivers a fully functional, production-ready AI platform at no direct cost to the institution during development, a strong return on institutional investment.

In conclusion, LitPath AI represents a high-value, zero-cost investment for DOST-STII during the development phase. The project directly addresses a well-documented institutional problem with a well-defined technical solution, a capable development team, and a clear June-July 2026 delivery target. With the entire development completed at no cost and post-handover operational expenses remaining minimal and manageable, approval of this project is strongly recommended.

07 Stakeholder Management Strategy

Introduction

This Stakeholder Management Strategy outlines how we will manage the people and organizations involved in or affected by the LitPath AI: Smart Pathfinder for Theses and Dissertations project.

The main goal of this strategy is to ensure that all stakeholders, from government agency leaders and technical staff to faculty advisers and student developers, are properly identified, understood, and engaged throughout the project's lifecycle. Effective stakeholder management helps us gain support, reduce resistance, and balance competing interests. Ignoring stakeholder needs can lead to misunderstandings, delays, or project failure.

Using this document, we will:

- 1) Identify and classify all project stakeholders
- 2) Determine each stakeholder's level of power, interest, and influence
- 3) Define how we will manage and communicate with each stakeholder group

By doing this early in initiation phase, we can:

- 1) Gain input from influential stakeholders to guide project planning
- 2) Minimize conflicts caused by differing objectives



- 3) Maximize the resources, support, and cooperation needed to complete LitPath AI successfully

Early identification and open communication are critical. Some stakeholders may benefit from the project, while others may have concerns, such as budget allocation, changing requirements, system integration issues, or unfamiliar interface changes. By managing stakeholder relationships proactively, we can address these concerns early and keep the project on track.

In summary, this Stakeholder Management Strategy will help the LitPath AI project team work effectively with DOST-STII, APC faculty, student developers, and the end users, ensuring that everyone's interests are understood and balanced as we build a smarter, faster research discovery tool for theses and dissertations.

Identify Stakeholders

The LitPath AI Project Team, consisting of student developers from APC, will conduct an internal brainstorming session to identify stakeholders for the project. This session will be based on available project documents, client communications, and faculty guidance. Because this is a Project-Based Learning (PBL) setup with limited direct access to DOST-STII personnel, the initial stakeholder identification will be done by the developer team. The session will be divided into two parts. The first part will focus on internal stakeholders within APC, including student project developers and SoCIT project advisers. The second part will focus on external stakeholders outside APC, which may include DOST-STII (Science Research Specialists) and end users of the research discovery tool.

Once the developer team has completed its initial stakeholder list, the list will be reviewed and validated by the project advisers to ensure completeness and accuracy. If any stakeholders were missed, they will be added at this validation step. Furthermore, if new stakeholders emerge during the project lifecycle, they will be added and managed accordingly. Based on actual project interactions, additional DOST-STII personnel such as a technical specialist, the head librarian, or the department head may occasionally attend meetings and should be considered stakeholders if they meet the criteria below.

To decide whether a person or organization should be considered a stakeholder, the project team will ask five questions.

- 1) Will the person or organization be directly or indirectly affected by this project?
- 2) Does the person or organization hold a position from which they can influence the project?
- 3) Does the person have an impact on the project's resources, including material, personnel, or funding?
- 4) Does the person or organization have any special skills or capabilities the project will require?
- 5) Does the person potentially benefit from the project, or are they in a position to resist this change?



Any individual or organization that meets one or more of these criteria will be identified as a stakeholder.

Key Stakeholders

For the LitPath AI project, the following stakeholders have been identified as key stakeholders. From DOST-STII, Assistant Secretary Napoleon K. Juanillo, Jr. PhD is a key stakeholder because he has high influence over budget allocations and project approval. The two Science Research Specialists, Khasian Eunice Romulo (Primary POC) and Jonathan Abalon (Alternate POC), are key stakeholders because they define requirements, approve deliverables, communicate with the Office of the Solicitor General, and have direct influence over project success.

From APC, the PBL Coordinator Manuel Sebastian S. Sanchez is a key stakeholder because he sources projects and manages the partnership with DOST-STII. The SoCIT Faculty or Project Advisers are key stakeholders because they provide project guidance, technical mentorship, approve deliverables, and evaluate academic outcomes. The student developers are also key stakeholders because they are directly responsible for system development and project delivery. The end user is not considered a key stakeholder for the purposes of intensive management but will be kept informed as users.

Stakeholder Analysis

Once all LitPath AI stakeholders have been identified, the project team will categorize and analyze each stakeholder. The purpose of this analysis is to determine the stakeholders' level of power or influence, plan the management approach for each stakeholder, and to determine the appropriate levels of communication and participation each stakeholder will have on the project.

The project team will categorize stakeholders based on their organization or department. Once all stakeholders have been categorized, the project team will utilize a power and interest matrix to illustrate the potential impact each stakeholder may have on the project. Based on this analysis the project team will also complete a stakeholder analysis matrix which illustrates the concerns, level of involvement, and management strategy for each stakeholder.

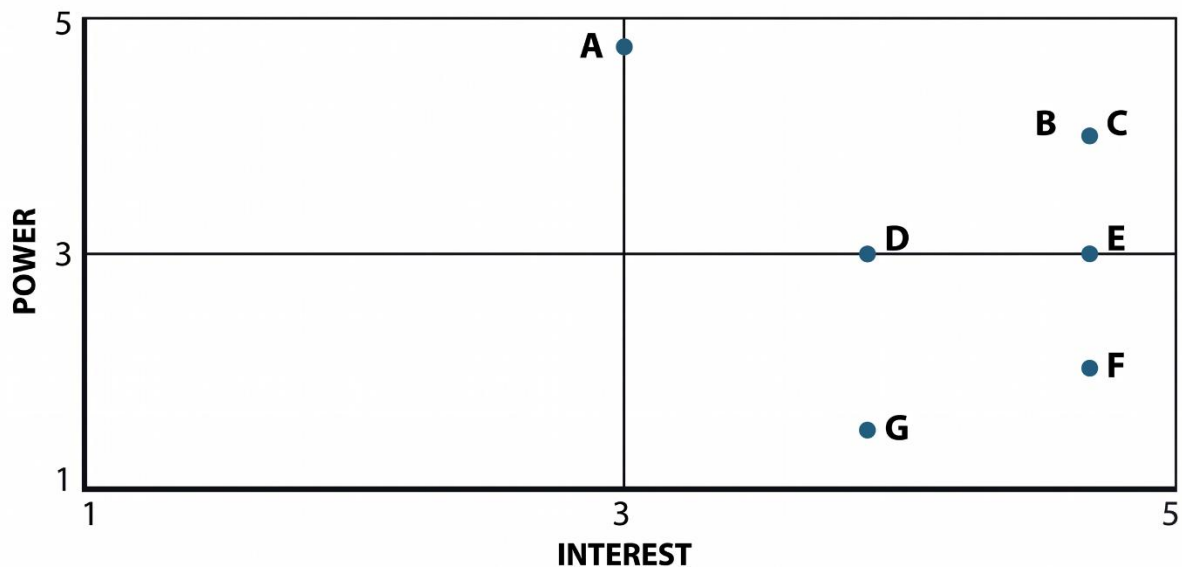
The chart below will be used to establish stakeholders and their levels of power and interest for use on the power and interest chart as part of the stakeholder analysis.

Key	Organization	Name	Power (1-5)	Interest (1-5)
A	DOST-STII	Assistant Secretary Napoleon K. Juanillo, Jr. PhD	5	3
B	DOST-STII	Science Research Specialist II - Khasian	4	5



		Eunice Romulo (Primary POC)		
C	DOST-STII	Science Research Specialist II - Jonathan Abalon (Alternate POC)	4	5
D	APC	PBL Coordinator - Manuel Sebastian S. Sanchez	3	4
E	APC	SoCIT Faculty / Project Advisers	3	5
F	APC	Student Developers	2	5
G	External	End Users	1	4

Below is the power and interest chart for the LitPath AI stakeholders. Each letter represents a stakeholder in accordance with the key in the chart above.



POWER: Refers to a stakeholder's level of direct or indirect influence and authority over project decisions, resources, and final outcomes.

INTEREST: Refers to the degree of importance or priority a stakeholder places on the project's goals, deliverables, and success, which affects their desire to be involved.

Quadrant Descriptions:



- **Manage Closely (High Power, High Interest): (B, C)** These stakeholders are critical for success and require maximum attention and active engagement to understand and satisfy their needs.
- **Keep Satisfied (High Power, Low/Mid Interest): (A)** These stakeholders have significant authority but lower engagement. Keep them content with major milestones and decisions to prevent them from becoming impediments.
- **Keep Informed (Low Power, High Interest): (D, E, F, G)** These stakeholders are highly invested in the outcome but have limited influence. Regular updates and opportunities for input are key to maintaining their support and managing their expectations.
- **Monitor (Low Power, Low/Mid Interest): (Empty)** These stakeholders have minimal interest and minimal impact. Monitor them for any changes in their position or status with minimal effort.

The stakeholder analysis matrix will be used to capture stakeholder concerns, level of involvement, and management strategy based on the stakeholder analysis and power and interest matrix above. The stakeholder analysis matrix will be reviewed and updated throughout the project's duration to capture any new concerns or stakeholder management strategy efforts.

Name	Department / Company	Position	Objectives, Requirements, Interests	Project Contribution	Resistance	Power and Interest
Mr. Napoleon Juanillo, Jr. PhD	DOST-STII	Assistant Secretary for Technology Transfer, Communications and Commercialization and Officer-in-Charge, Office of the Director	Research commercialization and partnership development	Provides projects to be worked on by the students	Concerned about budget allocations	High
Ms. Khasian Romulo	DOST-STII	Science Research Specialist II	Efficient research discovery and modernize search capabilities	Requirements definition, supports work, and approves deliverables	Concerned about changing initial requirements mid-project	High
Mr. Jonathan Abalon	DOST-STII	Science Research Specialist II	Improve user satisfaction and increase research utilization	System access and technical support	Concerned about the problems that might arise during system integration	High
Mr. Sean Sanchez	APC	Project Based Learning Coordinator	Educational innovation and partnership development	Project sourcing	Concerned about partner organizations modifying project scope	High
Mr. Gonzalo Guumogda	APC	SoCIT Faculty/ Project Adviser	Student academic success and skill development	Project guidance, technical mentorship and approves deliverables	Concerned about timeline delays of the deliverables	High
Mr. Jose Quesada	APC	SOCIT Professor	Student academic success and learning outcomes	Technical mentorship and academic evaluation	Concerned about insufficient time for mentorship due to multiple student commitments	Medium
End Users	N/A	End Users	Faster research discovery and friendly search interface	User feedback and system testing	Concerned about the interface change of the system	Medium
HecTech Team	APC	Project Developers	Academic achievement and successful project delivery	System development and client communication	Concerned about technical challenges and time constraints	High



07B Stakeholder Analysis

Stakeholder Analysis

Name	Department / Company	Position	Objectives, Requirements, Interests	Project Contribution	Resistance	Power and Interest
Mr. Napoleon Juanillo, Jr. PhD	DOST-STII	Assistant Secretary for Technology Transfer, Communications and Commercialization and Officer-in-Charge, Office of the Director	Research commercialization and partnership development	Provides projects to be worked on by the students	Concerned about budget allocations	High
Ms. Khasian Romulo	DOST-STII	Science Research Specialist II	Efficient research discovery and modernize search capabilities	Requirements definition, supports work, and approves deliverables	Concerned about changing initial requirements mid-project	High
Mr. Jonathan Abalon	DOST-STII	Science Research Specialist II	Improve user satisfaction and increase research utilization	System access and technical support	Concerned about the problems that might arise during system integration	High
Mr. Sean Sanchez	APC	Project Based Learning Coordinator	Educational innovation and and partnership development	Project sourcing	Concerned about partner organizations modifying project scope	High
Mr. Gonzalo Guumogda	APC	SoCIT Faculty/ Project Adviser	Student academic success and skill development	Project guidance, technical mentorship and approves deliverables	Concerned about timeline delays of the deliverables	High
Mr. Jose Quesada	APC	SOCIT Professor	Student academic success and learning outcomes	Technical mentorship and academic evaluation	Concerned about insufficient time for mentorship due to multiple student commitments	Medium
End Users	N/A	End Users	Faster research discovery and friendly search interface	User feedback and system testing	Concerned about the interface change of the system	Medium
HecTech Team	APC	Project Developers	Academic achievement and successful project delivery	System development and client communication	Concerned about technical challenges and time constraints	High

[07B Stakeholder Analysis Excel file](#)

08 Statement of Work (SOW)

Introduction

LitPath AI: Smart Pathfinder for Theses and Dissertations is an AI-powered, standalone web application developed for the DOST-STII Library. Operating independently from the existing Online Public Access Catalog (OPAC), the project addresses critical limitations found in the current system, such as exact-match keyword searches that often return irrelevant results and slow response times that hinder research efficiency. The goal is to implement a dedicated, AI-powered search engine that significantly improves search precision and the overall discovery of the library's collection of graduate-level academic materials.



Scope of Work

The scope of the LitPath AI project includes the delivery of the following tangible outputs:

In-Scope:

1. **AI-Powered Search Engine:** Development of a semantic search engine utilizing Retrieval-Augmented Generation (RAG) and Gemini AI for advanced query understanding and re-ranking.
2. **Standalone Web Application:** A fully functional, independent user interface that operates separately from the existing DOST-STII OPAC.
3. **Client-Provided Database:** An independent backend database populated exclusively with the sample thesis and dissertation papers provided directly by DOST-STII for the system's search repository.
4. **Automated Citation Generator:** A built-in tool that automatically formats citations for selected research materials in APA, MLA, Chicago Manual of Style, and IEEE formats.
5. **Research Management & Feedback Tools:** A bookmarking feature allowing users to save and organize their discovered research materials, alongside an integrated feedback tool to capture user ratings and comments.
6. **Admin Analytics Dashboard:** A dedicated dashboard providing DOST-STII library administrators with visibility into search trends, usage patterns, and user interactions.

Out-of-Scope:

1. **Digitization:** Full-text access to or the physical digitization of non-digitized library materials.
2. **Collection Management:** Direct management, deletion, or alteration of the library's primary collection records or physical materials.

Period of Performance

The project spans the academic year 2025-2026. Key development milestones began with initial sprint planning in November 2025, with the roadmap extending through three academic terms, targeting final production deployment and handover by June–July 2026. Project execution is constrained by a part-time academic schedule, limiting capacity to 10 hours per week per team member.

Place of Performance

- **Development Phase:** Work is conducted on the development team's local environments.
- **Testing Phase:** Internal and unit testing will be conducted on a Proxmox server environment.
- **Deployment:** The final standalone system will be hosted in the client's (DOST-STII) designated environment. Cloud hosting and AI/LLM API costs will be fully covered by DOST-STII.



Work Requirements

As part of the Website Redesign Project APC-HecTech will be responsible for performing tasks throughout various stages of this project. The following is a list of these tasks which will result in the successful completion of this project:

Kickoff:

- APC-HecTech will create and present detailed project plan including schedule, WBS, testing plan, implementation plan, training plan, and transition plan
- APC-HecTech will present project plan to DOST-STII/ScINET-PHIL Library for review and approval

Design Phase:

- Work with DOST-STII/ScINET-PHIL to gather requirements and establish metrics.
- Create site design based on collected requirements.
- Develop site design proposal for DOST-STII/ScINET-PHIL review and approval.
- Present written status at weekly meeting.

Build Phase:

- APC-HecTech will complete all coding for approved site design
- APC-HecTech will provide DOST-STII/ScINET-PHIL with a detailed testing plan
- APC-HecTech will include all content provided by DOST-STII/ScINET-PHIL on redesigned web site
- APC-HecTech will conduct testing in both their iLab as well as in a limited beta release
- APC-HecTech will resolve any coding and site issues identified in testing
- APC-HecTech will compile a testing report to present to SCG for review/approval
- Present written status at weekly meeting

Implementation Phase:

- APC-HecTech will implement the newly redesigned web site on DOST-STII/ScINET-PHIL servers
- APC-HecTech will begin providing 24x7 web site support at this point forward until the end of the period of performance
- Present written status at weekly meeting

Training Phase:

- APC-HecTech will provide training in accordance with approved training plan provided in the kickoff
- Present written status at weekly meeting

Project Handoff/Closure:



- APC-HecTech will provide DOST-STII/ScINET-PHIL with all documentation in accordance with the approved project plan
- APC-HecTech will present project closure report to DOST-STII/ScINET-PHIL for review and approval
- APC-HecTech will complete the project requirements checklist showing that all project tasks have been completed
- APC-HecTech will conclude 24x7 web support at 11:59pm on the final day of the period of performance
- Present written status at weekly meeting

Schedule/Milestones

The following milestones mark the key checkpoints of the LitPath AI project:

Milestone	Target Date	Status
Project Scoping	April 2025	Completed
Information Gathering	April 2025	Completed
Completion of Project Deliverables & Documentation	June 2025	Completed
Design Thinking Process (Stages 1–5)	Aug 2025	Completed
Core UI Development (System UI, Citation Generator, Bookmark)	Sep – Oct 2025	Completed
AI Search Engine, Generate Overview, Feedback Mechanism & Admin Dashboard	Oct 2025 – Jan 2026	Completed
Feature Refinement & Client Feedback Integration	Feb – Mar 2026	Completed
Continued Development & Performance Optimization (incl. codebase restructuring)	Mar – Apr 2026	In Progress
Proxmox Server Deployment & Internal Testing	Apr 2026	In Progress
Bug Fixes & Usability Testing	May – Jun 2026	Not Started
Final Deployment	Jun – Jul 2026	Not Started

Acceptance Criteria

For the LitPath AI Project, the acceptance of all deliverables will reside with the Project Sponsors, Khasian Eunice Romulo and Jonathan Abalon, representing DOST-STII.

Once a project phase is completed, the HecTech team will provide their report for review. The Project Sponsors will either sign off on the approval for the next phase to begin or provide actionable feedback. Schedule changes exceeding a 10% threshold variance must also be explicitly approved by the Project Sponsors.

Final acceptance of the project requires that:

1. The project delivers a 100% working standalone application with all defined features (AI search, citation generator, bookmarking, admin dashboard) fully completed.
2. The system achieves at least 70% relevance accuracy for its top 10 results.



3. The system achieves an average user satisfaction score of at least 3.5 out of 5 from the test group.

Once all project tasks have been completed, the project will enter the handoff stage. The HecTech team will provide their final report and documentation package. The formal sign-off by the Project Sponsors will acknowledge acceptance of all project deliverables.

Other Requirements

- Database Constraints: The project team will utilize an independent database populated with sample papers provided by DOST-STII. Direct access to the live DOST-STII production database is not required and will not be provided.
- Data Privacy: Compliance with the Data Privacy Act of 2012 (RA 10173) is required for administrative accounts; public search functions must remain anonymous.
- Security: All connections must be secured via HTTPS/SSL.
- Personnel & Leadership: Development is executed exclusively by the four-member HecTech team with no external outsourcing. Project leadership and management responsibilities are shared equally among the team members: Tracie Tomon, Joshua Sta Ana, Jenine Elaine Dulay, and Charijoy Cempron.



02 Planning Phase

02 Communication Management Plan

Introduction

This Communications Management Plan sets the communications framework for the LitPath AI project. It will serve as a guide for communications throughout the life of the project from its current development phase through final deployment in June–July 2026 and will be updated as communication needs change. This plan identifies and defines the roles of all persons involved in the project. It includes a communications matrix which maps the communication requirements of this project, an in-depth guide for conducting meetings, and a project team directory providing contact information for all stakeholders directly involved in the project.

Communications Management Approach

Effective communication is one of the most critical factors in the success of the LitPath AI project. Given that the project involves multiple stakeholders including the DOST-STII client and sponsor, the development team, the PBL instructor, and end users, considerable thought has been given to how communications will be managed across all phases of the project.

The Project Manager, Tracie Tomon, is authorized to interface with management, negotiate for resources, and communicate with all stakeholders to ensure the timely and successful completion of LitPath AI. The Project Manager will take a proactive role in ensuring that all communications are clear, timely, and aligned with the project's objectives and milestone schedule.

Given the academic and client-facing nature of this project, communications will be structured around two key tracks:

- Internal team communications — handled through weekly team meetings, sprint boards in OpenProject, and a group messaging platform (Messenger or Microsoft Teams) for day-to-day coordination among the four HecTech members.
- External stakeholder communications — handled through Viber, scheduled client meetings with DOST-STII, and documentation submissions to the PBL instructor aligned with milestone completion dates.

By maintaining this dual-track approach, the team ensures that development progress stays transparent and that stakeholders such as the Project Sponsor, Khasian Eunice Romulo, remain informed without being burdened by operational-level detail.

The communications requirements are documented in the Communications Matrix in this plan, which serves as the definitive guide for what to communicate, who communicates it, how frequently, and to whom. As with any project, updates or changes to this plan may be necessary as the project progresses, particularly with upcoming milestones such as the April 2026 internal testing, May–June 2026 usability testing, and the June–July 2026 final deployment. Any changes to this plan will be reviewed and approved by the Project Sponsor before taking effect, and the updated plan will be redistributed to the project team and all stakeholders.



Communications Management Constraints

All LitPath AI project communication activities will occur within the project's approved schedule and resource allocations. Because LitPath AI does not operate with a formal standing budget, the primary constraints are time and the availability of the four-person HecTech development team. AI/LLM API costs are anticipated and will be shouldered by DOST-STII, while system hosting is managed internally by the team using on-premises infrastructure.

The Project Manager is responsible for ensuring all communication activities are carried out by team members without exceeding available time or resources. The project team must exercise care in handling confidential information related to this project, including system architecture details, DOST-STII database structures, and user data. Distribution of such information is strictly controlled and must be authorized by the Project Manager or Project Sponsor prior to external release.

All formal project communications will use the standard templates and formats established by the team. Only the Project Manager or the Project Sponsor may authorize the distribution of sensitive information related to the DOST-STII library database, system architecture, or user data. The Project Manager is responsible for obtaining this approval before any such information is distributed externally.

Stakeholder Communication Requirements

As part of identifying all project stakeholders, the Project Manager will communicate with each stakeholder to determine their preferred frequency and method of communication. This feedback will be maintained by the Project Manager in the project's Stakeholder Register. Standard project communications will occur in accordance with the Communication Matrix; however, depending on the identified stakeholder communication requirements, individual communication is acceptable and within the constraints outlined for this project.

In addition to identifying communication preferences, stakeholder communication requirements must identify the project's communication channels and ensure that stakeholders have access to these channels. If project information is communicated via secure means or through internal team resources, all stakeholders, internal and external, must have the necessary access to receive project communications.

Once all stakeholders have been identified and communication requirements are established, the project team will maintain this information in the project's Stakeholder Register and use this, along with the project communication matrix, as the basis for all communications.

Roles

Project Sponsor

Khasian Eunice Romulo

Science Research Specialist II / Unit Head, Digital Information and Processing Delivery Unit, DOST-STII. She authorized the project by signing the Project Charter and is the primary client. As sponsor, she is responsible for approving changes to scope and milestone dates, and for shouldering cloud and AI/LLM operational costs. Communications to the sponsor should be presented in summary format unless she requests more detail.



Project Manager

Tracie Tomon

The Project Manager is authorized to interface with management, negotiate for resources, delegate responsibilities, and communicate with all stakeholders. She is responsible for developing the project plan, monitoring schedule and scope, assigning tasks, overseeing system integration, testing, and documentation, and organizing final outputs for defense and presentation. The Project Manager is the primary communicator for the project.

Key Stakeholders

Key Stakeholders for LitPath AI include the PBL instructor who evaluates milestone deliverables, the DOST-STII Library staff who will use and maintain the system, and student researchers and professionals who are the primary end users of the platform. These stakeholders require timely updates on milestone completions, usability testing results, and deployment readiness, and their feedback is incorporated into the project's continuous improvement cycle.

Change Control Board

The Change Control Board for LitPath AI consists of the Project Sponsor (Khasian Eunice Romulo) and the Project Manager (Tracie Tomon). This board reviews requests for changes to project scope, technical architecture, or milestone dates. Technical design documents, user impact analyses, and revised implementation strategies are the types of communication this group requires. No change may be implemented without the board's approval.

Customer

The DOST-STII Library is the customer and will be involved in reviewing prototypes, approving design stages, and formally accepting the final deliverables. The customer will be kept informed of project status, including any potential impacts to the schedule or the system itself. Communications with the customer are primarily handled through Viber and scheduled face-to-face or online meetings.

Project Team

The Project Team is comprised of the four HecTech members responsible for all design, development, testing, and documentation work packages: Tracie Tomon, Charijoy Cempron, Jenine Elaine Dulay, and Joshua Sta Ana. The team requires detailed communication through daily interactions, weekly team meetings, and sprint board updates in OpenProject. Each member is expected to keep task statuses current and flag blockers immediately to the Project Manager.

Steering Committee

The Steering Committee for this project is represented by the PBL instructor and DOST-STII management, who provide strategic oversight and ensure that the project's direction aligns with academic requirements and institutional goals. The Steering Committee requires communication on matters that will change the scope of the project and its deliverables and will be updated at key milestone checkpoints.

Technical Lead

Joshua Sta Ana

Responsible for ensuring all technical aspects of the project are addressed, including the AI search engine architecture, system turnover with the DOST-STII OPAC, and technical documentation. Requires close coordination with the Project Manager and the rest of the development team.



Project Team Directory

The following table presents contact information for all people identified in this communications management plan. The email addresses and phone numbers on this table will be used to communicate with these people.

Role	Name	Title	Organization/ Department	Email
Project Sponsor	Khasian Eunice Romulo	Science Research Specialist II / Unit Head	DOST-STII	(via Viber)
Project Manager	Tracie Tomon	Group Leader	HecTech / Asia Pacific College	tctomon@student.apc.edu.ph
Project Team	Charijoy Cempron	Developer	HecTech / Asia Pacific College	ctcempron@student.apc.edu.ph
Project Team	Jenine Elaine Dulay	Developer	HecTech / Asia Pacific College	jcdulay@student.apc.edu.ph
Project Team	Joshua Sta. Ana	Developer	HecTech / Asia Pacific College	jstaana2@student.apc.edu.ph
PBL Instructor	Jose Eugenio L. Quesada	Instructor	Asia Pacific College	jlquesada@apc.edu.ph

Communication Methods and Technologies

The project team will determine the communication methods and technologies based on several factors including stakeholder communication requirements, available technologies (internal and external), and the needs of the LitPath AI project.

HecTech maintains a SharePoint platform which all project members use to provide updates, archive various reports, and conduct project communications. This platform enables the Project Sponsor and DOST-STII staff, as well as other stakeholders with compatible technology, to access project data and communications at any point in time. SharePoint also provides the ability for stakeholders and project team members to collaborate on project work and communication.

For stakeholders who do not have the ability to access SharePoint, relevant documents will be shared via Viber or a direct SharePoint link by the Project Manager. The Project Manager is responsible for ensuring all project communications and documentation are uploaded to SharePoint and always kept current.

HecTech uses OpenProject as its primary project management platform. All project team members are responsible for developing, maintaining, and communicating schedules, work packages, backlogs, and sprint progress using this platform. The Gantt chart is the preferred format for communicating schedules to stakeholders. The project schedule will be maintained on both OpenProject and the SharePoint platform.



All project communication and documentation, in addition to being maintained on SharePoint and OpenProject, will be archived in the HecTech shared project directory.

The following communication tools are used by the project team:

- Face-to-face meetings — for client meetings with DOST-STII, design reviews, and formal milestone check-ins.
- Video conferencing (MS Teams) — for remote team meetings and instructor check-ins when in-person is not possible.
- Viber — for formal communications with the Project Sponsor and DOST-STII staff; used for distributing reports, documentation, and milestone updates.
- Microsoft Teams — for informal day-to-day communication among the four HecTech team members.

Communications Matrix

The following table identifies the communications requirements for this project.

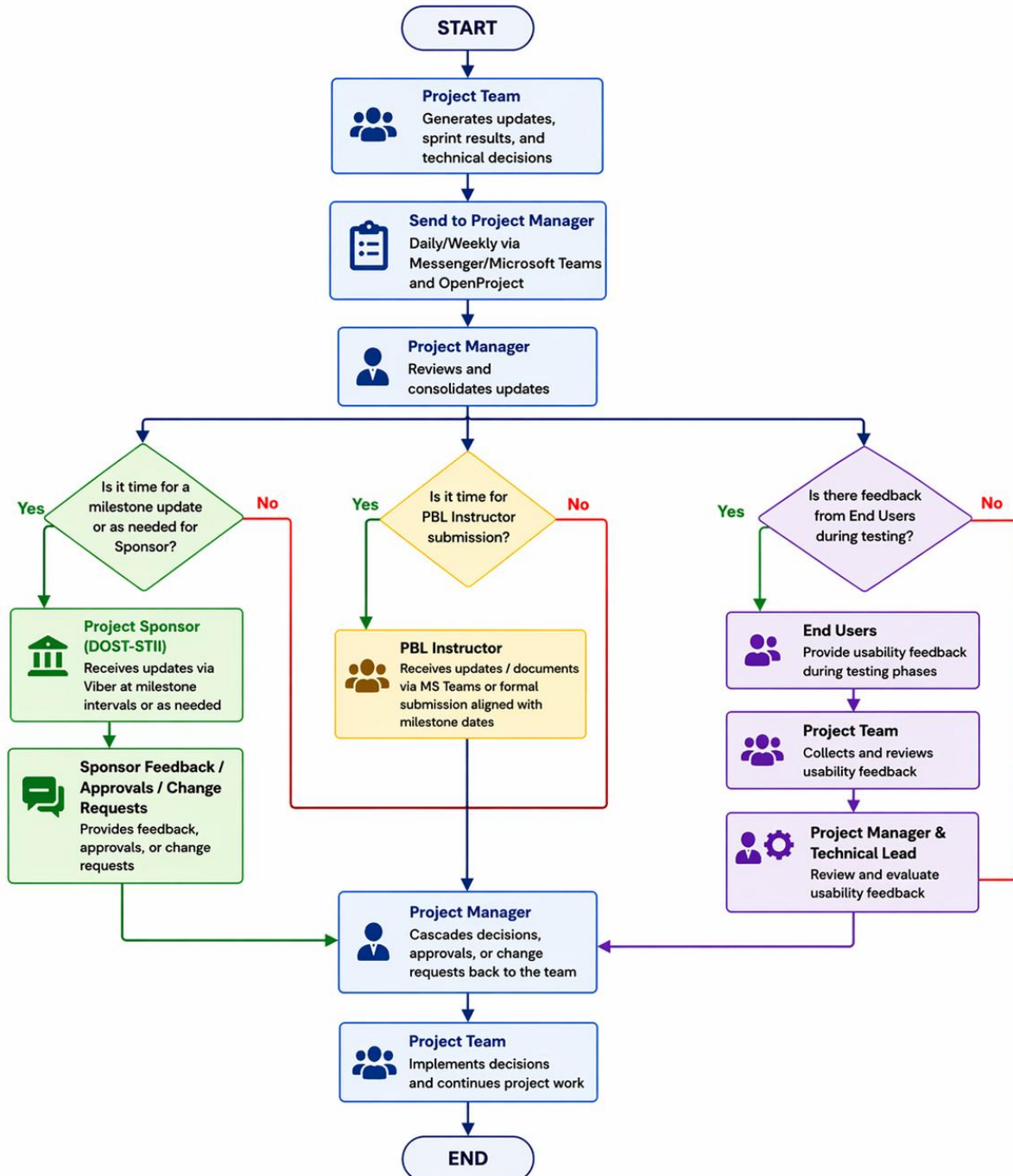
Communication Type	Objective of Communication	Medium	Frequency	Audience	Owner	Deliverable	Format
Sprint Planning	Define sprint goals, scope, and tasks for the team	Face to face or Microsoft Teams	Once per sprint	Project Team	Project Manager	Sprint Plan, Meeting Minutes	Video recording stored on Microsoft Teams channel and SharePoint
Daily Stand-up	Provide daily progress updates and identify blockers	Face to face or Microsoft Teams	Daily during active sprints	Project Team	Project Manager	Stand-up Meeting Minutes	Video recording stored on Microsoft Teams channel and SharePoint
Backlog Refinement	Review and update backlog items for upcoming work	Face to face or Microsoft Teams	Once per sprint	Project Technical Staff	Project Manager	Updated Product Backlog	Video recording stored on Microsoft Teams channel, OpenProject, and SharePoint
Sprint Review	Present completed work and gather feedback from stakeholders	Face to face or Microsoft Teams	Once per sprint	Project Team, Project Adviser, Client	Project Manager	Sprint Demo, Meeting Minutes	Video recording stored on Microsoft Teams channel and SharePoint
Sprint Retrospective	Evaluate sprint performance and identify improvements	Face to face or	Once per sprint	Project Team	Project Manager	Retrospective Meeting Minutes	Video recording stored on



		Microsoft Teams					Microsoft Teams channel and SharePoint
Project Status Reports	Report on overall project progress, issues, and updates	Face to face or Microsoft Teams	Per sprint	Project Sponsor, Project Adviser	Project Manager	Project Status Report	Soft copy sent via Viber
Client Coordination Meeting	Discuss project progress and align with client expectations	Face to face or Microsoft Teams or Viber	As needed	Project Manager, Client	Project Manager	Meeting Minutes	Soft copy archived in SharePoint

Communication Flowchart

The communication flowchart below was created to aid in project communication. This flowchart provides a framework for the project team to follow for this project. However, there may be occasions or situations which fall outside of the communication flowchart where additional clarification is necessary. In these situations, the Project Manager is responsible for discussing communication with the Project Sponsor and deciding on how to proceed.



Guidelines for Meetings

Meeting Agenda

Meeting Agenda will be distributed 5 business days in advance of the meeting. The agenda should identify the presenter for each topic along with a time limit for that topic. The first item on the agenda should be a review of action items from the previous meeting.



Meeting Minutes

Meeting minutes will be distributed within 2 business days following the meeting. Meeting minutes will include the status of all items on the agenda along with new action items and the Parking Lot list.

Action Items

Action Items are recorded in both the meeting agenda and minutes. Action items will include both the action items along with the owner of the action item. Meetings will start with a review of the status of all action items from previous meetings and end with a review of all new action items resulting from the meeting. The review of the new action items will include identifying the owner of each action item.

Meeting Chairperson

The Chairperson is responsible for distributing the meeting agenda, facilitating the meeting and distributing the meeting minutes. The Chairperson will ensure that the meeting starts and ends on time and that all presenters adhere to their allocated time frames.

Note Taker

The Note Taker is responsible for documenting the status of all meeting items, maintaining a Parking Lot item list and taking notes of anything else of importance during the meeting. The Note Taker will give a copy of their notes to the Chairperson at the end of the meeting as the Chairperson will use the notes to create the Meeting Minutes.

Timekeeper

The Timekeeper is responsible for helping the facilitator adhere to the time limits set in the meeting agenda. The Timekeeper will let the presenter know when they are approaching the end of their allocated time. Typically, a quick hand signal to the presenter indicating how many minutes remaining for the topic is sufficient.

Parking Lot

The Parking Lot is a tool used by the facilitator to record and defer items which aren't on the meeting agenda; however, it merits further discussion later or through another forum.

A parking lot record should identify the owner of the item as that person will be responsible for ensuring follow-up. The Parking Lot list is to be included in the meeting minutes.

Communication Standards

Standardization helps simplify the complexities of project management communications and ensures that all team members and stakeholders have a consistent understanding of what to expect.

For this project, HecTech will utilize standard templates and formats for all formal project communications. Formal project communications are detailed in the project's communication matrix and include:

- **Kickoff Meeting** — HecTech will use standard templates for meeting agenda and meeting minutes. Any slides presented will use the HecTech standard slideshow template.



- **Weekly Team Meetings** — HecTech will use standard templates for meeting agenda and meeting minutes. Any slides presented will use the HecTech standard slideshow template.
- **Technical Design Meetings** — HecTech will use standard templates for meeting agenda and meeting minutes. Any slides presented will use the HecTech standard slideshow template.
- **Milestone Status Updates / Client Review Meetings** — HecTech will use standard templates for meeting agenda and meeting minutes. Status reports and presentations will use the HecTech standard slideshow and report templates.
- **Project Status Reports** — HecTech will use the standard project status report document, available on the SharePoint shared drive, to provide project status updates to the sponsor and instructor.
- **Informal project communications** — such as day-to-day messages via Messenger or Discord — should be professional and effective but do not require a standard template or format.

All documents will follow the HecTech file naming convention and will be archived on the SharePoint platform. Version control will be maintained by the Project Manager.

Communication Escalation Process

As issues or complications arise regarding project communications, it may become necessary to escalate if a resolution cannot be achieved within the project team. While escalations are a normal part of project management, there must be a documented process that defines how those escalations will take place.

Efficient and timely communication is the key to successful project completion. As such, it is imperative that any disputes, conflicts, or discrepancies regarding project communications are resolved in a way that is conducive to maintaining the project schedule, ensuring the correct communications are distributed, and preventing ongoing difficulties. HecTech will use the following escalation model to provide a framework for resolving communication issues. The table below defines the priority levels, decision authorities, and timeframes for resolution.

Priority	Definition	Decision Authority	Timeframe for Resolution
Priority 1	Major impact to project or business operations. If not resolved quickly there will be a significant adverse impact on revenue and/or schedule.	Project Sponsor	Within 4 hours
Priority 2	Medium impact on project or business operations which may result in some adverse impact to revenue and/or schedule.	Project Sponsor	Within one business day
Priority 3	Slight impact may cause some minor scheduling difficulties with the project but no impact on business operations or revenue.	Project Manager	Within two business days
Priority 4	It has an insignificant impact on the project but there may be a better solution.	Project Manager	Work continues and any recommendations are submitted via the project change control process



**** NOTE:** Any communication including sensitive and/or confidential information, particularly information related to the DOST-STII library database, system architecture, or user data will require escalation to the Project Sponsor for approval prior to external distribution.

Glossary of Communication Terminology

Term	Definition
Communication	The effective sending and receiving of information. Ideally, the information received should match the information sent. It is the responsibility of the sender to ensure this takes place.
Stakeholder	Individuals or groups involved in the project or whose interests may be affected by the project's execution or outcome.
Communications Management Plan	Portion of the overall Project Management Plan which details how project communications will be conducted, who will participate in communications, frequency of communications, and methods of communications.
Escalation	The process which details how conflicts and issues will be passed up the management chain for resolution as well as the timeframe to achieve resolution.
Communications Matrix	A table that maps each type of project communication to its objective, medium, frequency, audience, owner, and deliverable format.
Sprint Board	The visual task management board in OpenProject used by the HecTech team to track work packages, backlogs, and sprint progress.
Milestone	A significant scheduled checkpoint in the project that marks the completion of a major phase or deliverable.

SPONSOR ACCEPTANCE

Approved by the Project Sponsor:

Khasian Eunice Romulo
Science Research Specialist II / Unit Head
(Digital Information and Processing Delivery Unit)

Date: April 1, 2026



03 Risk Management Plan

Introduction

As with any technology project involving AI integration, and third-party API dependencies, the project operates within an environment of inherent uncertainty. The purpose of this Risk Management Plan is to establish the framework by which the LitPath AI project team will identify, assess, and develop strategies to mitigate or avoid risks that may threaten the project's objectives, deliverables, and deadlines.

LitPath AI is considered a high-risk project with an overall risk score of 10.95 on a scale from 1 to 25, derived from the average of all 15 identified risk scores. A risk score below 5 is a low-risk project; a score between 5 and 10 is a medium risk project, and a score above 10 is a high-risk project.

Before risk management activities formally began, it was imperative that a foundation of structured project information be established. The following project elements were completed and defined prior to developing this Risk Management Plan:

- Define work scope, schedule, resources, and cost elements through
 - Statement of Work
 - Scope Management Plan
 - Schedule Management Plan
 - Cost Management Plan
- Define minimum and maximum baseline thresholds
 - Schedule
 - Resources
 - Cost
- Baseline reporting requirements
 - Format
 - Frequency of distribution
 - Distribution list
- Define Risk Management Roles and Responsibilities
 - Project Manager chairs the risk assessment meetings
 - Project team participates in risk assessment meetings and members serve as meeting recorder and timekeeper
 - Key stakeholders participate in risk assessment meetings

Top Three Risks

The top three high probability and high impact risks to the LitPath AI project are:

Gemini AI API Unavailability or Rate Limiting

Due to the project's dependency on the Gemini AI API for semantic query understanding, AI-generated overviews, and result re-ranking, any extended service outage, rate limiting, or unexpected deprecation of the API would render the system's core AI search functionality inoperable, directly threatening the project's primary deliverable. The Project



Manager will mitigate this risk by implementing response caching for frequent queries, monitoring Gemini AI service status continuously, and coordinating with DOST-STII to ensure API cost coverage is in place before production deployment.

Search Relevance Optimization

Due to the complexity of semantic search across specialized academic literature, the system's RAG pipeline requires continuous refinement to achieve the desired relevance benchmarks. Failure to resolve low-relevance query patterns prior to formal usability testing may result in rework that affects deployment timelines. To mitigate this risk, the Project Manager and Technical Lead will conduct iterative internal relevance evaluations using predefined test queries, continuously refine LLM prompts, and perform internal pre-testing before formal usability testing commences.

LLM Hallucination

Due to the probabilistic nature of large language models, the Gemini AI API may generate inaccurate or misleading outputs that could affect the system's reliability as an academic research tool and risk rejection during acceptance testing. To mitigate this risk, the Project Manager and QA Lead will implement source-grounded prompting, display citations with AI-generated outputs, and conduct hallucination testing using verified dataset entries during the QA cycle.

Risk Management Approach

The LitPath AI project follows a structured risk management approach aligned with its sprint-based development. Risk identification was conducted through a single formal risk assessment meeting held during the project planning phase. The HecTech team identified, scored, and prioritized risks across Technical, Schedule, Resource, External, and Compliance categories during the assessment meeting, with high-impact risks assigned to Risk Managers for active mitigation. These risks are monitored and reported during bi-weekly sprint reviews via Microsoft Teams when relevant to the current sprint window, while the Project Manager continuously updates the Risk Register as risks evolve. Upon project completion, the overall effectiveness of the risk management process is evaluated, with lessons learned documented and archived to support future HecTech projects.

Risk Identification

Risk identification for the LitPath AI project was conducted through a risk assessment meeting held during the project planning phase, with supplementary input gathered through an expert interview with the Project Clients and a review of similar projects. The Project Manager chaired the risk assessment meeting.

Expert Interview

One expert interview was conducted for the LitPath AI project. It was held with the DOST-STII client representatives to identify operational, compliance, and deployment risks from the client's perspective including data privacy requirements under the Data Privacy Act of 2012 (RA 10173), server hosting constraints, and AI/LLM API cost coverage responsibilities. Risks identified and resolved through this interview, such as adjusting the authentication architecture to comply with RA 10173, were addressed directly in the system design. Remaining risks were logged in the Risk Register.



Risk Assessment Meeting

A risk assessment meeting was held with all four HecTech team members and attended by the Project Clients. The Crawford Slip method was used, giving each participant 10 minutes to independently record as many risks as possible on notepads. The Project Manager then facilitated a group discussion in which risks were read aloud, consolidated, and categorized. Identified risks were added to the Risk Register.

Historical Review of Similar Projects

The project team reviewed post-project documentation and known failure patterns from comparable AI-powered academic search and web application projects to identify common risks, including AI API reliability issues, last-mile deployment failures, user acceptance shortfalls, and scope creep introduced by client feedback during refinement phases. These historical insights informed both the identification process and the development of mitigation strategies, particularly for the Gemini AI dependency risk and the usability testing acceptance threshold risk.

Risk Qualification and Prioritization

Once risks were identified, a probability and impact factor was assigned to each risk to allow the Project Manager to prioritize mitigation and avoidance strategies. Risks that are more likely to occur and have a significant impact on the project's schedule, cost, scope, or acceptance criteria were classified as the highest priority, while those that are less likely or have limited impact were assigned a lower priority.

Each risk was assigned a probability rating from 1 to 5 and an impact rating from 1 to 5, where 1 represents the lowest likelihood or negligible effect, and 5 represents a near-certain occurrence or a severe effect that directly blocks project delivery. The resulting risk score calculated as probability multiplied by impact falls on a scale from 1 to 25 and is used to categorize risks as Low (1–4), Medium (5–9), High (10–19), or Critical (20–25).

Probability and impact ratings were determined through a structured voting process conducted during the risk assessment meeting. For each identified risk, all four HecTech team members independently assigned a probability rating and an impact rating based on the following defined criteria:

Probability Rating Criteria:

- **1 — Very Low:** The risk event is highly unlikely with no historical precedent in the project context and existing controls making occurrence improbable.
- **2 — Low:** The risk event is unlikely but possible with limited exposure and conditions largely within the team's control.
- **3 — Moderate:** The risk event is possible due to external dependencies or known project constraints outside the team's direct control.
- **4 — High:** The risk event is likely because of strong dependence on external factors or recurring conditions that increase probability.
- **5 — Very High:** The risk event is very likely or near certain due to critical dependencies and the absence of effective mitigation measures.



Impact Rating Criteria:

- **1 — Very Low:** Negligible effect on project objectives with no significant impact on schedule, cost, or sprint capacity.
- **2 — Low:** Minor effect that can be resolved within the sprint cycle with minimal schedule adjustment.
- **3 — Moderate:** Noticeable effect on schedule, cost, or quality requiring corrective action during the sprint cycle.
- **4 — High:** Significant effect that threatens a project milestone or acceptance criterion and requires escalation to the Project Sponsor.
- **5 — Critical:** Severe effect that directly prevents project delivery, formal acceptance, or deployment.

After all team members submitted their ratings, the scores were averaged to determine the final probability and impact values for each risk. The overall risk score was then calculated by multiplying the average ratings, resulting in a score from 1 to 25. This method reflects the collective assessment of the project team and allows clearer differentiation between risks through decimal-based scoring. Individual ratings, averaged values, and computed scores are documented in the Voting Table.

The Project Manager used a probability-impact matrix to position each risk according to its assessed ratings. The matrix applied color coding to indicate severity levels namely green for Low, yellow for Medium, orange for High, and red for Critical risks. After all risks were plotted, the results were documented and the team proceeded to mitigation and avoidance planning for priority risks.

Risk Monitoring

Risk monitoring for the LitPath AI project is a continuous process embedded in bi-weekly sprint reviews managed through Microsoft Teams, where all priority risks are tracked with assigned Risk Manager responsible for monitoring status and trigger conditions within their occurrence window. During each review, only active risks relevant to the current sprint are discussed to maintain efficiency, with updates covering status (open, triggered, closed), observed triggers, and response actions taken. The Project Manager then updates the Risk Register and escalates significant issues to the Project Clients via Viber while predefined trigger conditions such as API latency thresholds or sustained drops in sprint velocity guide for timely responses.

Risk Mitigation and Avoidance

The Project Manager, with the full participation of the HecTech development team, has developed a specific avoidance or mitigation response for each risk identified and prioritized in the Risk Register. As additional risks are identified during subsequent sprints, they will be scored, assigned a Risk Manager, and logged into the Risk Register with a corresponding response strategy.

All risk responses for LitPath AI will be managed and controlled within the constraints of time, scope, and cost. Scope and time are the firm constraints for this project. Cost is the flexible



constraint of last resort like if a critical risk cannot be mitigated within the existing sprint budget and the PHP 40,000.00 contingency reserve, the Project Manager may request additional funding from the Project Sponsor to cover supplementary resources, extended cloud hosting, or additional AI/LLM API usage subject to a formal Change Request and Sponsor approval. The PHP 4,000.00 Management Reserve held by the Project Sponsor may also be drawn upon in extreme cases, as permitted under the Cost Management Plan.

Risk Register

The LitPath AI Risk Register is the central log for all 15 identified risks. For each risk, the register captures: risk title, risk description, category (Technical, Schedule, Resource, External, Compliance, or Cost), averaged probability rating (1–5), averaged impact rating (1–5), overall risk score (1–25), risk ranking, assigned Risk Manager, mitigation or avoidance strategy, and trigger conditions.

The Risk Register was created during the project risk assessment meeting facilitated by the Project Manager. Team members independently assigned probability and impact ratings, which were averaged to produce the final scores and rankings. Prior to each risk's anticipated occurrence period, the Project Manager confirms that mitigation measures and trigger monitoring are active. Assigned Risk Managers then provide status updates during bi-weekly sprint reviews within the relevant timeframe.

The Risk Register is maintained as a living document updated after each bi-weekly review meeting and upon any trigger event and is accessible to all team members and the Project Sponsor via the project's shared document repository on SharePoint and Google Drive. The Risk Register is maintained as Appendix A to this Risk Management Plan.

SPONSOR ACCEPTANCE

Approved by the Project Sponsor:

Khasian Eunice Romulo
Science Research Specialist II / Unit Head
(Digital Information and Processing Delivery Unit)

Date: April 30, 2026



Appendix A

Risk Register: [02 Project Planning 08 Risk-Register.xlsx](#)

Risk Identification		Risk Category	Qualitative Rating				Risk Response		
Risk Title	Risk Description		Probability	Impact	Risk Score	Risk Ranking	Risk Response	Trigger	Risk Owner
API Unavailability	Due to LIPATH AI's dependency on the Gemini AI API for semantic query processing and AI-generated overviews, an extended service outage, rate limiting, or API deprecation by Google could disable the system's core search functionality and prevent the project from meeting its primary acceptance criteria.	Technical / External	4	5	20	1	Implement response caching for high-frequency queries, establish a fallback degraded-mode search using ChromaDB vector similarity without Gemini AI, coordinate with DOST-STII to confirm API cost coverage before production deployment.	Gemini AI API returns persistent errors, exceeds rate limits, or announces deprecation during development or testing phases.	Charlojy Cempron
Search Relevance Failure	Due to the complexity of semantic search over specialized academic literature, the system's RAG pipeline requires continuous iterative refinement to produce results that consistently meet the team's internal relevance benchmarks. Failure to address low-relevance query patterns before formal usability testing could result in rework cycles that consume sprint capacity originally reserved for deployment activities.	Technical	3.25	5	16.25	2	Conduct iterative internal relevance evaluation throughout development using a defined test query set, refine LLM prompts continuously, perform pre-usability relevance testing at least two weeks before formal usability testing begins.	Internal relevance testing reveals consistent failures across the defined test query set, or early informal user feedback surfaces recurring irrelevant results.	Charlojy Cempron
LLM Hallucination	Due to the inherent probabilistic nature of large language models, the Gemini AI API may generate inaccurate, fabricated, or misleading summaries, citations, or search overviews that appear credible to users, undermining the system's reliability as an academic research tool and potentially causing DOST-STII to reject the deliverable during acceptance testing.	Technical / AI	3	5	15	3	Implement source-grounded prompting that constrains Gemini AI responses strictly to retrieved document content, display source citations alongside every AI-generated output, conduct targeted hallucination testing as part of the QA cycle using known dataset entries to verify output accuracy.	QA testing identifies AI-generated summaries or citations that do not correspond to actual content in the indexed dataset, or usability testing participants report misleading or fabricated search results.	Joshua Sta Ana
API Latency / Slow Response	Due to LIPATH AI's reliance on external API calls to Gemini AI for query processing and result generation, slow API response times under load or degraded network conditions could cause search query response times to exceed the ten seconds threshold, failing the system's performance acceptance criterion even when the API is technically available.	Technical / External	3.5	4	14	4	Implement timeout handling with defined maximum wait thresholds, use asynchronous API calls to prevent blocking, cache responses for frequent or repeated queries, display partial results while full AI overview loads to maintain perceived performance.	API response times consistently exceed 10 seconds during internal load testing, or usability testing participants report noticeable delays between query submission and results display.	Charlojy Cempron
Dataset Provision Delay	Due to the project's reliance on DOST-STII to provide a representative sample dataset of the relevant disquisitions, any delay or inadequacy in the dataset handover could prevent the team from completing RAG pipeline validation, relevance testing, and system acceptance within the project timeline.	External	3.25	4	13	5	Confirm dataset delivery schedule with DOST-STII at the start of each major testing phase, request a preliminary sample early to identify format or volume issues, define minimum dataset requirements in writing with the Project Sponsor.	DOST-STII has not provided the required dataset within two weeks of the agreed handover date, or the dataset is insufficient in volume or metadata quality for testing.	Tracie Tomon
Cloud Hosting Delay	Due to the planned transition from the local Promox development environment to a cloud hosting environment managed by DOST-STII for final production deployment, delays in confirming the hosting setup or cost approvals could prevent the system from going live by the June-July 2026 deadline.	External / Schedule	3	4	12	6	Initiate cloud hosting discussions with DOST-STII no later than May 2026, document hosting requirements in the Statement of Work, maintain the Promox server as a fallback demo environment, escalate any delays immediately to the Project Sponsor.	DOST-STII has not confirmed the production hosting environment or cost coverage by May 15, 2026, or the cloud environment setup encounters technical blockers during pre-deployment testing.	Tracie Tomon
Database Inconsistency	Due to the system's use of Supabase (PostgreSQL) for relational data and ChromaDB for vector embeddings, a failure in data consistency between the two systems could corrupt search results, break the bookmarking feature, or prevent accurate citation generation.	Technical	2.75	4	11	7	Implement automated consistency checks between Supabase and ChromaDB after each data ingestion cycle, maintain backup storage on the HDD per the system design, establish a documented data recovery procedure, test database integrity in every sprint QA cycle.	Search results return mismatched or duplicate records, bookmarked items fail to load, ChromaDB embeddings become out of sync with Supabase document records.	Joshua Sta Ana
Data Loss in Ingestion	Due to the complexity of the data ingestion and embedding generation pipeline, failures during batch processing — such as interrupted ingestion jobs, malformed document records, or embedding errors — may result in theses or dissertations being silently omitted from the indexed dataset, causing incomplete search results without any visible system error.	Technical	3	3.25	9.75	8	Implement comprehensive ingestion logging with record-level success and failure tracking, add post-ingestion validation that compares source document count against indexed record count, implement retry mechanisms for failed embedding generations, alert the team if record count discrepancies exceed a defined threshold.	Post-ingestion validation reveals a mismatch between the number of source documents provided and the number of records successfully indexed in ChromaDB or Supabase.	Charlojy Cempron
Scope Creep	Due to DOST-STII's ability to submit feedback and change requests during the Feature Refinement and usability testing phases, uncontrolled additions beyond the agreed Statement of Work could consume the team's limited sprint capacity and jeopardize the June-July 2026 deadline.	Scope	3	3	9	9	Enforce the formal scope change control process for all client-initiated change requests, document and confirm all feature clarifications in writing before development, require Project Sponsor approval for any changes affecting schedule or budget.	DOST-STII requests a new feature or significant modification outside the defined scope in the Statement of Work during the testing or deployment phase.	Tracie Tomon
Data Privacy Non-Compliance	Due to the Data Privacy Act of 2012 (RA 10173) requirements applicable to LIPATH AI, any failure to implement proper data handling for user accounts, search history, and admin access could result in a compliance violation that blocks formal project acceptance or requires significant post-deployment rework.	Compliance	2	4.25	8.5	10	Ensure all user authentication uses secure login with role-based access control and session management with secure tokens, keep public search anonymous, enforce HTTPS/SOCS on all connections, conduct a compliance review led by the Security Lead before usability testing.	Security audit identifies unprotected user data, unsecured API endpoints, or missing authentication controls during internal testing.	Joshua Sta Ana
Dependency / Library Compatibility	Due to the project's reliance on a Python-based AI stack including SentenceTransformers, ChromaDB, LangChain type utilities, and Supabase client libraries, updates or breaking changes to any of these dependencies could introduce runtime errors, degrade system performance, or require unplanned retooling that consumes sprint capacity.	Technical	2.25	3.75	8.44	11	Pin all dependency versions in the project's requirements file and lock file, test all updates in the staging environment before applying to the development or production environment, monitor dependency changelogs for breaking changes before each sprint.	System errors occur after a library update or environment change, or a dependency announces a breaking change in an upcoming release that affects a core system component.	Charlojy Cempron
Key Resource Unavailability	Due to the four-member Hactech team having no external outsourcing capacity, the unexpected unavailability of a key technical resource due to illness, personal emergency, or academic withdrawal could significantly disrupt sprint delivery and system integration milestones.	Resource	2	4	8	12	Maintain up-to-date technical documentation for all system components, cross-train team members on critical modules within sprint capacity, redistribute tasks among remaining members in the event of a temporary absence, escalate to Project Sponsor if absence exceeds one sprint.	A team member is unable to perform assigned sprint tasks for more than three consecutive working days due to illness, personal emergency, or academic withdrawal.	Tracie Tomon
Search Performance Degradation	Due to the system's use of SentenceTransformers for embedding generation and ChromaDB for vector search, performance degradation under concurrent user load or large dataset volume could cause search query response times to exceed the required ten seconds threshold, failing the system performance acceptance criterion.	Technical	2.25	3	6.75	13	Conduct load and performance testing during the Usability Testing phase, optimize ChromaDB indexing and embedding retrieval pipelines, implement query result caching for high-frequency searches, address all performance bottlenecks before final deployment.	Internal load testing reveals average search response times exceeding ten seconds under simulated concurrent user conditions, or usability testing participants report noticeable delays.	Charlojy Cempron
Local Server Hardware Failure	Due to the project's use of a locally hosted Promox server for development and internal testing, hardware failure such as SSD failure or power interruption could result in loss of code, database records, or configuration that delays sprint delivery.	Technical	2	3.25	6.5	14	Maintain all source code in a remote Git repository with regular commits after every work session, back up the Supabase database and ChromaDB embeddings to the 500 GB HDD, document all server configuration settings to enable rapid environment recovery.	The Promox server becomes inaccessible for more than 24 hours, or a team member reports data loss or file corruption on the local development environment.	Joshua Sta Ana
API Cost Overrun	Due to the project's use of the Gemini AI API without a formal written cost commitment from DOST-STII, any unexpected spike in API usage during testing or deployment could result in costs that exceed the project's contingency reserve and interrupt system functionality.	Cost / External	2	3	6	15	Confirm DOST-STII's formal commitment to cover all cloud hosting and AILLM API costs in writing before transitioning to production, monitor API usage costs throughout testing, report any projected overruns to the Project Sponsor immediately.	DOST-STII requests a delay in API budget commitment, or Gemini AI API usage costs during testing exceed the informally estimated monthly threshold.	Tracie Tomon



04 Scope Management Plan

Introduction

The Scope Management Plan provides the scope framework for this project. This plan documents the scope management approach; roles and responsibilities as they pertain to project scope; scope definition; verification and control measures; scope change control; and the project's work breakdown structure. Any project communication which pertains to the project's scope should adhere to the Scope Management Plan.

This project involves the design, development, and testing of LitPath AI: Smart Pathfinder for Theses and Dissertations, an AI-powered search engine built to enhance access to the DOST-STII Library's collection of graduate theses and dissertations. The system is developed as a web-based application hosted on the HecTech team's own server infrastructure, integrating with the existing DOST-STII Online Public Access Catalog (OPAC) without replacing it. This includes the design of the user interface, all backend and frontend programming, AI integration, and system testing and validation. Development is carried out entirely by the four-member HecTech team, with no external outsourcing anticipated.

Scope Management Approach

For the LitPath AI project, scope management is the primary responsibility of the Project Manager, Tracie Tomon. The scope for this project is defined by the Scope Statement, the Work Breakdown Structure (WBS), and the WBS Dictionary, all of which are grounded in the requirements collected through stakeholder interviews with DOST-STII, the Design Thinking process, and the Project Charter authorized by the Project Sponsor, Khasian Eunice Romulo.

The Project Manager, Sponsor, and stakeholders will establish and approve documentation for measuring project scope, which includes deliverable quality checklists and work performance measurements tracked through OpenProject.

Proposed scope changes may be initiated by the Project Manager, any member of the HecTech development team, or the Project Sponsor. All change requests must be submitted to the Project Manager, who will evaluate the request and conduct an impact assessment. Upon initial acceptance by the Project Manager, the scope change will be submitted to the Project Sponsor for final approval. Once approved, the Project Manager will update all project documents and communicate the scope change to all team members and stakeholders.

The Project Sponsor, Khasian Eunice Romulo, is responsible for the formal acceptance of all final project deliverables and the overall project scope.

Roles and Responsibilities

The Project Manager, Sponsor and team will all play key roles in managing the scope of this project. As such, the project sponsor, manager, and team members must be aware of their responsibilities in order to ensure that work performed on the project is within the established scope throughout the entire duration of the project. The table below defines the roles and responsibilities for the scope management of this project.



Name	Role	Responsibilities
Khasian Eunice Romulo	Client / Project Sponsor	- Approve or deny scope change requests as appropriate - Evaluate need for scope change requests - Accept project deliverables
Tracie Tomon	Project Manager/ Team lead	- Measure and verify project scope - - Facilitate and validate scope change requests - Conduct and participate in impact assessments - Organize and lead change control meetings - Facilitate team-level change review process - Communicate outcomes of scope change requests to stakeholders and team - Update project documents upon approval of scope changes -
Jenine Elaine Dulay	Team Member	- Participate in defining change resolutions - Evaluate the need for scope changes and communicate them to the project manager as necessary
Charijoy Cempron	Team Member	- Participate in defining change resolutions - Evaluate the need for scope changes and communicate them to the project manager as necessary
Joshua Sta Ana	Team Member	- Participate in defining change resolutions - Evaluate the need for scope changes and communicate them to the project manager as necessary

Scope Definition

The scope for the LitPath AI project was defined through a comprehensive requirements collection process rooted in both the client's needs and the existing system's limitations. First, a thorough analysis was performed on the DOST-STII Library's current Online Public Access Catalog (OPAC) system based on feedback gathered through stakeholder interviews with DOST-STII librarians, staff, student researchers, and DOST employees during the Design Thinking Empathize stage. From this information, the project team identified the core problems: exact keyword matching, limited filtering options, slow search performance, and the lack of AI-powered features and citation tools.

Building on this, the project requirements were documented, and the scope was formally established through the Project Charter (authorized March 29, 2026) and the Scope Statement included in the final project paper. The project description and deliverables were further refined based on input from the client (DOST-STII), the PBL instructor, and the team's own expert judgment in software design, AI integration, and user interface development.

The tools and techniques used to define the project scope include stakeholder interviews, the 5 WHYS analysis, persona development, How-Might-We exercises, Crazy 8s ideation, Desirability-Viability-Feasibility analysis, and prototype testing — all carried out across the five stages of the Design Thinking process.



Project Scope Statement

Product Scope Description

LitPath AI is an AI-powered search engine designed to improve access to the DOST-STII Library's collection of graduate theses and dissertations. It enhances the existing OPAC system by offering AI-powered query analysis, an automated citation generator, bookmarking, research history, follow-up questions, and an admin dashboard for library staff to monitor usage analytics.

Product Acceptance Criteria

The project will be accepted as complete when the following criteria are met:

1. The system achieves a minimum of 70% relevance accuracy for its top 10 search results, validated through user feedback and expert review.
2. The system achieves an average user satisfaction score of at least 3.5 out of 5 from usability testing participants (student researchers and DOST employees).
3. The system consistently processes search queries and displays results up to the industry standard
4. The system is fully integrated with the existing DOST-STII OPAC.
5. All key deliverables have been formally reviewed and accepted by the Project Sponsor.

Project Deliverables

1. AI-powered search engine — a fully functional backend system providing semantic search and content-based recommendations, delivered with an API for integration into the existing OPAC.
2. User Interface — a responsive web interface, a built-in automated citation generator supporting APA, MLA, Chicago, and IEEE styles and bookmark feature.
3. User Feedback Tool — a mechanism for capturing star ratings and thumbs up/down and comments from users to measure satisfaction and validate search relevance.
4. Admin Dashboard — an interface for DOST-STII library staff to track search trends and monitor system usage.
5. Manual and Final Report — complete technical documentation, a user guide, and a final report verifying that all project objectives have been met.

Project Constraints

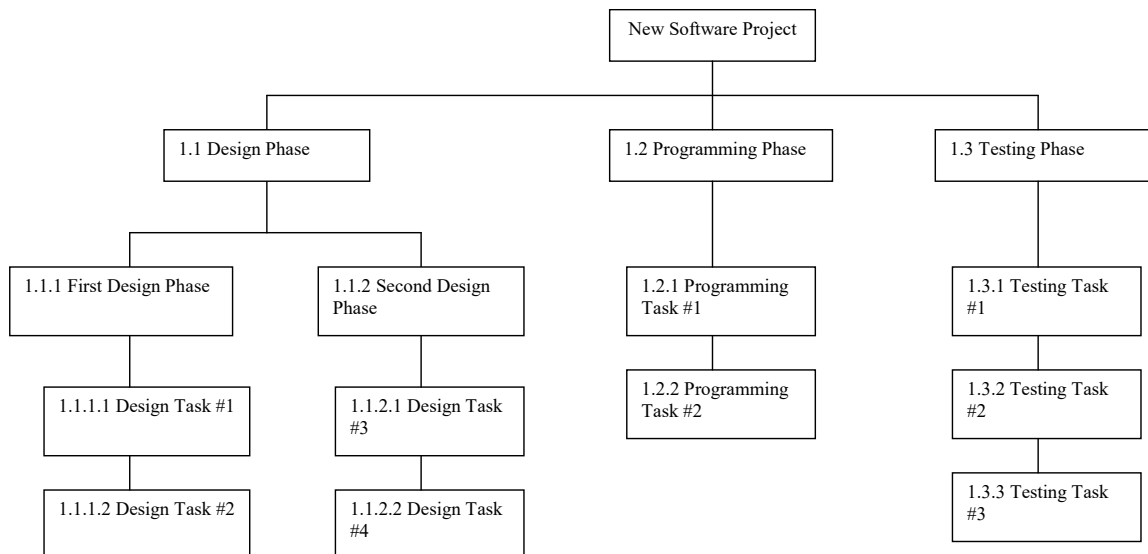
1. The project must be deployed by June–July 2026 in alignment with the academic timeline.
2. The system will be hosted on a local server maintained by the HecTech team, eliminating external hosting costs. Any AI/LLM API expenses during production will be coordinated with DOST-STII, as no development budget has been allocated to the team.
3. Development is limited to the four-member HecTech team, and no external outsourcing is anticipated.

Work Breakdown Structure

To effectively manage the work required to complete this project, it will be subdivided into individual work packages which will not exceed 40 hours of work. This will allow the Project Manager to more effectively manage the project's scope as the project team works on the tasks necessary for project completion. The project is broken down into three phases: the design phase; the programming phase; and the testing phase. Each of these phases is then subdivided further down to work packages which will require no more than 40 hours of work and no less than 4 hours of work (see WBS structure below).



To effectively manage the work required to complete LitPath AI, the project has been subdivided into individual work packages assigned to each team member and tracked in OpenProject. Work packages are organized by project phase and are scoped to be manageable within each sprint cycle.



In order to more clearly define the work necessary for project completion the WBS Dictionary is used. The WBS Dictionary includes an entry for each WBS element. The WBS Dictionary includes a detailed description of work for each element and the deliverables, budget and resource needs for that element. The project team will use the WBS Dictionary as a statement of work for each WBS element.

AY 2025-2026 T3 Sprint 01 | Feedback Management System

Sprint Period: 03/23/2026 – 04/03/2026 **Total Points:** 80 **Total Features:** 12

Level	WBS Code	Element Name	Definition
1	1	Feedback Management System	All work to implement and deliver features for the Feedback Management System under AY 2025-2026 T3 Sprint 01.
2	1.1	AY 2025-2026 T3 Sprint 01	Sprint covering the period 03/23/2026 – 04/03/2026 with a total of 80 story points across 12 feature items.
3	1.1.1	Feature 1.0 User Account Registration System	Implement the user account registration system including sign-up forms, input validation, account creation logic, and confirmation workflows. (8 points, Issue #4674)



Level	WBS Code	Element Name	Definition
3	1.1.2	Feature 2.0 LLM Source Accuracy Optimization / Enhance Prompt	Optimize the accuracy of LLM-generated responses by refining source selection logic and enhancing prompt engineering techniques to improve output quality and relevance. (10 points, Issue #4679)
3	1.1.3	Feature 3.0 Welcome Page & Form UI Modernization	Redesign and modernize the welcome page and associate form interfaces to improve user experience, visual consistency, and accessibility standards. (5 points, Issue #4689)
3	1.1.4	Feature 4.0 Display Philippine Standard Time (PST)	Update all date and time displays throughout the application to reflect Philippine Standard Time (PST, UTC+8), ensuring consistent localized timestamps across the system. (5 points, Issue #4690)
3	1.1.5	Feature 5.0 Clickable Modal Interface (Feedback Manager)	Implement a clickable modal interface within the Feedback Manager module to allow users to view detailed feedback entries in an overlay dialog without leaving the current page context. (5 points, Issue #4692)
3	1.1.6	Feature 6.0 UI Decluttering (Feedback Table View)	Streamline and declutter the Feedback Table View by removing redundant elements, reorganizing columns, and improving the overall layout for enhanced readability and usability. (7 points, Issue #4693)
3	1.1.7	Feature 7.0 Data Pagination (Feedback Manager)	Introduce data pagination functionality to the Feedback Manager to efficiently handle large datasets, enabling users to navigate through records in manageable pages. (7 points, Issue #4694)
3	1.1.8	Feature 8.0 Targeted Dormant Material Export	Develop an export feature that allows administrators to selectively export dormant materials from the system based on defined criteria, supporting data management and archival processes. (8 points, Issue #4695)
3	1.1.9	Feature 9.0 Failed Query Keyword Tracking	Implement a tracking mechanism to capture and log keywords from failed search queries, enabling the team to identify content gaps and improve system coverage over time. (8 points, Issue #4696)
3	1.1.10	Feature 10.0 "Least Viewed" Metric Tracking	Build a metric tracking system that identifies and monitors the least-viewed content items, providing insights to inform content curation and resource prioritization decisions. (8 points, Issue #4697)
3	1.1.11	Feature 11.0 Column Merger & Analysis Dropdown Refinement	Merge related data columns and refine the analysis dropdown menus to reduce redundancy, improve navigation clarity, and streamline the analytical workflow for end users. (4 points, Issue #5100)



Level	WBS Code	Element Name	Definition
3	1.1.12	Feature 12.0 Explicit Date Formatting & "Feedback Type" Column Renaming	Standardize date formatting across the application with explicit format definitions and rename the Feedback Type column for improved clarity and consistency in the data presentation layer. (5 points, Issue #5101)

All features have a status of Closed as of the sprint end date.

Scope Verification

As the LitPath AI project progresses, the Project Manager will verify interim project deliverables against the original scope as defined in the Scope Statement, WBS, and WBS Dictionary. Once the Project Manager confirms that a deliverable meets the requirements defined in the project plan, the Project Manager and Project Sponsor will meet for formal acceptance.

During these milestone review meetings, the Project Manager will present the deliverable to the Project Sponsor, Khasian Eunice Romulo, for formal acceptance. The Project Sponsor will accept the deliverable by signing a project deliverable acceptance document. Key verification checkpoints are aligned with the milestone schedule:

- Completion of the UI and core features (Oct–Nov 2025) ✓
- Completion of AI Search Engine and advanced features (Oct 2025–Jan 2026) ✓
- Feature refinement and client feedback integration (Feb–Mar 2026) ✓
- Internal testing and temporary deployment (Apr 2026) — In-progress
- Usability testing results (May–Jun 2026) — Not started
- Final deployment and handover (Jun–Jul 2026) — Not started

This incremental verification approach ensures that project work remains within scope on a consistent basis throughout the project lifecycle rather than being evaluated only at the end.

Scope Control

The Project Manager and the HecTech development team will work together to control the scope of the LitPath AI project. The team will use the WBS Dictionary as the statement of work for each WBS element, ensuring that only the described work is performed and that defined deliverables are produced for each element. The Project Manager will oversee the team and the progression of the project to ensure this scope control process is followed throughout all phases.

If a change to the project scope is needed, the following process must be carried out:

1. Any team member, the Project Manager, or the Project Sponsor may identify and initiate a scope change request.
2. The change request must be submitted to the Project Manager in writing (via Viber or a formal change request document stored on SharePoint).
3. The Project Manager will review the request and conduct an impact assessment, evaluating effects on schedule, resources, and the June–July 2026 deployment deadline.
4. If the Project Manager denies the request as outside the intent of the project, the requestor will be notified with a rationale.



5. If the Project Manager approves the request at the initial level, a change control meeting will be convened with the team and Project Sponsor to review the change further.
6. Upon approval by the Project Sponsor, Khasian Eunice Romulo, the Project Manager will update all project documents (Scope Statement, WBS, WBS Dictionary, OpenProject tasks) and communicate the approved scope change to all team members and stakeholders.

SPONSOR ACCEPTANCE

Approved by the Project Sponsor:

Khasian Eunice Romulo
Science Research Specialist II / Unit Head
(Digital Information and Processing Delivery Unit)

Date: April 1, 2026

05 Work Breakdown Structure (WBS)

Introduction

The Work Breakdown Structure presented here represents all the work required to complete this project.

Outline View

1. DOST LitPath AI – APC 2025-2026 T1 MI231 G07 HecTech
 - 1.3 3.0 Executing Phase
 - 1.3.1 3.1 Usability Testing
 - 1.3.3 3.3 Pre-deployment
 - 1.3.5 3.5 Release & Finalization (System Dry-run)
 - 1.3.11 Feature 11.0 Targeted Dormant Material Export
 - 1.4 4.0 Monitoring and Controlling



- 1.4.1 4.1 Tracking Progress
- 1.4.2 4.2 Comparing it to the Plan
- 1.4.3 4.3 Identifying any Deviations or Issues
- 1.4.4 4.4 Measuring Performance
- 1.4.5 4.5 Performance Reporting
- 1.4.6 4.6 Risk and Issue Management
- 1.4.7 4.7 Quality Assurance and Control

1.5 5.0 Closing Phase

- 1.5.1 5.1 Completing All Tasks
 - 1.5.1.1 5.1.1 User Acceptance Test Completion
- 1.5.2 5.2 Obtaining Final Approvals
- 1.5.3 5.3 Archiving Project Documentation
- 1.5.4 5.4 Conducting a Final Review
- 1.5.5 5.5 Project Turnover

Hierarchical Structure

Level	WBS Code	Element Name
1	1	DOST LitPath AI – APC 2025-2026 T1 MI231 G07 HecTech
2	1.3	3.0 Executing Phase
3	1.3.1	3.1 Usability Testing
3	1.3.3	3.3 Pre-deployment
3	1.3.5	3.5 Release & Finalization (System Dry run)
3	1.3.11	Feature 11.0 Targeted Dormant Material Export
2	1.4	4.0 Monitoring and Controlling
3	1.4.1	4.1 Tracking Progress
3	1.4.2	4.2 Comparing it to the Plan
3	1.4.3	4.3 Identifying any Deviations or Issues
3	1.4.4	4.4 Measuring Performance
3	1.4.5	4.5 Performance Reporting
3	1.4.6	4.6 Risk and Issue Management
3	1.4.7	4.7 Quality Assurance and Control



2	1.5	5.0 Closing Phase
3	1.5.1	5.1 Completing All Tasks
4	1.5.1.1	5.1.1 User Acceptance Test Completion
3	1.5.2	5.2 Obtaining Final Approvals
3	1.5.3	5.3 Archiving Project Documentation
3	1.5.4	5.4 Conducting a Final Review
3	1.5.5	5.5 Project Turnover



Tabular View

4/16/26, 2:26 AM

All open | Work Packages | APC_2025_2026_T1_MI231_G07 HecTech: DOST LitPath AI | OpenProject

APC_2025_2026_T1_MI231_G0... / Work packages / Default: All open

← Work packages

All open

ID ↑	🔍 SUBJECT	TYPE	STATUS	ASSIGNEE	PRIORITY
855	▼ 3.0 Executing Phase	PHASE	🟡 In progress	CC Charijoy Cempron	🔵 Normal
1856	3.1 Usability testing	MILESTONE	🟢 New	CC Charijoy Cempron	🔵 Normal
1858	3.3 Pre-deployment	MILESTONE	🟢 New	CC Charijoy Cempron	🔵 Normal
1859	3.5 Release & Finalization (System dry...	MILESTONE	🟡 To be scheduled	CC Charijoy Cempron	🔵 Normal
4695	Feature 11.0 Targeted Dormant Materi...	FEATURE	🟢 New	TT Tracie Tomon	🔵 Normal
856	▼ 4.0 Monitoring and Controlling	PHASE	🟡 To be scheduled	JD Jenine Elaine Dulay	🔵 Normal
861	4.1 Tracking progress	TASK	🟢 New	TT Tracie Tomon	🔵 Normal
866	4.2 Comparing it to the plan	TASK	🟢 New	MC Marielle Kloie Concepcion	🔵 Normal
867	4.3 Identifying any deviations or issues	TASK	🟢 New	MC Marielle Kloie Concepcion	🔵 Normal
868	4.4 Measuring performance	TASK	🟢 New	CC Charijoy Cempron	🔵 Normal
1860	4.5 Performance Reporting	TASK	🟢 New	CC Charijoy Cempron	🔵 Normal
1861	4.6 Risk and Issue Management	TASK	🟢 New	JD Jenine Elaine Dulay	🔵 Normal
1862	4.7 Quality Assurance and Control	TASK	🟢 New	TT Tracie Tomon	🔵 Normal
857	▼ 5.0 Closing Phase	PHASE	🟡 To be scheduled	-	🔵 Normal

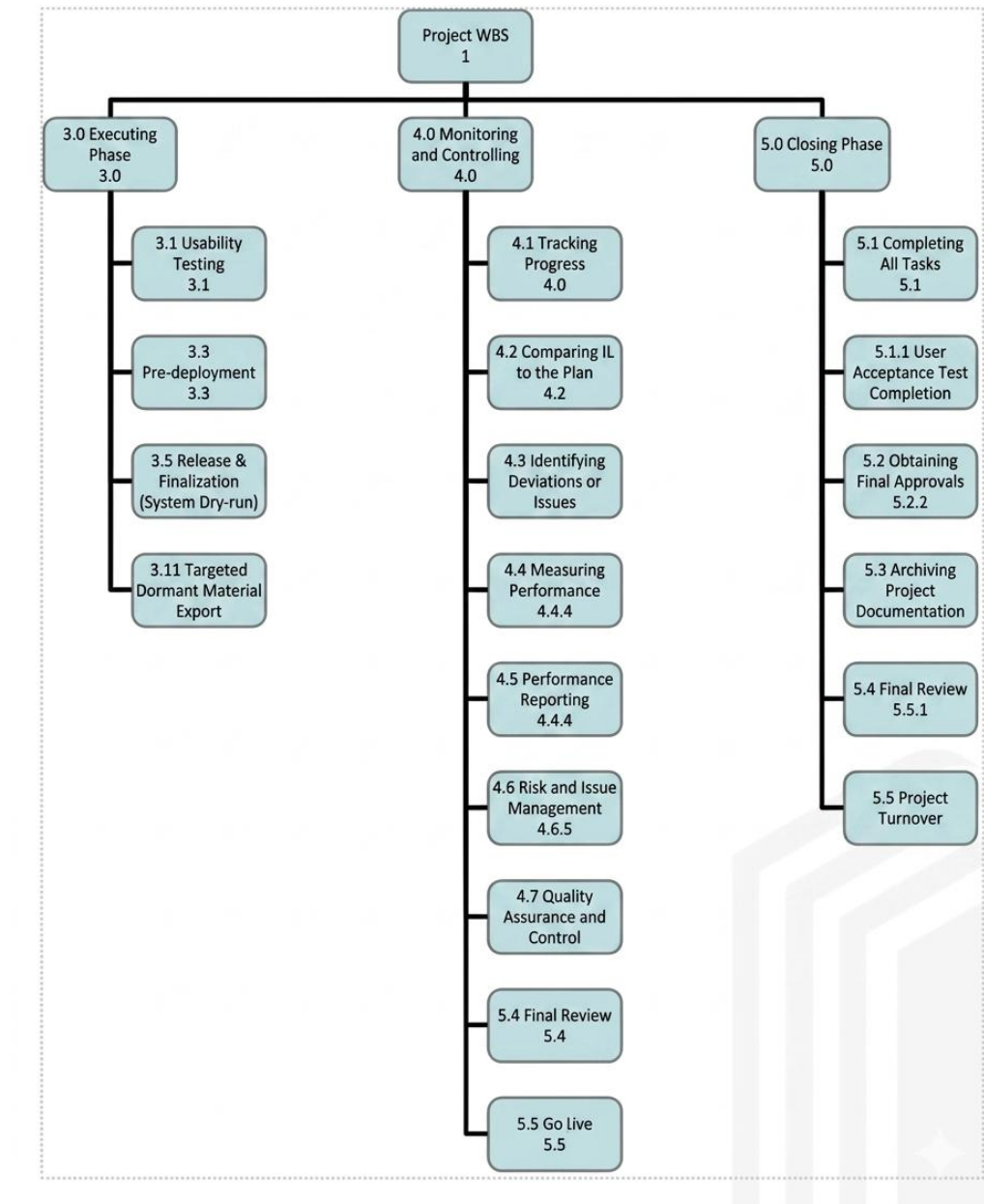
4/16/26, 2:26 AM

All open | Work Packages | APC_2025_2026_T1_MI231_G07 HecTech: DOST LitPath AI | OpenProject

862	▼ 5.1 Completing all tasks	SUMMARY TASK	🟢 New	-	🔵 Normal
891	5.1.1 User Acceptance Test Comple...	TASK	🟢 New	-	🔵 Normal
863	5.2 Obtaining final approvals	SUMMARY TASK	🟢 New	-	🔵 Normal
864	5.3 Archiving project documentation	SUMMARY TASK	🟢 New	-	🔵 Normal
865	5.4 Conducting a final review	SUMMARY TASK	🟢 New	-	🔵 Normal
869	5.5 Project Turnover	SUMMARY TASK	🟢 New	-	🔵 Normal



Tree Structure View





WBS Dictionary

AY 2025-2026 T3 Sprint 01

Sprint Period: 03/23/2026 – 04/03/2026 Total Points: 80 Total Features: 12

Level	WBS Code	Element Name	Definition
1	1	Feedback Management System	All work to implement and deliver features under AY 2025-2026 T3 Sprint 01.
2	1.1	AY 2025-2026 T3 Sprint 01	Sprint covering the period 03/23/2026 – 04/03/2026 with a total of 80 story points across 12 feature items.
3	1.1.1	Feature 1.0 User Account Registration System	Implement the user account registration system including sign-up forms, input validation, account creation logic, and confirmation workflows. (8 points, Issue #4674)
3	1.1.2	Feature 2.0 LLM Source Accuracy Optimization / Enhance Prompt	Optimize the accuracy of LLM-generated responses by refining source selection logic and enhancing prompt engineering techniques to improve output quality and relevance. (10 points, Issue #4679)
3	1.1.3	Feature 3.0 Welcome Page & Form UI Modernization	Blur the picture in the welcome page and associated form interfaces to improve user experience, visual consistency, and accessibility standards. (5 points, Issue #4689)
3	1.1.4	Feature 4.0 Display Philippine Standard Time (PST)	Update all date and time displays throughout the application to reflect Philippine Standard Time (PST, UTC+8), ensuring consistent localized timestamps across the system on both user and admin side. (5 points, Issue #4690)
3	1.1.5	Feature 5.0 Clickable Modal Interface (Feedback Manager)	Turn the Feedback Manager into a clickable modal interface to allow admin to view detailed feedback entries in an overlay dialog without leaving the current page context. (5 points, Issue #4692)
3	1.1.6	Feature 6.0 UI Decluttering (Feedback Table View)	Declutter the Feedback Table View by removing redundant elements, removing columns, and improving the overall layout for enhanced readability and usability. (7 points, Issue #4693)
3	1.1.7	Feature 7.0 Data Pagination (Feedback Manager)	Introduce data pagination functionality to the Feedback Manager in the Admin Dashboard to efficiently handle large datasets, enabling



			users to navigate through records in manageable pages. (7 points, Issue #4694)
3	1.1.8	Feature 8.0 Targeted Dormant Material Export	Develop an export feature that allows administrators to selectively export dormant materials from the system based on defined criteria, supporting data management and archival processes. (8 points, Issue #4695)
3	1.1.9	Feature 9.0 Failed Query Keyword Tracking	Implement a tracking mechanism to capture and log keywords from failed search queries, enabling the team to identify content gaps and improve system coverage over time. (8 points, Issue #4696)
3	1.1.10	Feature 10.0 "Least Viewed" Metric Tracking	Build a metric tracking system that identifies and monitors the least-viewed content items, providing insights to inform content curation and resource prioritization decisions. (8 points, Issue #4697)
3	1.1.11	Feature 11.0 Column Merger & Analysis Dropdown Refinement	Merge related data columns and refine the analysis dropdown menus to reduce redundancy, improve navigation clarity, and streamline the analytical workflow for admin users. (4 points, Issue #5100)
3	1.1.12	Feature 12.0 Explicit Date Formatting & "Feedback Type" Column Renaming	Standardize date formatting across the application with explicit format definitions and rename the Feedback Type column for improved clarity and consistency in the data presentation layer. (5 points, Issue #5101)

All features have a status of Closed as of the sprint end date.

Glossary of Terms

Level of Effort (LOE):

Level of Effort (LOE) refers to the amount of work, time, and resources required to complete a specific task or activity. It is typically used for support-type tasks that do not produce a tangible deliverable but are necessary for project completion.

WBS Code (WBS Number):

A WBS Code, also commonly referred to as a WBS Number, is a unique identifier assigned to each element in a Work Breakdown Structure. It indicates the hierarchical position of a component within the WBS and is used for tracking, organization, and reference purposes.

Work Package:



A Work Package is the lowest level of a WBS branch. It represents a deliverable or a specific unit of work that can be assigned, scheduled, and tracked. Work Packages are defined clearly enough to allow estimation of cost, duration, and resources.

WBS Component:

A WBS Component refers to any element within a Work Breakdown Structure at any level of the hierarchy. It may represent a Work Package or a higher-level grouping of work. All elements in the WBS are considered WBS Components regardless of their level.

WBS Element:

A WBS Element is any individual item within the Work Breakdown Structure that may exist at any level of the hierarchy. It can represent a delivery, a grouping of tasks, or a Work Package. A WBS Element may contain other WBS Elements or Work Packages depending on its level in the structure.

06 Schedule Management Plan

Introduction

The LitPath AI project is a standalone AI-powered web application developed for the DOST-STII Library. It helps researchers search for theses and dissertations using smart search technology, including vector embeddings and large language models, instead of the traditional keyword-only OPAC system.

This Schedule Management Plan explains how the project team plans, tracks, and manages the project timeline from start to finish. It covers how the schedule is created, how progress is monitored, and how any changes to the timeline are handled.

It is important to note that LitPath AI is developed by a team of dedicated full-time students. Development activities are primarily conducted during weekday evenings (typically 4 to 6 hours per session), with additional progress made during occasional weekend morning sessions. The schedule is thoughtfully aligned with academic commitments, ensuring a sustainable and consistent workflow. Task estimates are planned to reflect this balanced approach, allowing the team to maintain productivity while effectively managing their academic responsibilities.

Schedule Management Approach

The project schedule is managed using OpenProject, a web-based project management tool that supports Gantt charts, task tracking, milestones, and team assignments. OpenProject allows the team to plan tasks, assign them to team members, set due dates, and track completion in real time.



The schedule is built starting from the deliverables defined in the project's Work Breakdown Structure (WBS). From there, the team follows these steps:

- **Activity Definition** — The team identifies the specific tasks needed to complete each deliverable. For example, under backend development, tasks include setting up the vector search engine, integrating the LLM API, and testing AI outputs.
- **Activity Sequencing** — Tasks are arranged in the correct order based on which ones must be completed first. Dependencies are clearly marked in OpenProject.
- **Duration Estimating** — The team estimates how long each task will take based on its complexity and the team's available working hours. Given that members are students, estimates account for evening-only weekday sessions (4 to 6 hours) and occasional half-day weekend work.
- **Resource Assignment** — Tasks are assigned to team members based on their roles. Jenine and Charijoy handle development tasks, while Joshua handles quality assurance and testing.

Once a draft schedule is created, the whole team reviews it to confirm that task assignments, timelines, and sequencing are realistic given their availability. After the team agrees, the Project Sponsor reviews and approves the schedule before it is finalized (baselined).

a. Working Hours and Capacity

Since all team members are full-time students, the project follows a part-time working schedule. The table below summarizes the team's typical working capacity:

Timeframe	Availability & Application	Assigned Value
Weekly	Part-time allocation per project resource	10 hours per week
Monthly	Based on a standard 4-week month	40 hours per month
Costing	Standard flat rate applied in OpenProject	Php 500.00 per hour

All task duration estimates and resource costs in the schedule are based on this established capacity. For example, a task requiring 20 hours of effort would realistically take a single team member a minimum of 2 full weeks to complete, as they cannot exceed the 10-hour weekly limit. This distinction is critical when reading the OpenProject timeline, distributing workload, and calculating the project budget.

b. Project Milestones

The following milestones mark the key checkpoints of the LitPath AI project:

Milestone	Target Date	Status
Project Scoping	April 2025	Completed
Information Gathering	April 2025	Completed
Completion of Project Deliverables & Documentation	June 2025	Completed
Design Thinking Process (Stages 1–5)	Aug 2025	Completed
Core UI Development (System UI, Citation Generator, Bookmark)	Sep – Oct 2025	Completed
AI Search Engine, Generate Overview, Feedback Mechanism & Admin Dashboard	Oct 2025 – Jan 2026	Completed
Feature Refinement	Feb – Mar 2026	Completed
Continued Development & Performance Optimization (incl. codebase restructuring)	Mar – May 2026	In Progress



Proxmox Server Deployment & Internal Testing	May 2026	In Progress
Bug Fixes & Usability Testing	May – Jun 2026	Not Started
Final Deployment	Jun – Jul 2026	Not Started

c. Roles and Responsibilities in Schedule Development

Person Assigned	Role	Schedule Responsibilities
Tracie Tomon	Project Manager	Leads schedule development; creates and maintains the OpenProject schedule; validates with team; obtains sponsor approval; updates the schedule after each bi-weekly review.
Jenine Elaine Dulay	Developer	Participates in defining and estimating development tasks; performs assigned tasks; reports actual progress and any blockers to the Project Manager.
Charijoy Cempron	Developer	Participates in defining and estimating development tasks; performs assigned tasks; reports actual progress and any blockers to the Project Manager.
Joshua Sta Ana	QA / Tester	Defines and estimates testing tasks; executes test plans; documents bugs; reports QA outcomes to the Project Manager during bi-weekly reviews.
Khasian Eunice Romulo	Project Sponsor	Reviews and approves the final schedule before it is baselined; approves schedule change requests that exceed the defined threshold.
DOST-STII	Stakeholder / Client	Reviews the schedule against operational and deployment needs. All feedback is routed to the Project Sponsor, who then relays the updates and feedback to the project team via Viber.

Schedule Control

Schedule control means making sure the project stays on track throughout its lifecycle. This section explains how the team monitors progress, communicates updates, and responds to delays.

Monitoring Frequency

The project schedule is reviewed and updated every two weeks (bi-weekly). During each update, task owners report the following through the Microsoft Teams group chat or during a Teams meeting:

- Actual start dates
- Actual finish dates
- Completion percentage for tasks that are still in progress
- Any blockers or issues affecting the timeline

These updates are then entered into OpenProject by the Project Manager to keep the schedule current and visible to all team members.

Communication Channels

The team uses the following communication tools, depending on the audience:

Channel	Used For	Audience
---------	----------	----------



Microsoft Teams (group chat)	Day-to-day coordination, task updates, sharing files, and quick questions	Project team (Tracie, Jenine, Charijoy, Joshua)
Microsoft Teams (meeting/call)	Bi-weekly schedule reviews, sprint planning, and issue discussions	Project team
Viber	Client updates, feedback sharing, and deployment coordination	Project Manager and DOST-STII client representative
OpenProject	Official schedule tracking, task assignments, milestone updates, and progress visibility	Entire project team

Roles and Responsibilities in Schedule Control

Role	Responsibilities
Project Manager	Facilitates bi-weekly review meetings on Teams; collects and records task updates; evaluates schedule variances; submits change requests when needed; communicates schedule status to the sponsor and to the client via Viber.
Development Team	Attends bi-weekly reviews via Teams; reports actual progress and completion percentages; flags potential delays early, especially when academic deadlines from other subjects are coming up; participates in finding solutions to schedule conflicts.
Project Sponsor	Stays informed of schedule status through regular reports from the Project Manager; reviews and approves schedule change requests that exceed the defined threshold.

Schedule Changes and Thresholds

Not all delays or adjustments automatically require a formal change request. The team uses a 10% threshold rule to decide when a change needs to go to the Project Sponsor for approval.

Change Threshold

A formal Schedule Change Request must be submitted to the Project Sponsor if either of the following is true:

- A single task or work package is expected to take 10% more or 10% less time than originally planned.
- The overall project timeline is expected to increase or decrease by 10% or more from the baselined schedule.

Any adjustments that fall below these thresholds may be approved directly by the Project Manager without escalating to the Project Sponsor.

Change Request Process

Step	Who	What Happens
1	Team Member	Spots a potential delay or schedule issue and raises it in the Microsoft Teams group chat or during the bi-weekly review.
2	Project Manager + Team	Meet via Teams to review the issue; identify which tasks are affected; calculate the variance; and explore options such as reassigning tasks, reducing scope, or requesting more time.
3	Project Manager	Determines if the change exceeds the 10% threshold. If yes, a formal Schedule Change Request is prepared and submitted to the Project Sponsor for approval.



4	Project Sponsor	Reviews and approves or rejects the change request.
5	Project Manager	Updates the schedule in OpenProject and notifies the team via Teams chat. Communicates any impact to the Project Sponsor via Viber. Saves the approved change request in the project records.

Scope Change

Occasionally, approved changes to what the project must deliver (scope changes) may significantly affect the schedule. For LitPath AI, this could happen when the DOST-STII client requests a new or revised feature, or when a technical discovery during development requires a different approach than originally planned.

If an approved scope change is expected to significantly affect the timeline, the Project Manager may request that the schedule be re-baselined — meaning the official reference schedule is reset to reflect the updated plan.

The process for handling scope-driven schedule changes is as follows:

- The Project Sponsor formally approves the scope change.
- The Project Manager and team evaluate how the change affects the current schedule and available working hours.
- If the impact is significant, the Project Manager submits a re-baselining request to the Project Sponsor.
- The Project Sponsor reviews and approves the request.
- The Project Manager updates the schedule in OpenProject and communicates the changes to the team via Teams and to the client via Viber.

Examples of scope changes that may trigger re-baselining in LitPath AI include adding a new feature to the AI search engine beyond what was originally agreed, significant redesign of the admin dashboard, or changes to the deployment environment that require additional setup and testing time. To prevent uncontrolled scope changes, all feature clarifications with the client must be documented and confirmed before development begins.

SPONSOR ACCEPTANCE

Approved by the Project Sponsor:

Khasian Eunice Romulo
Science Research Specialist II / Unit Head
(Digital Information and Processing Delivery Unit)

Date: April 1, 2026

07 Requirements Management Plan

Introduction

The Requirements Management Plan establishes how requirements will be collected, analyzed, documented, and managed throughout the lifecycle of the LitPath AI project. LitPath AI is a standalone AI-powered web application that provides advanced, AI-driven academic thesis and dissertation search capabilities for the DOST-STII Library. While it operates independently as its own platform, it relies on the DOST-STII Library's existing thesis and dissertation database as its primary data source, maintaining a read-only connection to ensure no modifications are made to library records.

Requirements are divided into two categories: project requirements (non-technical requirements related to delivery, integration, and compliance) and product requirements (technical specifications for system functionality, performance, and quality). Inputs to this plan include the LitPath AI Project Charter, SRS document, the Stakeholder Management Strategy, and the Stakeholder Analysis Matrix. The Project Sponsor is Khasian Eunice M. Romulo, Science Research Specialist II / Unit Head of the Digital Information and Processing Delivery Unit, DOST-STII. The Project Manager is Tracie Tomon (HecTech Group Leader), authorized to negotiate resources, delegate responsibilities, and ensure timely completion.

Requirements Management Approach

The requirements management approach is the methodology the project team will use to identify, analyze, document, and manage the project's requirements. For LitPath AI, this approach is broken down into four areas: requirements identification, requirements analysis, requirements documentation, and ongoing requirements management.

Requirements Identification: The LitPath AI team collected requirements through analysis of the DOST-STII OPAC system's limitations, benchmarking tools like UP Diliman, Tuklas and Google Scholar, and consultations with the project client and faculty advisers. Methods included document reviews, client meetings, and iterative SRS development to ensure all requirements were captured.

Requirements Analysis: The team categorized requirements into functional and non-functional groups and evaluated each based on priority, feasibility, and technical constraints. A Work Breakdown Structure (WBS) was used to organize these requirements into manageable components and map them to corresponding deliverables. Traceability was established through use cases and business rules, while accountability, metrics, and acceptance criteria were defined to ensure clear completion standards.

Requirements Documentation: All identified and analyzed requirements were documented with unique IDs for functional requirements and supporting use cases and business rules. These were integrated into the project plan and tracked to ensure proper implementation and reporting.

Ongoing Requirements Management: Throughout the project lifecycle, the Project Manager ensures continuous monitoring of requirement status and addresses any issues raised by the team. Any changes follow the established change control process, and final approval of all requirements is secured from the Project Sponsor before project closure.

Configuration Management

To effectively manage the LitPath AI project, communication must be managed and controlled. While every effort has been made to identify requirements thoroughly during project initiation and planning, situations may arise that require changes. In these situations, it is important to utilize configuration management to consider proposed changes, establish a review and approval process, implement approved changes, and communicate those changes to all stakeholders.

Documentation and Version Control: All project documentation is version-controlled and tracked through the SRS revision history. The LitPath AI SRS has been through 15 documented revision cycles (Draft v1.0 through Draft v3.0), with each revision attributed to specific team members and dated. Any proposed changes to project requirements must be reviewed by the HecTech team and have written approval from the Project Sponsor, Ms. Khasian Eunice Romulo, before any documentation changes are made. Once approved, the Project Manager is responsible for communicating the change to all project stakeholders.

Change Control: Any proposed changes to project requirements must be carefully considered before approval and implementation, as such changes are likely to impact project scope, time, and cost. To ensure consistency, all requests must adhere to the formal Change Control Process, beginning with the submission of a LitPath AI Change Request Form to the Project Manager and subsequent logging in the Change Request Register.

Changes undergo a preliminary impact evaluation by the Project Manager, Technical Lead, and System Architect. Critically, any changes affecting the read-only connection to the DOST-STII database or the Gemini AI service integration must also be reviewed with the alternate POC, Mr. Jonathan Abalon, who is specifically responsible for system access and technical support and has identified system integration risks as a primary concern.

Following evaluation, requests are submitted to the Change Control Board (CCB) for formal review. The CCB Chair renders the final decision; specifically, any change that affects milestone dates must be explicitly approved by the Project Sponsor. Once approved, changes are tracked as tasks in OpenProject, all relevant project documentation (such as the SRS) is updated, and the final status is communicated to stakeholders via Microsoft Teams or Viber.

Requirements Prioritization Process

Prioritizing requirements is an important part of requirements management for LitPath AI, as the project operates under a fixed deployment deadline of June–July 2026 and limited development resources. The HecTech project team, in coordination with key stakeholders, established priorities for all project requirements using a three-level scale.

Requirement priorities were established in coordination with the key stakeholders and managed closely per the Stakeholder Management Strategy: Ms. Khasian Eunice Romulo and Mr. Jonathan Abalon (High Power, High Interest). Their primary concerns are search relevance improvement and system integration stability, which directly informed the High priority classification of core functional requirements. The chart below illustrates the three priority levels and defines how requirements are grouped:

Priority Level	Definition
High	Must be delivered for the system to function which is directly tied to core client needs and project charter deliverables.



Medium	Important features that significantly improve usability but are not blockers for deployment
Low	Additional features that can be implemented later if necessary

As the project moves forward, and additional constraints are identified, the project team and stakeholders may meet to determine which requirements must be achieved, which can be re-baselined, or which can be omitted. These determinations will be made collaboratively based on the priority levels above. Any approved changes will be reflected in updated project documentation and communicated to all stakeholders.

Product Metrics

Product metrics are an important part of determining the success of LitPath AI. The following metrics are drawn from the LitPath AI Project Charter and SRS and represent the quantitative benchmarks the system must meet or exceed.

Search Performance:

- Search relevance accuracy must reach at least 70% relevance in the top 10 results
- Search response time must be up to the industry standard, improved from the current 2–3 minutes in the existing OPAC system

User Satisfaction:

- The system must achieve a user satisfaction score of at least 3.5 out of 5 for search feature usability, to be confirmed through usability testing during the Bug Fixes and Usability Testing phase (May–June 2026)

Functional Coverage:

- Citation formats supported: APA 7th Edition, MLA 9th Edition, Chicago Manual of Style, and IEEE (4 formats)
- All functional requirements must be delivered and verified before final deployment

System Reliability and Security:

- Each authenticated user may have only one active session at a time
- Passwords must be stored as hashed values and all sessions must be managed with secure tokens

Requirements Traceability Matrix

The requirements traceability matrix ensures that all LitPath AI deliverables meet the requirements established in the Project Charter and SRS v3.0. It provides a thread from each requirement through its corresponding use case, charter reference, and priority level, ensuring that all features can be traced from inception through to testing and deployment.

Project Name	LitPathAI	Project Manager	Tracie Tomon	
Req #	UC Ref	Requirement Description	Test Case Ref	Test Cases
F10	UC-01	The system shall support user authentication with	TC_MA	TC_MA_001 to TC_MA_030



		email and password through login.		
F11	UC-01	The system shall manage user sessions with secure tokens. Each authenticated user can have only one active session at a time.	TC_MA	TC_MA_008, TC_MA_039
F12	UC-01	The system shall allow authenticated users to change their passwords.	TC_MA	TC_MA_031 to TC_MA_050
F13	UC-01	The system shall allow authenticated users to deactivate their accounts.	TC_MA	TC_MA_049
F15	UC-01	The system shall support multiple user roles: User and Admin with appropriate access permissions.	TC_MA	TC_MA_006, TC_MA_007
F1	UC-02	The system shall allow users to enter natural language research queries in a search input field.	TC_ST/D	TC_ST/D_001 to TC_ST/D_010
F2	UC-02	The system shall perform semantic search across indexed thesis documents using vector embeddings and return relevant results ranked by similarity.	TC_ST/D	TC_ST/D_006 to TC_ST/D_025
F4	UC-02	The system shall display thesis metadata for each search result including title, author, publication year, abstract, degree, subject/s, and university.	TC_ST/D	TC_ST/D_026 to TC_ST/D_050
F4	UC-03	The system shall display thesis metadata for each	TC_VTD	TC_VTD_001 to TC_VTD_021



		search result including title, author, publication year, abstract, degree, subject/s, and university.		
F5	UC-04	The system shall allow users to bookmark/save thesis documents for later reference.	TC_BMK	TC_BMK_001 to TC_BMK_013
F6	UC-04	The system shall persist bookmarks for authenticated users in the database.	TC_BMK	TC_BMK_007 to TC_BMK_010
F7	UC-05	The system shall generate properly formatted citations in APA, MLA, Chicago, and IEEE citation styles.	TC_CTD	TC_CTD_001 to TC_CTD_012
F8	UC-06	The system shall maintain research history tracking all user search sessions with timestamps.	TC_VSH	TC_VSH_001 to TC_VSH_010
F9	UC-06	The system shall allow users to reload and revisit previous research sessions from history.	TC_VSH	TC_VSH_011 to TC_VSH_020
F14	UC-07.1	The system shall allow users to provide feedback on search results including relevance ratings (star ratings) and comments.	TC_EF	TC_EF_001 to TC_EF_050
F17	UC-07.2	The system shall allow admins to respond to feedback and generate reports for usage analytics.	TC_MF	TC_MF_001 to TC_MF_075
F17	UC-08	The system shall allow admins to respond to feedback and generate reports	TC_UA	TC_UA_001 to TC_UA_085



		for usage analytics.		
F3	UC-09	The system shall generate AI-powered overview summaries synthesizing information from retrieved thesis documents with inline reference citations.	TC_GAA	TC_GAA_001 to TC_GAA_025
F16	UC-09	The system shall extract and index metadata from thesis documents including title, author, year, abstract, degree, and subjects.	TC_GAA	TC_GAA_026 to TC_GAA_050

SPONSOR ACCEPTANCE

Approved by the Project Sponsor:

Khasian Eunice Romulo
Science Research Specialist II / Unit Head
(Digital Information and Processing Delivery Unit)

Date: April 30, 2026



08 Risk Register

Risk Identification			Qualitative Rating				Risk Response		
Risk Title	Risk Description	Risk Category	Probability	Impact	Risk Score	Risk Ranking	Risk Response	Trigger	Risk Owner
API Unavailability	Due to LIPATH AI's dependency on the Gemini AI API for semantic query processing and AI-generated overviews, an extended service outage, rate limiting, or API deprecation by Google could disable the system's core search functionality and prevent the project from meeting its primary acceptance criteria.	Technical / External	4	5	20	1	Implement response caching for high-frequency queries; establish a fallback degraded-mode search using ChromaDB vector similarity without Gemini AI; coordinate with DOST-STI to confirm API cost coverage before production deployment.	Gemini AI API returns persistent errors, exceeds rate limits, or announces deprecation during development or testing phases.	Charjoij Cempron
Search Relevance Failure	Due to the complexity of semantic search over specialized academic literature, the system's RAG pipeline requires continuous iterative refinement to produce results that consistently meet the team's internal relevance benchmarks. Failure to address low-relevance query patterns before formal usability testing could result in rework cycles that consume sprint capacity originally reserved for deployment activities.	Technical	3.25	5	16.25	2	Conduct iterative internal relevance evaluation throughout development using a defined test query set; refine LLM prompts continuously; perform pre-usability relevance testing at least two weeks before formal usability testing begins.	Internal relevance testing reveals consistent failures across the defined test query set, or early informal user feedback surfaces recurring irrelevant results.	Charjoij Cempron
LLM Hallucination	Due to the inherent probabilistic nature of large language models, the Gemini AI API may generate inaccurate, fabricated, or misleading summaries, citations, or search overviews that appear credible to users, undermining the system's reliability as an academic research tool and potentially causing DOST-STI to reject the deliverable during acceptance testing.	Technical / AI	3	5	15	3	Implement source-grounded prompting that constrains Gemini AI responses strictly to retrieved document content; display source citations alongside every AI-generated output; conduct targeted hallucination testing as part of the QA cycle using known dataset entries to verify output accuracy.	QA testing identifies AI-generated summaries or citations that do not correspond to actual content in the indexed dataset, or usability testing participants report misleading or fabricated search results.	Joshua Sta Ana
API Latency / Slow Response	Due to LIPATH AI's reliance on external API calls to Gemini AI for query processing and result generation, slow API response times under load or degraded network conditions could cause search query response times to exceed the industry standard, failing the system's performance acceptance criterion even when the API is technically available.	Technical / External	3.5	4	14	4	Implement timeout handling with defined maximum wait thresholds; use asynchronous API calls to prevent blocking; cache responses for frequent or repeated queries; display partial results while full AI overview loads to maintain perceived performance.	API response times consistently exceed the industry standard during internal load testing, or usability testing participants report noticeable delays between query submission and results display.	Charjoij Cempron
Dataset Provision Delay	Due to the project's reliance on DOST-STI to provide a representative sample dataset of theses and dissertations, any delay or inadequacy in the dataset handover could prevent the team from completing RAG pipeline validation, relevance testing, and system acceptance within the project timeline.	External	3.25	4	13	5	Confirm dataset delivery schedule with DOST-STI at the start of each major testing phase; request a preliminary sample early to identify format or volume issues; define minimum dataset requirements in writing with the Project Sponsor.	DOST-STI has not provided the required dataset within two weeks of the agreed handover date, or the dataset is insufficient in volume or metadata quality for testing.	Tracie Tomon
Cloud Hosting Delay	Due to the planned transition from the local Proxmox development environment to a cloud hosting environment managed by DOST-STI for final production deployment, delays in confirming the hosting setup or cost approvals could prevent the system from going live by the June-July 2026 deadline.	External / Schedule	3	4	12	6	Initiate cloud hosting discussions with DOST-STI no later than May 2026; document hosting requirements in the Statement of Work; maintain the Proxmox server as a fallback demo environment; escalate any delays immediately to the Project Sponsor.	DOST-STI has not confirmed the production hosting environment or cost coverage by May 15, 2026, or the cloud environment setup encounters technical blockers during pre-deployment testing.	Tracie Tomon
Database Inconsistency	Due to the system's use of Supabase (PostgreSQL) for relational data and ChromaDB for vector embeddings, a failure in data consistency between the two systems could corrupt search results, break the bookmarking feature, or prevent accurate citation generation.	Technical	2.75	4	11	7	Implement automated consistency checks between Supabase and ChromaDB after each data ingestion cycle; maintain backup storage on the HDD per the system design; establish a documented data recovery procedure; test database integrity in every sprint QA cycle.	Search results return mismatched or duplicate records; bookmarked items fail to load; ChromaDB embeddings become out of sync with Supabase document records.	Joshua Sta Ana
Data Loss Ingestion	Due to the complexity of the data ingestion and embedding generation pipeline, failures during batch processing — such as interrupted ingestion jobs, malformed document records, or embedding errors — may result in theses or dissertations being silently omitted from the indexed dataset, causing incomplete search results without any visible system error.	Technical	3	3.25	9.75	8	Implement comprehensive ingestion logging with record-level success and failure tracking; add post-ingestion validation that compares source document count against indexed record count; implement retry mechanisms for failed embedding generations; alert the team if record count discrepancies exceed a defined threshold.	Post-ingestion validation reveals a mismatch between the number of source documents provided and the number of records successfully indexed in ChromaDB or Supabase.	Charjoij Cempron
Scope Creep	Due to DOST-STI's ability to submit feedback and change requests during the Feature Refinement and usability testing phases, uncontrolled additions beyond the agreed Statement of Work could consume the team's limited sprint capacity and jeopardize the June-July 2026 deadline.	Scope	3	3	9	9	Enforce the formal scope change control process for all client-initiated change requests; document and confirm all feature modifications in writing before development; require Project Sponsor approval for any changes affecting schedule or budget.	DOST-STI requests a new feature or significant modification outside the defined scope in the Statement of Work during the testing or deployment phase.	Tracie Tomon
Data Privacy Non-Compliance	Due to the Data Privacy Act of 2012 (RA 10173) requirements applicable to LIPATH AI, any failure to implement proper data handling for user accounts, search history, and admin access could result in a compliance violation that blocks formal project acceptance or requires significant post-deployment rework.	Compliance	2	4.25	8.5	10	Ensure all user authentication uses secure login with role-based access control and session management with secure tokens; keep public search anonymous; enforce HTTPS/SSL on all connections; conduct a compliance review led by the Security Lead before usability testing.	Security audit identifies unprotected user data, unsecured API endpoints, or missing authentication controls during internal testing.	Joshua Sta Ana
Dependency / Library Compatibility	Due to the project's reliance on a Python-based AI stack including SentenceTransformers, ChromaDB, LangChain-type utilities, and Supabase client libraries, updates or breaking changes to any of these dependencies could introduce runtime errors, degrade system performance, or require unplanned refactoring that consumes sprint capacity.	Technical	2.25	3.75	8.44	11	Pin all dependency versions in the project's requirements file and lock file; test all updates in the staging environment before applying to the development or production environment; monitor dependency changelogs for breaking changes before each sprint.	System errors occur after a library update or environment change, or a dependency announces a breaking change in an upcoming release that affects a core system component.	Charjoij Cempron
Key Resource Unavailability	Due to the four-member HeTech team having no external outsourcing capacity, the unexpected unavailability of a key technical resource due to illness, personal emergency, or academic withdrawal could significantly disrupt sprint delivery and system integration milestones.	Resource	2	4	8	12	Maintain up-to-date technical documentation for all system components; cross-train team members on critical modules within sprint capacity; redistribute tasks among remaining members in the event of a temporary absence; escalate to Project Sponsor if absence exceeds one sprint.	A team member is unable to perform assigned sprint tasks for more than three consecutive working days due to illness, personal emergency, or academic withdrawal.	Tracie Tomon
Search Performance Degradation	Due to the system's use of SentenceTransformers for embedding generation and ChromaDB for vector search, performance degradation under concurrent user load or large dataset volume could cause search query response times to exceed the industry standard, failing the system performance acceptance criterion.	Technical	2.25	3	6.75	13	Conduct load and performance testing during the Usability Testing phase; optimize ChromaDB indexing and embedding retrieval pipelines; implement query result caching for high-frequency searches; address all performance bottlenecks before final deployment.	Internal load testing reveals average search response times exceeding industry standard under simulated concurrent user conditions, or usability testing participants report noticeable delays.	Charjoij Cempron
Local Server Hardware Failure	Due to the project's use of a locally hosted Proxmox server for development and internal testing, hardware failure such as SSD failure or power interruption could result in loss of code, database records, or configuration that delays sprint delivery.	Technical	2	3.25	6.5	14	Maintain all source code in a remote Git repository with regular commits after every work session; back up the Supabase database and ChromaDB embeddings to the 500 GB HDD; document all server configuration settings to enable rapid environment recovery.	The Proxmox server becomes inaccessible for more than 24 hours, or a team member reports data loss or file corruption on the local development environment.	Joshua Sta Ana
API Cost Overrun	Due to the project's use of the Gemini AI API without a formal written cost commitment from DOST-STI, any unexpected spike in API usage during testing or deployment could result in costs that exceed the project's contingency reserve and interrupt system functionality.	Cost / External	2	3	6	15	Confirm DOST-STI's formal commitment to cover all cloud hosting and AI/LLM API costs in writing before transitioning to production; monitor API usage costs throughout testing; report any projected overruns to the Project Sponsor immediately.	DOST-STI requests a delay in API budget commitment, or Gemini AI API usage costs during testing exceed the informally estimated monthly threshold.	Tracie Tomon



Risk matrix

Risk Matrix						
		IMPACT				
		Very Low (1)	Low (2)	Moderate (3)	High (4)	Very High (5)
PROBABILITY	Very High (5)	Medium	High	High	Critical	Critical
	High (4)	Low	Medium	High	#4 API Latency / Slow	#1 API Unavailability
	Moderate (3)	Low	Medium	#8 Data Loss in Ingestion	#5 Dataset Provision Delay	#2 Search Relevance Failure
	Low (2)	Low	Low	#13 Search Performance	#10 Data Privacy Non-	High
	Very Low (1)	Low	Low	Low	Low	Medium

LEGEND

Critical	Score 20–25
High	Score 10–19
Medium	Score 5–9
Low	Score 1–4

RISK SUMMARY

Rank	Risk Title	P avg	I avg	Score	Level
1	API Unavailability	4	5	20	Critical
2	Search Relevance Failure	3.25	5	16.25	High
3	LLM Hallucination	3	5	15	High
4	API Latency / Slow Response	3.5	4	14	High
5	Dataset Provision Delay	3.25	4	13	High
6	Cloud Hosting Delay	3	4	12	High
7	Database Inconsistency	2.75	4	11	High
8	Data Loss in Ingestion	3	3.25	9.75	Medium
9	Scope Creep	3	3	9	Medium
10	Data Privacy Non-Compliance	2	4.25	8.5	Medium
11	Dependency / Library Compat.	2.25	3.75	8.44	Medium
12	Key Resource Unavailability	2	4	8	Medium
13	Search Performance Degrad.	2.25	3	6.75	Medium
14	Local Server Hardware Failure	2	3.25	6.5	Medium
15	API Cost Overrun	2	3	6	Medium



Risk assessment Voting Table

Risk Assessment Voting Table														
Each team member independently assigned Probability (P) and Impact (I) ratings (scale 1–5) during the risk assessment meeting. Final scores are averaged.														
Risk Identification			Probability Votes (1–5)					Impact Votes (1–5)					Final Score	
Rank	Risk Title	Category	P - Tracie	P - Jenine	P - Charlijoy	P - Joshua	P Average	I - Tracie	I - Jenine	I - Charlijoy	I - Joshua	I Average	Final Score	Risk Level
1	API Unavailability	Technical / External	4	4	5	3	4	5	5	5	5	5	20	Critical
2	Search Relevance Failure	Technical	3	3	4	3	3.25	5	5	5	5	5	16.25	High
3	LLM Hallucination	Technical / AI	3	3	3	3	3	5	5	5	5	5	15	High
4	API Latency / Slow Response	Technical / External	3	4	4	3	3.5	4	4	4	4	4	14	High
5	Dataset Provision Delay	External	3	3	3	4	3.25	4	4	4	4	4	13	High
6	Cloud Hosting Delay	External / Schedule	3	3	3	3	3	4	4	4	4	4	12	High
7	Database Inconsistency	Technical	3	3	2	3	2.75	4	4	4	4	4	11	High
8	Data Loss in Ingestion	Technical	3	3	3	3	3	3	3	4	3	3.25	9.75	Medium
9	Scope Creep	Scope	3	3	3	3	3	3	3	3	3	3	9	Medium
10	Data Privacy Non-Compliance	Compliance	2	2	2	2	2	4	4	4	5	4.25	8.5	Medium
11	Dependency / Library Compatibility	Technical	2	2	3	2	2.25	4	4	3	4	3.75	8.44	Medium
12	Key Resource Unavailability	Resource	2	2	2	2	2	4	4	4	4	4	8	Medium
13	Search Performance Degradation	Technical	2	2	3	2	2.25	3	3	3	3	3	6.75	Medium
14	Local Server Hardware Failure	Technical	2	2	2	2	2	3	3	3	4	3.25	6.5	Medium
15	API Cost Overrun	Cost / External	2	2	2	2	2	3	3	3	3	3	6	Medium

Risk assessment Voting Table

Probability & Impact Rating Scale				
Rating	Probability Descriptor	Probability Criteria	Impact Descriptor	Impact Criteria
1	Very Low	Risk event is highly unlikely; no historical precedent within the project context and existing controls make occurrence improbable.	Very Low	Negligible effect on project objectives; easily absorbed within existing sprint capacity with no schedule or cost impact.
2	Low	Risk event is unlikely but possible; limited exposure or conditions are largely within the team's control.	Low	Minor effect; resolvable within sprint capacity with minimal schedule adjustment.
3	Moderate	Risk event is possible; some exposure exists due to external dependencies or known project constraints outside the team's direct control.	Moderate	Noticeable effect on schedule, cost, or quality; requires corrective action within the sprint cycle.
4	High	Risk event is likely; strong dependency on uncontrolled external factors or recurring conditions increase probability.	High	Significant effect; threatens a project milestone or acceptance criterion and requires escalation to the Project Sponsor.
5	Very High	Risk event is very likely or near-certain; critical dependency exists with no current mitigation in place.	Critical	Severe effect; directly blocks project delivery, formal acceptance, or deployment.



Risk Score Color Legend

Risk Score Color Legend (Score = P avg × I avg, scale 1–25)	
Low	Score 1.00 – 4.99
Medium	Score 5.00 – 9.99
High	Score 10.00 – 19.99
Critical	Score 20.00 – 25.00

10 Project Quality Management Plan

Introduction

The Quality Management Plan is an integral part of the LitPath AI project management plan. The purpose of this plan is to describe how quality will be managed throughout the software development lifecycle (SDLC), from initial architecture design through deployment, testing, and post-launch maintenance. It includes the processes and procedures for ensuring quality planning, assurance, and control are consistently executed. All stakeholders, developers, and QA personnel should be familiar with how quality will be planned, assured, and controlled.

The Quality Management Plan for the LitPath AI project will establish the activities, processes, and procedures for ensuring a secure, accurate, and high-performing software product upon the conclusion of the project. The purpose of this plan is to:

Ensure quality is planned into every SDLC phase and AI pipeline component.

1. Define how quality will be managed across development, testing, and deployment.
2. Define quality assurance activities focused on process compliance, code reviews, and architecture validation.
3. Define quality control activities focused on functional testing, AI accuracy validation, performance benchmarking, and security verification.
4. Define acceptable quality standards aligned with DOST-STII guidelines, Philippine Data Privacy standards, and software engineering best practices.

Quality Management Approach

This section describes the approach the organization will use for managing quality throughout the project's life cycle. Quality must always be planned into a software project to prevent rework, technical debt, security vulnerabilities, and inaccurate AI output. Quality will be considered from both a **product** (software functionality, AI relevance, UX, performance) and **process** (SDLC adherence, code quality, documentation, testing coverage) perspective.

The quality management approach for LitPath AI will ensure quality is planned for both the product and processes. To be successful, this project will meet its quality objectives by utilizing an integrated quality approach to define quality standards, measure quality, and continuously improve through Agile retrospectives and AI pipeline evaluations.



Product Quality for LitPath AI will be defined by DOST-STII functional specifications, AI relevance thresholds, security compliance (OWASP, DPA 2012), and user experience standards. The focus is on delivering a stable, accurate, and responsive web application that meets established quality standards and user satisfaction.

Process Quality for LitPath AI will focus on the development of workflows, version control practices, CI/CD pipeline execution, documentation compliance (aligned with the DOST-STII SDS template), and testing protocols. Establishing process quality standards will ensure all development activities conform to organizational engineering standards.

The project team will work with the QA Tester and Project Lead to define and document all organizational and project-specific quality standards for both product and processes. All quality documentation will become part of the LitPath AI Project Plan and will be transitioned to DOST-STII IT operations upon successful project completion.

Metrics will be established and used to measure quality throughout the project life cycle. The QA Tester will be responsible for working with the development team to define these metrics, conduct measurements, and analyze results. These product and process measurements will be used as key criteria in determining project success and must be reviewed by the Project Sponsor. Metrics will include:

1. **Schedule:** Sprint delivery rate, milestone adherence.
2. **Resources:** Developer/QA allocation efficiency, cloud/API utilization.
3. **Cost:** Budget vs. actual spending (hosting, Gemini API usage, development tools).
4. **Process Performance:** Code review turnaround time, CI/CD pipeline success rate, documentation completeness vs. SDS template.
5. **Product Performance:** API response time (industry standard, <5s streaming), AI relevance score (cosine similarity threshold <1.5), system uptime (≥99.5%), test coverage (≥80% unit/integration).
6. **Customer Satisfaction:** CSM survey average rating (≥4.0/5.0), feedback resolution rate, failed query rate.

Quality improvements will be identified by any team member. Each recommendation will be reviewed to determine cost vs. benefit, impact on AI accuracy, system performance, or user experience. If implemented, the Project Lead will update project documentation, and the QA Tester will update test suites and QA logs.

Quality Requirements / Standards

This section describes how the project team and QA group will identify and document quality requirements and standards, and how compliance will be demonstrated.

Product Quality:

Product quality standards will be determined by the development team, QA Tester, and DOST-STII stakeholders. Standards will be based on:

- DOST-STII functional and non-functional requirements.
- AI/ML accuracy benchmarks (reranking validation, citation accuracy, query rewriting precision).
- Security & compliance: OWASP Top 10, Philippine Data Privacy Act (DPA) 2012, secure token authentication, password hashing (PBKDF2).



- Performance standards: Streaming response latency, concurrent user handling, database query optimization.

The product will be deemed compliant once it passes UAT, achieves $\geq 80\%$ automated test coverage, maintains AI relevance scores within acceptable thresholds, and receives sign-off from DOST-STII stakeholders after a 5-day beta trial period.

Process Quality:

Process quality standards will focus on SDLC adherence, Git branching strategy, peer code reviews, CI/CD automation, and documentation alignment with the DOST-STII SDS template. Unique steps in AI pipeline integration (RAG, vector indexing, LLM prompting) will require specific validation protocols.

Process compliance will be achieved when all pull requests undergo mandatory peer review, CI/CD pipelines pass without critical failures, all API endpoints are documented, and the SDS v1.0 is fully approved by the project sponsor.

Quality Assurance

Quality Assurance focuses on the **processes** used in developing, testing, and deploying the LitPath AI product. An iterative QA process will be used throughout the SDLC. This includes measuring process metrics, analyzing development data, and continuously improving workflows through sprint retrospectives and pipeline audits.

The Project Lead and QA Tester will perform assessments at planned intervals to ensure all development processes are correctly implemented. Key process metrics include code review completion rate, CI/CD build success rate, documentation adherence, and AI pipeline validation accuracy.

Process Action	Acceptable Process Standards	Process Phase	Assessment Interval
Code Review & Merge	100% PRs require ≥ 1 peer approval: zero critical linting/security issues	Development	Per sprint / per merge
CI/CD Pipeline Execution	$\geq 95\%$ automated build/test success rate; rollback protocol documented	Build & deploy	Daily / per commit
AI Pipeline Validation	RAG components (embedding, reranking, generation) pass benchmark tests with $\geq 85\%$ relevance accuracy	AI Integration	Weekly
Documentation Compliance	SDS, API docs, and architecture diagrams updated and approved per DOST-STII template	Planning & Design	Bi-weekly
Security & Data Privacy Audit	Token auth, password reset, and CSM consent flows pass OWASP & DPA checks	Testing & pre-launch	Monthly / Pre-release

The QA Tester will provide day-to-day quality management, conduct process audits weekly, monitor development metrics, and ensure all processes comply with project and organizational standards. If discrepancies are found, the QA Tester will meet with the Project Lead to review findings and initiate corrective actions.



The Project Lead will schedule regularly sprint reviews, management check-ins, and documentation audits. Agenda items will include process compliance, audit findings, and process improvement initiatives.

Process improvement is a core aspect of QA. All findings should result in documented workflow optimizations, test automation updates, or AI prompt refinements. Changes will be communicated to all stakeholders.

Quality Control

Quality Control applies to the **product** (LitPath AI application) and focuses on monitoring, testing, and recording results to assess performance and recommend necessary changes. It verifies that deliverables meet defined standards before release.

QC for LitPath AI will focus on functional accuracy, AI response quality, UI/UX consistency, API performance, and security resilience. All trial builds and staged releases will undergo structured testing by the QA Tester.

Product / Module	Physical/Performance Standards	Quality Assessment Activities	Assessment Intervals
Auth & Session Management	Token generation/expiration works correctly; PBKDF2 hashing verified; zero auth bypass vulnerabilities	Unit testing, security scanning, manual penetration tests	Per sprint / Pre-release
Search & RAG Pipeline	AI answers citation-backed; cosine similarity <1.5; reranking precision ≥80%; streaming latency <5s	AI evaluation benchmarks, load testing, manual query validation	Weekly / Pre-release
Admin Dashboard & Analytics	KPI calculations accurate; charts render correctly; date filters apply to all tables	E2E testing, data reconciliation with DB, cross-browser testing	Bi-weekly
CSM Feedback & Compliance	Form validates required fields; data privacy consent logged; responses stored securely in Supabase & PostgreSQL	Functional testing, DPA compliance review, UAT	Monthly / Post-deployment
API Endpoints (REST)	All endpoints return correct HTTP status codes; JSON schemas match docs; rate limiting applied	Automated API testing (Postman/PyTest), contract testing	Per deployment

The QA Tester will execute all test suites, log defects, and verify fixes. Developers will resolve critical/high-priority bugs within 48 hours. The QA Tester will ensure all physical and performance standards are met, perform audit trails, and maintain QC documentation.

The Project Lead will schedule product review meetings to discuss test results, defect trends, and product improvement initiatives. Meeting these standards ensures LitPath AI achieves high user satisfaction and aligns with DOST-STII operational expectations.



Quality Control Measurements

All LitPath AI products and processes must be measured and meet established standards. The logs below will be used by the development and QA teams to document findings, track resolutions, and serve as acceptance evidence.

Quality Assurance Log

Sprint/Date	Process Measured	Required Value	Actual Measured	Acceptable? (Y/N)	Recommendation	Date Resolved
	Code Review Compliance	100% PRs approved				
	CI/CD Success Rate	≥95%				
	AI Pipeline Relevance	≥85% accuracy				
	Documentation Update	SDS & API docs aligned				

Quality Control Log

Build/Date	Component Tested	Required Standard	Actual Result	Acceptable? (Y/N)	Recommendation	Date Resolved
	Auth Token Flow	24h expiry, 1 session/user				
	RAG Search Query	<2s response, relevant sources				
	Streaming Endpoint	<5s latency, no truncation				
	CSM Form Submission	DPA consent logged, DB sync				
	Admin KPI Dashboard	Accurate counts, filters work				

If actual measurements do not meet standards, corrective actions will be documented, prioritized in the backlog, and resolved before the next release candidate. Discrepancies will be reviewed in sprint retrospectives or ad-hoc quality meetings as necessary.

SPONSOR ACCEPTANCE

Approved by the Project Sponsor:

Khasian Eunice Romulo
Science Research Specialist II / Unit Head
(Digital Information and Processing Delivery Unit)

Date: April 1, 2026



13 Cost Management Plan

Introduction

The Cost Management Plan clearly defines how the costs of this project will be managed throughout the project's lifecycle. It sets the format and standards by which project costs are measured, reported, and controlled.

Specifically, this plan:

- Identifies who is responsible for managing costs
- Identifies who has the authority to approve changes to the project or its budget
- Defines how cost performance is quantitatively measured and reported
- Establishes report formats, frequency, and the audience for each report

The Project Manager is responsible for managing and reporting on the project's cost throughout the duration of the project. During sprint reviews and monthly project status meetings, the Project Manager will present and review the project's cost performance for the preceding period. Performance is measured using Earned Value Management (EVM). The Project Manager is responsible for accounting for cost deviations and presenting the Project Sponsor with options for getting the project back on budget. The Project Sponsor has the authority to make changes to the project to bring it back within budget.

Cost Management Approach

This project uses OpenProject as its Project Management Information System (PMIS). Costs are managed at the Work Package level, which provides the highest level of granularity and visibility into cost performance. This approach is aligned with PMBOK best practices, as the availability of a PMIS enables cost tracking down to individual tasks and features.

Control Accounts are established per sprint budget. Earned Value calculations for each sprint measure and manage the financial performance of the project. Cost estimates are detailed within individual work packages, with overall cost management accuracy maintained at the sprint budget level.

Credit for work is assigned at the work package level. Work started on a work package grants 50% credit; the remaining 50% is credited upon full completion. Costs are rounded to the nearest peso and work hours to the nearest whole hour.

Cost variances of +/- 0.1 in the CPI or SPI change the cost status to cautionary (yellow in project status reports). Variances of +/- 0.2 escalate to alert status (red), requiring corrective action from the Project Manager. All corrective actions require a formal change request approved by the Project Sponsor before implementation.

Project Assumptions & Sprint Configuration

Sprint Assumptions

The following assumptions form the basis for all budget and cost calculations in OpenProject for Term 3 Sprint 1:



Parameter	Value
Sprint Duration	2 weeks
Hours per week per member	10 hours
Hours per member per sprint	20 hours (10 hrs/week × 2 weeks)
Number of team members	4 members
Total planned hours	80 hours (20 hrs × 4 members)
Hourly rate (all members)	PHP 500.00 per hour
Planned labor cost	PHP 40,000.00
Planned unit costs	PHP 50,700.00
Total Planned Value (PV)	PHP 90,700.00

All team members are assigned the same hourly rate of PHP 500.00 in OpenProject under Administration → Labor Costs → Hourly Rate.

Sprint Budget Details

Planned Labor Costs

Each team member is allocated 20 hours over the 2-week sprint at PHP 500.00 per hour, for a total planned labor cost of PHP 40,000.00. The table below reflects the planned labor budget per member as configured in OpenProject:

Team Member	Planned Hours	Rate/Hour	Budget
Tracie Tomon	20 hrs	PHP 500.00	PHP 10,000.00
Jenine Elaine Dulay	20 hrs	PHP 500.00	PHP 10,000.00
Charijoy Cempron	20 hrs	PHP 500.00	PHP 10,000.00
Joshua Sta Ana	20 hrs	PHP 500.00	PHP 10,000.00
TOTAL	80 hrs		PHP 40,000.00

Planned Unit Costs

Unit costs represent non-labor expenditures added to the sprint budget in OpenProject. The Co-working Space (Hourly) cost is set to 80 hours, based on the team's total planned sprint hours for all members. All other unit costs reflect realistic 2-week sprint consumption:

Units	Unit Name	Cost Type	Rate	Budget
80.00	hours	Co-working space (Hourly)	PHP 100.00	PHP 8,000.00
1.00	month	Internet Monthly Subscription Php1K	PHP 1,000.00	PHP 1,000.00



1.00	month	Cloud Services Monthly Subscription	PHP 100.00	PHP 100.00
4.00	package	Contingency Reserve	PHP 10,000.00	PHP 40,000.00
8.00	units	Food Allowance	PHP 100.00	PHP 800.00
8.00	units	Transportation expenses Php 100	PHP 100.00	PHP 800.00
TOTAL PLANNED UNIT COSTS				PHP 50,700.00

Notes on key unit cost decisions:

- Co-working space (80 hrs): Set to match the total planned sprint hours across all 4 team members (20 hrs/member × 4 members = 80 hrs). At PHP 100/hr, this yields PHP 8,000.00 planned.
- Contingency Reserve (4 packages × PHP 10,000): Total of PHP 40,000.00, representing approximately 100% of the labor budget, serving as a buffer for unforeseen costs within the sprint.
- Food Allowance & Transportation (8 units each): Based on 1 team meeting per week × 2 weeks × 4 members = 8 total instances each for meals and transport at PHP 100 each.
- Internet and Cloud subscriptions are billed monthly, so 1 unit covers the full 2-week sprint period.

Actual Unit Costs — Co-working Space (Hourly)

The following table shows actual co-working space costs logged per work package in OpenProject, matched against Spent Time entries for each task and feature:

Work Package	Units	Actual Cost
Task #6327: 6.1.1 Audit and identify redundant columns...	0.69	PHP 68.50
Task #5270: Task 1 Improve LLM prompt handling	4.00	PHP 400.00
Task #5271: Task 2 Improve follow-up retrieval behavior	4.00	PHP 400.00
Task #6328: 6.1.2 Remove identified redundant columns...	3.00	PHP 300.00
Feature #4690: Feature 4.0 Display Philippine Std Time	4.00	PHP 400.00
Feature #4693: Feature 6.0 UI Decluttering (Feedback)	4.00	PHP 400.00
Feature #4696: Feature 8.0 Failed Query Keyword Tracking	16.00	PHP 1,600.00
Task #6336: Task 2 Dropdown Refinement (Frontend)	3.00	PHP 300.00
Task #6335: Task 1 UI/Table Update (Column Merger)	3.00	PHP 300.00
Task #6300: Task 1 Implement backend logic...	4.00	PHP 400.00



Task #6306: Task 1 Implement frontend pagination...	1.00	PHP 100.00
Feature #4689: Feature 3.0 Welcome Page & Form UI...	4.00	PHP 400.00
Feature #4694: Feature 7.0 Data Pagination	2.00	PHP 200.00
Feature #4674: Feature 1.0 User Account Registration	5.00	PHP 500.00
Feature #4679: Feature 2.0 LLM Source Accuracy...	8.00	PHP 800.00
TOTAL ACTUAL CO-WORKING COSTS	65.69	PHP 6,568.50

Measuring Project Costs

Overview

Performance of this project is measured using Earned Value Management (EVM). EVM integrates scope, schedule, and cost baselines to provide an objective measure of project performance. OpenProject captures Spent Time and unit costs, which feed directly into the EVM calculations below.

The following EVM metrics are tracked and reported for this sprint:

- Planned Value (PV) — the authorized budget assigned to all scheduled work for the sprint
- Earned Value (EV) — the budgeted value of work performed, based on percent complete
- Actual Cost (AC) — the realized cost incurred for work performed, extracted from OpenProject Spent Time
- Cost Variance (CV = EV – AC) — positive value means under budget
- Schedule Variance (SV = EV – PV) — positive value means ahead of schedule
- Cost Performance Index (CPI = EV / AC) — greater than 1.0 means under budget
- Schedule Performance Index (SPI = EV / PV) — equal to 1.0 means on schedule

EVM Computation — Term 3 Sprint 1

The computations below are based on actual OpenProject data. Sprint 1 ran from March 23 to April 3, 2026, and has been fully completed, all work packages were delivered. EV equals PV because 100% of planned work was accomplished.

Actual Cost (AC) of PHP 13,323.50 is composed of actual labor costs logged in OpenProject (PHP 6,755.00) plus actual co-working space costs logged per work package (PHP 6,568.50). This figure is verified against the OpenProject budget summary which shows Spent = PHP 13,323.50.

EVM Metric	Interpretation	Value
Planned Value (PV)	Total authorized budget for Sprint 1 (Mar 23 – Apr 3)	PHP 90,700.00



Earned Value (EV)	All work packages completed and delivered — 100%	PHP 90,700.00
Actual Cost (AC)	Labor logged (PHP 6,755.00) + Co-working logged (PHP 6,568.50)	PHP 13,323.50
Cost Variance (CV = EV – AC)	Under budget — spent less than the value earned	PHP 77,376.50
Schedule Variance (SV = EV – PV)	Perfectly on schedule — all planned work delivered	PHP 0.00
CPI (EV / AC)	Cost efficient — PHP 6.81 earned per PHP 1.00 spent	6.8076
SPI (EV / PV)	Perfectly on schedule — sprint completed as planned	1.0000

EVM Interpretation

Cost Variance (CV) = PHP 77,376.50: The sprint finished significantly under budget. The team delivered PHP 90,700.00 worth of work while spending only PHP 13,323.50. The large savings reflect the nature of how actual costs are tracked in OpenProject, as explained in the note below.

Schedule Variance (SV) = PHP 0.00: The sprint was completed exactly on schedule. All planned work packages were delivered by April 3, 2026 — the end of the 2-week sprint window.

CPI = 6.8076: For every PHP 1.00 spent, the team delivered PHP 6.81 of value. While this is an unusually high CPI, it is explained in full in the note below. The SPI = 1.0000 confirms the sprint was completed exactly as scheduled.

NOTE ON HIGH CPI — Why Actual Cost is Lower than Planned

The CPI of 6.8076 is significantly above the normal range. This is not an error, but a direct result of how actual hours were tracked in OpenProject during Sprint 1.

Planned Value (PV) was set based on the full team capacity allocation: 4 members × 20 hours/member × PHP 500.00/hour = PHP 40,000.00 in planned labor, plus PHP 50,700.00 in planned unit costs, for a total PV of PHP 90,700.00. This represents the maximum budget authorized for the sprint.

Actual Cost (AC), however, reflects only the hours that team members explicitly logged against individual work packages in OpenProject (PHP 6,755.00 in actual labor), plus the co-working space costs logged per task (PHP 6,568.50). Not all allocated hours were logged as spent time entries in OpenProject, even though the full work was completed and delivered.

This discrepancy arises because the team is composed of full-time students who completed sprint deliverables within their available time but did not always log every hour of effort in OpenProject at the task level. The planned budget represents a cost ceiling, while the actual cost reflects only what was formally recorded in the system.

In future sprints, the team will aim to log all actual hours against work packages in real time to improve the accuracy of EVM calculations and bring the CPI closer to the expected range.



Cost Performance Thresholds

The following thresholds define when escalation and corrective action are required:

Metric	Yellow (Caution)	Red (Alert)
SPI	0.8 – 0.9 or 1.1 – 1.2	< 0.8 or > 1.2
CPI	0.8 – 0.9 or 1.1 – 1.2	< 0.8 or > 1.2

Yellow range (0.8–0.9 or 1.1–1.2): The Project Manager must report the reason for the variance at the next status meeting. No formal change request is required but monitoring is intensified.

Red range (< 0.8 or > 1.2): The Project Manager must report the reason and provide a detailed corrective action plan to management within five business days. The current CPI of 1.27 is above 1.2 in the positive direction, meaning the sprint finished under budget, a favorable outcome requiring no corrective action.

Reporting Format

Cost performance reporting is included in the Sprint Review Report and the Monthly Project Status Report. Each report contains a dedicated Cost Management section with the following

- All seven EVM metrics: PV, EV, AC, CV, SV, CPI, and SPI with current values
- Status color indicators (Green / Yellow / Red) per the performance thresholds above
- Commentary on any variances outside acceptable thresholds, with planned corrective actions
- A log of any Change Requests triggered by cost overruns or budget adjustments
- Updated Spent Time and unit cost data exported from OpenProject

Reports are generated at the end of each sprint and presented to the Project Sponsor and relevant stakeholders.

Cost Variance Response Process

If the CPI or SPI reaches a Control Threshold (less than 0.8 or greater than 1.2 in the negative direction), a Cost Variance Corrective Action Plan is required:

- Within 5 business days of the variance first being reported: the Project Manager presents corrective action options to the Project Sponsor
- Within 3 business days of option selection: the Project Manager presents a formal Cost Variance Corrective Action Plan
- The plan details all actions needed to bring the project back within acceptable cost and schedule performance, along with how effectiveness will be measured
- Upon acceptance, the plan becomes part of the project baseline, and all affected schedules and budgets are updated in OpenProject



Corrective actions may include increasing the project budget, reducing scope or quality, re-allocating team resources, or other measures approved by the Project Sponsor.

Cost Change Control Process

The cost change control process follows the established project change request process. All changes to the project budget, including adjustments to labor rates, unit costs, or overall budget ceilings, must be submitted via a formal Change Request and approved by the Project Sponsor before implementation.

All approved changes are reflected in updated budget records in OpenProject. No budget change takes effect until the Change Request has been formally reviewed and approved by the Project Sponsor.

Project Budget Summary

The total budget for Term 3 Sprint 1 is summarized below, reflecting the corrected and verified OpenProject configuration:

Budget Item	Amount
PLANNED COSTS	
Planned Labor Costs (80 hrs × PHP 500.00)	PHP 40,000.00
Co-working Space (80 hrs × PHP 100.00)	PHP 8,000.00
Internet Monthly Subscription (1 × PHP 1,000.00)	PHP 1,000.00
Cloud Services Monthly Subscription (1 × PHP 100.00)	PHP 100.00
Contingency Reserve (4 packages × PHP 10,000.00)	PHP 40,000.00
Food Allowance (8 units × PHP 100.00)	PHP 800.00
Transportation Expenses (8 units × PHP 100.00)	PHP 800.00
TOTAL PLANNED VALUE (PV)	PHP 90,700.00
ACTUAL COSTS	
Actual Labor Cost (logged in OpenProject)	PHP 6,755.00
Actual Co-working Space Cost (logged per work package in OpenProject)	PHP 6,568.50
TOTAL ACTUAL COST (AC)	PHP 13,323.50
EVM SUMMARY	
Cost Variance (CV = EV – AC) — Under Budget	PHP 77,376.50
Management Reserve (10% of Labor Budget — held by Sponsor)	PHP 4,000.00



The Management Reserve of PHP 4,000.00 (10% of labor budget) is held by the Project Sponsor and was not drawn upon during Sprint 1, as the team completed all work within budget. The sprint achieved a Cost Variance of +PHP 77,376.50, finishing significantly under budget with a CPI of 6.8076. As noted in Section 5, the high CPI is attributable to incomplete logging of actual hours in OpenProject rather than a true reduction in effort. The team will address this in subsequent sprints by logging all hours against work packages in real time.

SPONSOR ACCEPTANCE

Approved by the Project Sponsor:

Date: April 1, 2026

Khasian Eunice Romulo
Science Research Specialist II / Unit Head
(Digital Information and Processing Delivery Unit)

14 Configuration Management Plan

Introduction

The purpose of this Configuration Management Plan (CMP) is to describe how configuration management (CM) will be conducted throughout the lifecycle of the **LitPath AI** project. This plan documents how configuration management activities are managed, defines roles and responsibilities, establishes how configuration item (CI) changes are identified and controlled, and describes how all configuration-related information is communicated to project stakeholders. Without a formally documented configuration management plan, configuration items may be overlooked, improperly versioned, or inconsistently maintained, resulting in rework, deployment errors, or loss of system integrity. While configuration management is important for all projects, it is especially critical for **software and information technology projects** such as LitPath AI.

The LitPath AI project utilizes a modern, hybrid technology environment that combines **on-premise infrastructure hosted on Proxmox VE** with secure external access provided through **Cloudflare Tunnel**. The system integrates web-based application components, artificial intelligence services, relational and vector databases, and institutional research data owned by DOST-STII. This environment enables remote access, centralized administration, AI-powered search and analytics, and improved monitoring of system usage and performance. As a result, DOST-STII benefits from improved accessibility to academic resources, enhanced research efficiency, and reduced operational effort compared to traditional keyword-based catalog systems.

In order to effectively manage the LitPath AI project, a coordinated and well-defined Configuration Management Plan is required. This CMP establishes the configuration



management framework that the project team will use to identify, track, implement, audit, and communicate configuration items and configuration changes throughout the project lifecycle. By enforcing standardized configuration control, version management, and audit practices, this plan ensures system stability, supports secure and reliable deployments, and enables informed decision-making by the Project Manager and Project Sponsor.

Roles and Responsibilities

Clear roles and responsibilities are essential to ensure effective configuration management. All CM-related activities for LitPath AI are assigned to specific project roles as defined below.

Configuration Control Board (CCB)

The Configuration Control Board (CCB) is responsible for reviewing and approving all significant configuration changes.

CCB Members:

- Project Sponsor (Ms. Khasian Eunice Romulo)
- Project Manager (Ms. Tracie Tomon)
- DevOps Engineers (Charijoy Cempron, Joshua Sta.Ana,)
- Developers (Ms. Tracie Tomon, Ms. Jenine Dulay, Charijoy Cempron, Joshua Sta.Ana)
- QA Tester (Jenine Dulay, Joshua Sta.Ana)

CCB Responsibilities:

- Review and approve or reject Configuration Change Requests (CCRs)
- Evaluate the impact of changes on scope, schedule, cost, quality, and risk
- Ensure approved changes are recorded in the CMDB

Project Sponsor

The Project Sponsor is responsible for:

- Providing final approval for configuration changes that affect project scope, schedule, or cost
- Participating in CCB decision-making as required

Project Manager

The Project Manager is responsible for:

- Overall accountability for configuration management activities
- Identifying configuration items
- Communicating CM activities and decisions to stakeholders
- Re-baselining project artifacts when approved changes are implemented
- Participating in CCB reviews



Configuration Manager (DevOps Engineer)

The Configuration Manager is responsible for:

- Managing and maintaining the Configuration Management Database (CMDB)
- Defining and enforcing version control standards
- Managing infrastructure, deployment, and environment configurations
- Conducting configuration audits prior to releases

Developers (Frontend and Backend)

Developers are responsible for:

- Implementing approved configuration changes
- Ensuring changes comply with version control standards
- Documenting changes and their impacts
- Supporting testing and rollback procedures

QA Tester

The QA Tester is responsible for:

- Verifying that configuration changes meet approved requirements
- Performing functional and regression testing after changes
- Supporting configuration audits through validation activities

Configuration Control

Configuration Control is the process of systematically controlling and managing all configuration activities throughout the lifecycle of the LitPath AI project. To effectively manage configuration items (CIs), it is essential that a structured and standardized process is followed to ensure that only necessary and authorized configuration changes are implemented. Like all change management activities, configuration control decisions must be made with a clear understanding of the potential impact on project scope, schedule, cost, quality, and risk.

The LitPath AI project will utilize a standardized configuration control process for the duration of the project lifecycle to ensure that all configuration items are handled consistently, reviewed thoroughly, and communicated appropriately to all stakeholders. This approach supports system stability, traceability, and accountability, particularly given the project's use of AI components, hybrid infrastructure (Proxmox VE and Cloudflare Tunnel), and institutional data owned by DOST-STII.

As configuration items are identified by the project team, the Configuration Manager (DevOps Engineer) will assign a unique CI identifier and record the CI in the Configuration Management Database (CMDB) with an initial status of *Initiated*. Once recorded, the CI will be assigned to the appropriate responsible role or focus group (Frontend Developer, Backend Developer, QA Tester, or DevOps Engineer) based on the nature of the configuration item.

Authorized team members will be granted access to the CI through the CMDB to perform approved updates or modifications. Each change or edit must be clearly documented in the CMDB



log, including a description of the modification, rationale for the change, date, and responsible individual. This ensures full traceability of all configuration changes throughout the project lifecycle.

It is imperative that all software-related configuration changes undergo appropriate testing to validate that the implemented changes function as intended and do not introduce defects or regressions. The responsible developer or focus group is required to conduct unit, integration, or functional testing as applicable. The Configuration Manager, in coordination with the QA Tester, is responsible for confirming that testing has been completed, results are documented, and changes are properly recorded in the CMDB. The Configuration Manager is also responsible for assigning updated version numbers and revising the CI status to reflect its current state.

Configuration items often maintain dependencies or relationships with other CIs within the LitPath AI system, such as relationships between frontend components, backend services, databases, and infrastructure configurations. The Configuration Manager, Project Manager, and responsible developers will collaborate to identify and document these relationships. All CI dependencies and co-dependencies will be illustrated and maintained within the CMDB to ensure a comprehensive understanding of system interactions and potential downstream impacts of changes.

Any configuration changes identified by project team members or stakeholders must be formally documented using a Configuration Change Request (CCR) and submitted to the Configuration Control Board (CCB) for review. The CCB will analyze each CCR and approve or reject the request based on its assessed impact on scope, schedule, cost, quality, and risk. If a configuration change is approved, project baselines and requirements will be updated as necessary, and the Project Manager will ensure that all approved changes are communicated to relevant stakeholders. CCRs that are denied may be revised and resubmitted with additional justification or supporting information for reconsideration by the CCB.

No configuration change may be deployed to the production environment without formal CCB approval and proper documentation within the CMDB.

Configuration Management Database (CMDB)

The Configuration Management Database (CMDB) serves as the centralized repository for all configuration information related to LitPath AI.

The CMDB contains:

- Configuration item identifiers
- Version numbers and baselines
- Change history and audit trails
- Relationships between configuration items

CMDB Tools

The LitPath AI CMDB is implemented using:

- Git repositories for source code and documentation
- Secure storage for environment variables and secrets
- Deployment and infrastructure configuration records

CMDB Access Control

Access to the CMDB is controlled as follows:



- Read/Write Access: Project Manager, Configuration Manager, Developers
- Read-Only Access: Project Sponsor and DOST-STII stakeholders

The CMDB ensures that all team members are working with the most current approved configuration while maintaining historical records for traceability.

Configuration Status Accounting

Configuration status accounting is the process of collecting, processing, and reporting configuration information for all configuration items (CIs) at any given point in time. This activity includes the management and reporting of configuration data stored within the Configuration Management Database (CMDB), such as approved configuration documents, software components, infrastructure configurations, data assets, and their associated version numbers. Configuration status accounting also provides visibility into build reports, the status of submitted Configuration Change Requests (CCRs), and any discrepancies or issues identified through configuration audits.

For the LitPath AI project, configuration status accounting is critical to ensure transparency, traceability, and informed decision-making across the project lifecycle. It is essential that the Project Sponsor (DOST-STII) and the Project Manager have the ability to review the current configuration status at any time. To support this requirement, the Project Manager will include configuration status information as part of regular project reporting.

The Project Manager will submit weekly project status reports, which will include a dedicated configuration status section. This section will provide a consolidated view of the current state of configuration items and changes. Configuration status reporting will include, but is not limited to, the following categories:

1) Configuration Change Requests (CCRs)

- Aging: The length of time each CCR has remained open, from submission to resolution
- Distribution: The number of CCRs submitted, categorized by role or functional area (Frontend, Backend, DevOps, QA)
- Trending: Identification of areas where configuration changes are most frequently occurring, helping highlight risk or instability within specific components

2) Version Control Status

Configuration version status will be tracked and reported for all major CI categories, including:

- Software: Frontend applications, backend services, APIs, and AI components
- Infrastructure: Proxmox VE configurations, Cloudflare Tunnel settings, and deployment scripts
- Data: Database schemas, migration scripts, and AI embedding datasets
- Documentation: Project plans, technical documentation, and user/admin guides

3) Build and Release Reporting

Build reporting provides visibility into the composition and readiness of system releases and includes:

- Files: Confirmation that all required source code and configuration files are included in the build



- CI Relationships: Verification that dependencies and relationships between configuration items are correctly represented and compatible
- Incorporated Changes: A summary of approved CCRs and enhancements included in each build or release

https://asiapacificcollege.sharepoint.com/:w:/r/sites/HecTech2/_layouts/15/Doc.aspx?sourcedoc=%7BAEF50DB7-5F83-4961-B8D1-3C2E45A612EC%7D&file=04%20Project%20Monitoring%20and%20Controlling%2002%20Project-Status-Report.docx&action=default&mobileredirect=true

4) Audit Status

Configuration status accounting also tracks the results and status of configuration audits, including:

Physical Configuration Status: Verification that deployed components match documented and approved configurations

Functional Configuration Status: Verification that implemented features meet approved requirements and specifications

Prior to any new software release, the Configuration Manager (DevOps Engineer) will work closely with the Project Manager and responsible developers to ensure that all configuration items are updated to their latest approved release versions, all CCRs have been properly incorporated, and the CMDB accurately reflects the current system baseline.

Configuration Audits

Configuration audits are an essential component of both project management and configuration management for the LitPath AI project. The primary purpose of a configuration audit is to ensure that established configuration management processes and procedures are being followed as intended and to provide a structured opportunity to identify, document, and correct any deviations from these processes. While audits are sometimes viewed negatively, when applied appropriately, they serve as an effective quality assurance and management control mechanism that helps maintain system integrity, reliability, and compliance.

Configuration audits will be an ongoing activity throughout the LitPath AI project lifecycle. The objective of each configuration audit is to verify that all project team members are adhering to approved configuration management standards, version control practices, and documentation requirements. Formal configuration audits will be conducted prior to any major software release, as well as at the discretion of the Project Manager or Project Sponsor should they determine that an audit is necessary due to risk, scope changes, or quality concerns.

All configuration audits for the LitPath AI project will be performed by the Configuration Manager (DevOps Engineers). Throughout the project, the Configuration Manager works closely with the Frontend Developers, Backend Developers, QA Testers, and Project Manager to ensure that configuration management procedures are consistently applied. As part of each configuration audit, the Configuration Manager will perform the following activities:

- Establish a dedicated audit environment within the Configuration Management Database (CMDB)



- Copy the latest approved versions of all software components, databases, infrastructure configurations, and project documentation into the audit environment
- Verify that all configuration items are correctly versioned and that version control standards have been properly applied
- Analyze historical versions, commit logs, and timestamps of software, data, and documentation to confirm that all changes were formally approved, recorded, and traceable
- Deploy and test the latest software versions to ensure that approved requirements and functional specifications are being met
- Verify that all required artifacts (source code, configuration files, documentation, test results) are present, complete, and current within the CMDB
- Confirm that all approved Configuration Change Requests (CCRs) have been fully implemented and accurately recorded in the CMDB

Upon completion of each configuration audit, the Configuration Manager will compile a detailed audit findings report. For each identified discrepancy, the Configuration Manager will work collaboratively with the Project Manager and relevant team members to define the required corrective action(s), assign responsibility for each action, and establish a resolution timeline.

Once all findings have been reviewed, the Configuration Manager will document all discrepancies, corrective actions, and resolutions in a formal audit report. This report will be submitted to the Project Manager and Project Sponsor for review and approval. Audit results may be used to improve configuration management processes, reduce future risks, and support informed decision-making for subsequent project phases.

SPONSOR ACCEPTANCE

Approved by the Project Sponsor:

Date: April 30, 2026

Khasian Eunice Romulo
Science Research Specialist II / Unit Head
(Digital Information and Processing Delivery Unit)

15 Project Change Management Plan

Introduction

Change management is a critical component of any project. Changes must be evaluated and managed to ensure they remain within the project's scope, do not adversely affect the project's schedule or quality, and are properly communicated to all stakeholders when approved.



Without a structured process for submitting, reviewing, and approving changes, a project risks scope creep, resource misallocation, and delivery failure.

This Change Management Plan has been created for the LitPath AI: Smart Pathfinder for Theses and Dissertations project, developed by HecTech for the DOST-Science and Technology Information Institute (DOST-STII). It establishes the approach to managing change throughout the project lifecycle, defines what constitutes a change, describes the Change Control Board (CCB) and its composition, and details the end-to-end change control process.

All project stakeholders, including the DOST-STII client team, the HecTech development team, and academic advisers from Asia Pacific College, are expected to submit and manage change requests in accordance with this plan. All change requests and submissions will follow the process detailed herein.

Change Management Approach

The Change Management approach for the LitPath AI project ensures that all proposed changes are properly defined, reviewed, and agreed upon before implementation, and that all relevant stakeholders are informed of approved changes. This approach also ensures that only changes within the scope of this project are approved and carried out.

The Change Management approach consists of three core areas:

- Ensure changes are within scope and beneficial to the project — every proposed change must be evaluated against the project's defined objectives, deliverables, and constraints before it is considered for approval.
- Determine how the change will be implemented — once a change is approved, the project manager, in coordination with the relevant team members, will determine the impact on the sprint schedule, technical architecture, and documentation, and plan the implementation accordingly.
- Manage the change as it is implemented — approved changes will be tracked through OpenProject, communicated to all stakeholders via Microsoft Teams (internal team) and Viber (client communication), and reflected in updated project documentation.

The Change Management approach is distinct from the Change Management Process, which is detailed in Section 6. By consistently following this approach, the HecTech team will prevent unnecessary change from occurring and focus its resources only on changes that are beneficial and within project scope.

Definitions of Change

Several types of changes may be requested and considered throughout the LitPath AI project. Depending on the nature and extent of the proposed change, updates to project documentation and communication to stakeholders will be required to incorporate any approved changes into the project plan. The four types of changes recognized for this project are:

Scheduling Changes

Changes that impact the approved project schedule, including sprint timelines, milestone target dates, or the overall project completion date. Given the project's sprint-based cadence, scheduling changes may require re-sequencing tasks, fast-tracking certain activities, or re-

baselining the schedule in OpenProject. Examples specific to LitPath AI include delays in receiving the thesis dataset from DOST-STII, or extended testing periods resulting from user feedback during beta testing.

Scope Changes

Changes that alter the project's agreed deliverables, features, or boundaries as defined in the Software Requirements Specification (SRS) and Business Case. These may arise from unforeseen requirements identified during development or from client feedback. Scope changes may also have downstream effects on schedule and technical design. Examples include a request to add new citation formats beyond the four initially specified, or expanding the system to index additional document types beyond theses and dissertations.

Technical Changes

Changes that affect the system's architecture, technology stack, or AI components without necessarily changing the project scope or budget. Given LitPath AI's reliance on specific AI services and infrastructure, technical changes require careful evaluation by the System Architect and Security Lead before approval. Examples include replacing the Gemini AI API with a different language model, switching from ChromaDB to an alternative vector database, restructuring the Django backend, or modifying the Cloudflare deployment configuration.

Budget Changes

Changes that affect the project's cost structure. As LitPath AI was developed at zero cost to the HecTech team using free and open-source tools, budget changes during the development phase are unlikely. However, post-handover, DOST-STII may encounter budget decisions related to the Gemini AI API subscription tier or Cloudflare hosting plan. Any such changes that affect the project during the active development phase must be formally reviewed through this process.

The project manager must ensure that any approved changes are communicated to all project stakeholders. Additionally, as changes are approved, the project manager must ensure they are captured in the relevant project documentation, including the SRS, System Design Document, sprint board in OpenProject, and this Change Management Plan, and that updates are distributed to the team and stakeholders promptly.

Change Control Board

The Change Control Board (CCB) is the approval authority for all proposed change requests pertaining to the LitPath AI project. The purpose of the CCB is to review all submitted change requests, assess their potential impact on the project's risk, scope, schedule, and technical integrity, and to formally approve or deny each request. The CCB composition for the LitPath AI project is as follows:

Name	Organization	Position	CCB Role
Khasian Eunice Romulo	DOST-STII	Science Research Specialist II (Primary POC)	CCB Chair / Project Sponsor
Tracie Tomon	HecTech / APC	Project Manager	CCB Member



Joshua Sta Ana	HecTech / APC	Security Lead	CCB Co-Chair / Technical Lead
Charijoy Cempron	HecTech / APC	System Architect	CCB Member

As change requests are submitted to the Project Manager by the project team or client stakeholders, the Project Manager will log the requests in the change request register and the CCB will convene during the sprint review week, the third week following each two-week sprint, to review all pending change requests. This cadence aligns change review with the team's natural sprint reflection checkpoint, avoiding unnecessary overhead.

For a change request to be approved, the CCB Chair (Project Sponsor) must provide formal approval. If more information is needed for a particular request, the request will be deferred and returned to the requestor for clarification. If a change is deemed critical and time-sensitive, an ad hoc CCB review may be called via Microsoft Teams or Viber prior to the next scheduled sprint review meeting.

Roles and Responsibilities

The following defines the roles and responsibilities of all parties involved in the change management process for the LitPath AI project. It is essential that all stakeholders understand their respective responsibilities to ensure the change management process runs smoothly and consistently.

Project Sponsor (CCB Chair) - Khasian Eunice Romulo, DOST-STII

- Serve as the final approval authority for all change requests.
- Chair all scheduled and ad hoc CCB meetings.
- Approve or deny changes that affect project scope, schedule, or technical direction.
- Communicate client-side feedback and change needs to the Project Manager via Viber.

Project Manager (CCB Member) - Tracie Tomon, HecTech

- Receive and formally log all change requests from team members and client stakeholders in the change request register.
- Conduct a preliminary impact analysis on each change request prior to CCB review, covering effects on scope, schedule, technical architecture, and project risk.
- Seek clarification from change requestors on any incomplete or unclear submissions.
- Facilitate CCB meetings and coordinate scheduling of ad hoc reviews when needed.
- Update and re-baseline project documentation for all approved changes, including the sprint board in OpenProject.
- Communicate approved changes to all project stakeholders via Microsoft Teams (internal) and Viber (client).

Technical Lead / CCB Co-Chair - Joshua Sta Ana, HecTech

- Evaluate the technical feasibility and impact of all proposed changes, particularly those classified as technical changes affecting the AI components, backend architecture, or deployment configuration.
- Provide technical recommendations to the CCB prior to the approval decision.
- Coordinate with the System Architect (Charijoy Cempron) to assess architectural implications of proposed changes.



- Oversee the technical implementation of approved changes.

System Architect (CCB Member) - Charijoy Cempron, HecTech

- Assess the impact of proposed changes on the system architecture, RAG pipeline, database design, and AI component integrations.
- Provide architectural analysis and recommendations for the CCB.
- Ensure that approved technical changes are implemented in a manner consistent with the overall system design.

Project Team and Stakeholders

- Submit all change requests using the standard LitPath AI Change Request Form (see Appendix A).
- Provide complete and accurate information on change request forms, including justification, impact, and priority assessment.
- Be available to address questions or provide additional information regarding submitted change requests during CCB review.
- Internal team members (HecTech) submit requests via Microsoft Teams. Client stakeholders (DOST-STII) may submit requests via Viber or directly to the Project Manager.

Change Control Process

The Change Control Process for the LitPath AI project follows a consistent, repeatable workflow for all submitted change requests. The Project Manager bears overall responsibility for executing this process for each request. The process consists of the following steps:

1. **Identify and Submit the Change Request (Requestor)** - Any project stakeholder who identifies the need for a change must complete the LitPath AI Change Request Form (Appendix A) and submit it to the Project Manager. Internal team members submit via Microsoft Teams; DOST-STII stakeholders submit via Viber or direct message to the Project Manager. The form must include a clear description of the change, its type, justification, and an initial assessment of its impact.
2. **Log the Change Request (Project Manager)** - Upon receipt, the Project Manager assigns a unique Change Request ID and logs the request in the Change Request Register, recording the submission date, requestor, change type, and current status. This register is maintained throughout the project lifecycle and accessible to all CCB members.
3. **Evaluate the Change (Project Manager, Technical Lead, System Architect)** - The Project Manager conducts a preliminary analysis of the change's impact on scope, schedule, technical architecture, and project risk. For technical changes, the Technical Lead and System Architect are consulted to assess feasibility and architectural implications. If the submission is incomplete, the Project Manager returns it to the requestor for clarification before proceeding.
4. **Submit to CCB for Review (Project Manager)** - The Project Manager presents the change request, along with the preliminary impact analysis, to the CCB at the next scheduled sprint review week meeting. For critical changes, an ad hoc CCB session may be called via Microsoft Teams or Viber.



5. CCB Decision (CCB Chair and Members) - The CCB reviews all submitted information and discusses the proposed change. The CCB Chair (Project Sponsor) renders the final decision: Approved, Denied, or Deferred for more information. The decision and rationale are recorded in the Change Request Register.
6. Implement the Approved Change (Project Manager and Team) - If approved, the Project Manager updates all relevant project documentation, including the sprint board in OpenProject, SRS, System Design Document, and any other affected plans, and re-baselines the schedule if necessary. The implementation is assigned to the appropriate team member(s) and tracked as a task in OpenProject.
7. Communicate the Change (Project Manager) - The Project Manager notifies all project stakeholders of the approved change and its implementation status. Internal team communication is conducted via Microsoft Teams; client communication is conducted via Viber. If a change request is denied, the requestor is also notified with the CCB's rationale.

The following table summarizes the change control process briefly:

Step	Activity	Responsible Party	Channel / Tool
1	Identify and submit change request	Requestor (Team or Client)	Teams (internal) / Viber (client)
2	Log request in Change Request Register	Project Manager	OpenProject
3	Evaluate impact on scope, schedule, technical, risk	PM, Technical Lead, System Architect	Microsoft Teams
4	Submit to CCB with impact analysis	Project Manager	Sprint Review Meeting / Ad hoc via Teams or Viber
5	Review and decide (Approve / Deny / Defer)	CCB Chair and Members	Sprint Review Meeting
6	Implement approved change and update documentation	Project Manager and Team	OpenProject
7	Communicate decisions to all stakeholders	Project Manager	Teams (internal) / Viber (client)



Appendix A – Change Request Form

Project Name	LitPath AI	Change Request ID:	(Assigned by Project Manager)
Project Manager		Change Type: [Scope, Schedule, Technical, DB/AI, Other]	
Requestor		Date Requested	
Date Required			

1. Change Request Details

1.1 Change Title
1.2 Detailed Description (what is changing, added, or removed)
1.3 Justification / Reason for Change
1.4 Initial Assessment of Impact

2. Evaluation & Impact Analysis

Impact Area	Analysis & Notes	Reviewer/Approver Initials
Scope Impact		
Schedule Impact		
Technical Architecture / Feasibility		
DOST-STII DB Access & Gemini AI Integration		
Project Risk		

3. CCB Review & Decision

Approved ☐ Denied ☐ Deferred ☐
CCB Decision Rationale:
CCB Chair / Project Sponsor Name: _____ Signature: _____ Date: _____

4. Implementation Tracking (PM Only)

OpenProject Task ID: []
Document Updated (e.g., SRS, Schedule, Design): []
Stakeholder Communication Status: []

SPONSOR ACCEPTANCE

Approved by the Project Sponsor:

Khasian Eunice M. Romulo
Science Research Specialist II / Unit Head
(Digital Information and Processing Delivery Unit)

Date: April 30, 2026



16 Human Resource Plan

Introduction

Human resources management is an essential component of the LitPath AI project. The human resources management plan serves as a guide for managing the team's human resource activities from project initiation through closure. The human resources management plan includes:

- Roles and responsibilities of team members throughout the project
- Project organization charts
- Staffing management plan to include:
 - a. How resources will be acquired
 - b. Timeline for resources/skill sets
 - c. Training required to develop skills
 - d. How performance reviews will be conducted
 - e. Recognition and rewards system

The purpose of the human resources management plan is to achieve project success by ensuring that the right team members are assigned to the appropriate roles, any skill gaps are identified and addressed through training, team collaboration strategies are clearly established, and all project activities are effectively coordinated and managed.

Roles and Responsibilities

The roles and responsibilities for the LitPath AI project are critical to its success. All team members must have a clear understanding of their individual roles to effectively contribute to the project. For the LitPath AI project, the following six roles and responsibilities have been established:

Project Manager (PM), (1 position): responsible for the overall success of the LitPath AI project. The PM must authorize and approve all project expenditures and ensure that all work activities meet established quality criteria and fall within acceptable variances. The PM is responsible for reporting project status in accordance with the communications management plan and for evaluating the performance of all project team members. The PM must possess the following skills: leadership, project planning, budgeting, scheduling, and effective communication.

Frontend Developer (FD), (1 position): responsible for designing and implementing the client-side interface of the LitPath AI platform. The FD oversees building responsive and accessible UI components, ensuring cross-browser compatibility, and collaborating with the backend developer on API integration points. The FD will provide timely status updates to the PM as required by the communications management plan. The FD may not authorize any project expenditure without PM approval. The FD must be proficient in HTML, CSS, JavaScript, and a modern frontend framework such as React or Vue.

Backend Developer (BD), (1 position): responsible for developing and maintaining the server-side logic, RESTful APIs, and database systems that power LitPath AI. The BD ensures data security, system reliability, and efficient communication between the application and its data sources. The BD coordinates with the Frontend Developer on API contracts and reports status to the PM per the communications management plan. The BD may not authorize project expenditures without PM approval. The BD must be proficient in a server-side language (e.g.,



Python, Node.js), database management (SQL/NoSQL), and API development best practices. (add ai)

DevOps Engineer (DO), (1 position): responsible for managing the infrastructure, deployment pipelines, and operational reliability of the LitPath AI platform. The DO provisions and maintains cloud environments, automates build and deployment processes through CI/CD pipelines, and monitors system performance and uptime. The DO works closely with all developers to ensure consistent and stable environments across development, staging, and production. The DO will report infrastructure status to the PM via the communications management plan. The DO may not authorize project expenditures without PM approval. The DO must be proficient in cloud platforms (e.g., AWS, GCP, or Azure), containerization tools (Docker/Kubernetes), and CI/CD systems (e.g., GitHub Actions, Jenkins).

QA Tester (QA), (1 position): responsible for planning and executing all quality assurance activities for the LitPath AI project. The QA Tester designs and performs test cases covering functional, regression, and user acceptance testing across all features and components of the system. The QA Tester documents identified defects, tracks their resolution, and verifies fixes in coordination with the responsible developers. The QA Tester will report on testing progress and outcomes to the PM in accordance with the communications management plan. The QA Tester may not authorize project expenditure without PM approval. The QA Tester must be proficient in test planning, manual and automated testing tools, and defect tracking systems.

Project Organizational Charts

The following RACI chart shows the relationship between project tasks and team members. Any proposed changes to project responsibilities must be reviewed and approved by the project manager. Changes will be proposed in accordance with the project's change control process. As changes are made, all project documents will be updated and redistributed accordingly.

	Project Manager	Frontend Developer	Backend Developer	DevOps Engineer	QA Tester
Requirements Gathering	A	R	R	C	C
System Architecture Design	A	C	C	C	I
UI/UX Design	A	R	C	I	C
Frontend Development	A	R	I	I	R
Backend Development	A	I	R	I	R
Database Design & Management	A	I	R	I	C
Infrastructure & Deployment	A	I	C	R	C
CI/CD Pipeline Setup	A	I	I	R	I
System Monitoring	A	I	I	R	C
QA & Software Testing	A	C	C	C	R
User Acceptance Testing	A	C	C	I	R
Documentation	A	R	R	R	R
Project Monitoring & Reporting	A/R	I	I	I	I

Key:

R – Responsible for completing the work

A – Accountable for ensuring task completion/sign off

C – Consulted before any decisions are made



I – Informed of when an action/decision has been made

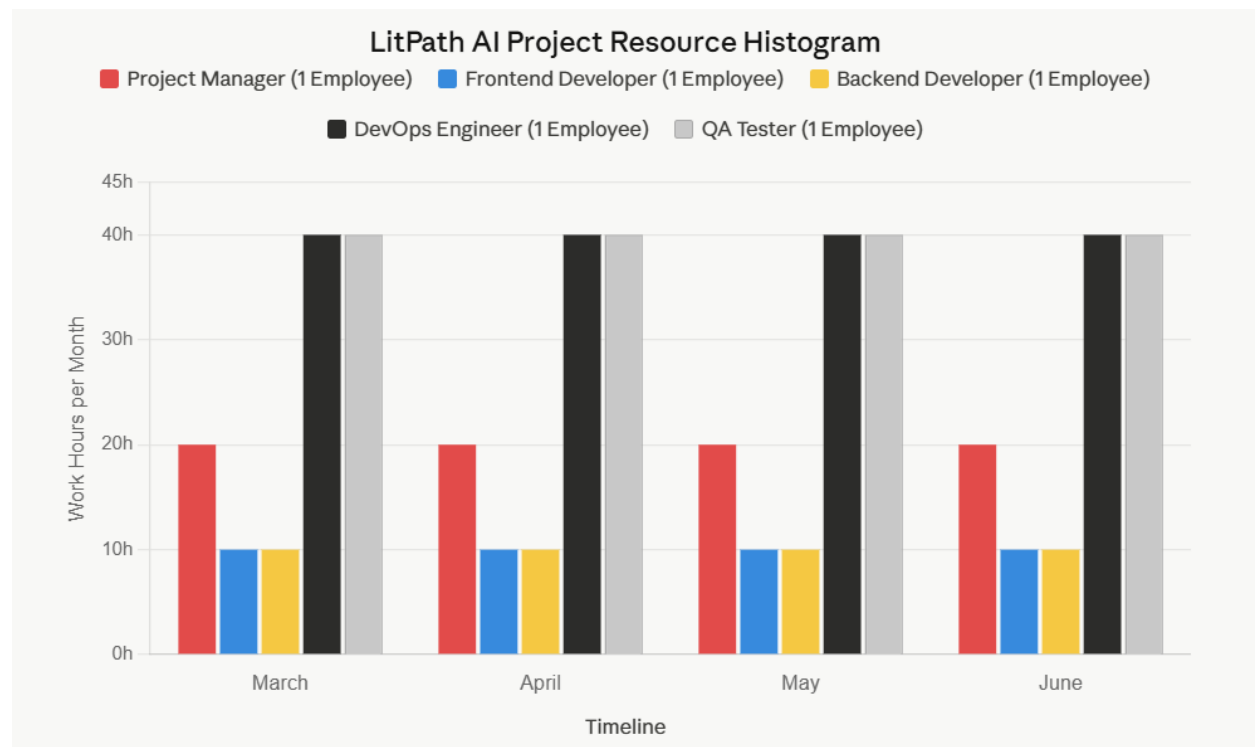
Staffing Management

Staff Acquisition:

For the LitPath AI project, the project staff will consist entirely of internal HecTech resources. There will be no outsourcing or contracting performed within the scope of this project. The Project Manager will coordinate with stakeholders to identify and assign the appropriate team members in accordance with the project organizational structure. All resources must be confirmed and approved before any project work may begin. The project team may operate in a hybrid or remote arrangement depending on availability and project needs.

Resource Calendars:

The LitPath AI project is a continuing development effort spanning four months. Since major development has already been completed, all resources remain on standby throughout the project duration. The resource table below illustrates the estimated monthly hours for each team member. As the project is in its continuation phase, DevOps and QA Tester carry the heaviest workload at full capacity to maintain system reliability and ensure quality, while the Frontend and Backend Developers are available at minimal hours for minor fixes and adjustments only.



Training:

There is currently no formal training scheduled for the LitPath AI project as all team members possess the required skill sets for their respective roles. However, if training requirements are identified during the project such as adoption of new tooling, emerging technologies, or compliance requirements, the funding will be provided from the project reserve budget. The Project Manager is responsible for identifying and escalating training needs in a timely manner.



Performance Reviews:

The project manager will review each team member's assigned work activities at the onset of the project and communicate all expectations of work to be performed. The project manager will then evaluate each team member throughout the project to assess how effectively they are completing their assigned tasks. Prior to closing the project, the project manager will provide a formal performance summary for each team member. The following criteria will be used in performance evaluations:

- Timely completion of assigned work packages and deliverables
- Quality of outputs and adherence to project standards
- Collaboration and communication with team members
- Responsiveness to feedback and issue resolution

Recognition and Rewards:

Although the scope of this project does not allow for ample time to provide cross-training or potential for monetary rewards, there are several planned recognition and reward items for project team members.

- Upon successful completion of the LitPath AI project, a celebration will be held to recognize the contributions of each team member.
- Upon successful completion of the project, any team member who satisfactorily completed all assigned work packages on time will receive a certificate of thanks from the Project Manager.
- Team members who successfully complete all their assigned tasks will have their contribution highlighted in the HecTech SharePoint.

SPONSOR ACCEPTANCE

Approved by the Project Sponsor:

Khasian Eunice Romulo
Science Research Specialist II / Unit Head
(Digital Information and Processing Delivery Unit)

Date: April 1, 2026



03 Execution Phase

Sprint 1 Ceremonies

Sprint Planning

To learn more about Scrum:

- Download the [Scrum Reference Card](#).
- Download an [Example ScrumMaster's Checklist](#).
- Find a [Scrum class](#) in your city.

Agenda:

- Watch video about Sprint Planning
- Assign product owner, scrum master and dev team
- Update OpenProject product backlog
- Create a new board AY 2025-2026 T3 Sprint 01
 - Create the following lists:
 - Committed Backlog Items
 - Tasks Assigned (Not Started)
 - In Progress
 - For Review/Testing
 - Reviewed/Testing
 - Done
 - Individual sprint backlog item(s) assigned to you
 - test cases
- Assign product owner, scrum master and dev team
 - Product owner - Kuya Josh
 - Scrum master - Tracie
 - dev team - All
- Facilitator - **tracie**
- secretary -

Participants:

- Tracie Tomon
- Joshua Sta Ana
- Jenine Elaine Dulay
- Charjony Cempron



Daily Standups

Sprint 1 Week 1 Day 01 Daily Scrum Meeting Individual Updated Report (March 25, 2026)

Added by Tracie Tomon 3 months ago

Answer the following questions:

1. What did you accomplish yesterday?
2. What can I commit today?
3. Anything blocking my progress?

JD RE: Sprint 1 Week 1 Day 01 Daily Scrum Meeting Individual Updated Report (March 25, 2026) - Added by Jenine Elaine Dulay 3 months ago

1. What did you accomplish yesterday?
 - Updated the background blur effect on the welcome page.
2. What can I commit today?
 - Start on the "UI decluttering" work package.
3. Anything blocking my progress?
 - Limited availability due to house chores and other subject-related assignments.

TT RE: Sprint 1 Week 1 Day 01 Daily Scrum Meeting Individual Updated Report (March 25, 2026) - Added by Tracie Tomon 3 months ago

1. What did you accomplish yesterday?
 - Set up Forum for the Daily Scrum reports. Reviewed some of the FMBOK 40 processes.
2. What can I commit today?
 - Prepare the agenda for tomorrow's consultation meeting with the advisers. Set up the schedules for the two meetings.
3. Anything blocking my progress?
 - Work schedule

CC RE: Sprint 1 Week 1 Day 01 Daily Scrum Meeting Individual Updated Report (March 25, 2026) - Added by Charjoy Cempron 3 months ago

1. What did you accomplish yesterday?
 - Created minutes of the meeting.
2. What can I commit today?
 - Create minutes of the meeting and start the registration feature.
3. Anything blocking my progress?
 - Other subjects.

JA RE: Sprint 1 Week 1 Day 01 Daily Scrum Meeting Individual Updated Report (March 25, 2026) - Added by Joshua Sta Ana 3 months ago

1. What did you accomplish yesterday?
 - Planned and started building an on-prem server for github cloning and environment setup
2. What can I commit today?
 - Continuing review of project documentations, recordings and work packages to update
 - Research for more deployment tools/platforms, DevOps solutions.
3. Anything Blocking my progress?
 - Personal and work schedules/ agendas.

Sprint 1 Week 2 Day 03 Daily Scrum Meeting Individual Updated Report (April 1, 2026)

Added by Tracie Tomon 2 months ago

Answer the following questions:

1. What did you accomplish yesterday?
2. What can I commit today?
3. Anything blocking my progress?

TT RE: Sprint 1 Week 2 Day 03 Daily Scrum Meeting Individual Updated Report (April 1, 2026) - Added by Tracie Tomon 2 months ago

1. What did you accomplish yesterday?
 - Started coding the Failed Query content for the admin dashboard and reviewed the 02 Planning documents to finalize the designations.
2. What can I commit today?
 - Continue working on system development and documentation, as well as updating the work packages.
3. Anything blocking my progress?
 - Holy Week schedule and work schedule.

CC RE: Sprint 1 Week 2 Day 03 Daily Scrum Meeting Individual Updated Report (April 1, 2026) - Added by Charjoy Cempron 2 months ago

1. What did you accomplish yesterday?
 - I found the stakeholder analysis file from MSVADO and updated OpenProject details except for budget and cost.
2. What can I commit today?
 - I will do the planning phase documents assigned to me and to help in the initiation phase document group work.
3. Anything blocking my progress?
 - Other subjects.

JD RE: Sprint 1 Week 2 Day 03 Daily Scrum Meeting Individual Updated Report (April 1, 2026) - Added by Jenine Elaine Dulay 2 months ago

1. What did you accomplish yesterday?
 - Yesterday, I started the Business Case Document.
2. What can I commit today?
 - I will continue to work on the Business Case Document.
3. Anything blocking my progress?
 - Holy week plans and other subject-related assignments.

JA RE: Sprint 1 Week 2 Day 03 Daily Scrum Meeting Individual Updated Report (April 1, 2026) - Added by Joshua Sta Ana 2 months ago

- What did you accomplish yesterday?
 - OPEN PROJECT - *ross matrix* features to log user stories and task budget costing - Created a separate draft or completion and posting to Open project
 - PRBOK requirements - started the 01 Initiation documentation specific to Project brief template and statement of work draft for checking
 - DEPLOYMENT - *rhomas* server already installed, configured for remote access.
- What can I commit today?
 - Will start the project feature development in VSCODE
- 01 Initiation documentation requirements continuation, assigned and group work
 - Project brief template
 - Statement of work
- Will start Ubuntu server instance creation to proxmox and configuration
 - Github cloning
 - CI/CD workflows to Github and Vscode
- Anything blocking my progress?
 - Holy week plans, limited internet bandwidth

Backlog Refinement

Name	Story Points
Chari	
Josh	20
Jenine	20
User Story Points	20

The right side of the screenshot shows a video call interface with participants: Jenine Elaine Dulay, Charjoy Cempron, Joshua Sta Ana, and Tracie Tomon. The top of the right panel displays the session title: "Term 3 Sprint 1 Product Backlog Refinement (March 28, 2026, 9:59 pm)". Below the title, it lists "Added backlogs" and "User story points". A detailed list of features is visible, including:

- Feature 1.0 User Account Registration System -
- Feature 2.0 LLM Source Accuracy Optimization / Enhance Prompt -
- Feature 3.0 Welcome Page & Form UI Modernization -
- Feature 4.0 Display Philippine Standard Time (PST) -
- Feature 5.0 Clickable Model Interface (Feedback Manager) -
- Feature 6.0 UI Decluttering (Feedback Table View) -
- Feature 7.0 Data Pagination (Feedback Manager) -
- Feature 8.0 Targeted Dormant Material Export -
- Feature 9.0 Failed Query Keyword Tracking -
- Feature 10.0 "Least Viewed" Metro tracking -
- Feature 11.0 Column Merger & Analysis Dropdown Refinement -
- Feature 12.0 Explicit Date Formatting & "Feedback Type" Column Renaming -



Sprint Review

The screenshot shows a web browser window with the URL `localhost:5173/account-profile`. The page displays an 'Account Profile' form with the following fields:

- Full Name: Tracie Tomon
- Username: tracie
- Email: tracetomon@gmail.com
- School Level: Postgraduate
- School Name: Asia Pacific College
- Client Type: Citizen
- Sex: Female
- Age Range: 21-25
- Region: NCR
- User Category: Student

Below the form is a 'Security' section with fields for 'Current Password', 'New Password', and 'Confirm Password'. A 'Back to Search' link is located at the top right of the form.

On the right side of the browser window, there is a vertical stack of video call participants:

- Tracie Tomon
- Jenine Elaine Dulay
- Charjoy Cempron
- Joshua Sta Ana

The Windows taskbar at the bottom shows the system clock as 2:03 pm on 13/04/2020.

Sprint Retrospective

The screenshot shows a Trello board titled 'Sprint Retrospective Term 3 Sprint 1'. The board is organized into four columns:

- START (Things the team should begin doing)**
 - ☐ Immediately update the work package status assigned
 - ☐ Start familiarizing OpenProject for tracking tasks
 - ☐ Start asking for help early when tasks become difficult to prevent delays or bigger issues
 - ☐ Start double-checking outputs before submission to avoid errors
 - ☐ Start prioritizing tasks based on urgency and importance
- STOP (Things the team should stop doing)**
 - ☐ Stop cramming tasks near the deadline
 - ☐ Closing all work packages at once
 - ☐ Assigning a document as group work without defined parts
 - ☐ Stop overcommitting to tasks beyond individual capacity
 - ☐ Copy pasting screenshot of chat responses into ppt
- CONTINUE (Things the team should keep doing)**
 - ☐ Continue providing updates in our teams GC
 - ☐ Continue working on assigned tasks proactively without needing reminders; initiative is highly appreciated
 - ☐ Continue contributing to both development and enhancements
 - ☐ Continue supporting team members when they're unable to fulfill their responsibilities
- ACTION ITEMS**
 - ☐ Add a card

On the right side of the Trello board, there is a list of 'Sprint Retrospective' steps:

1. Set the Stage
2. Gather Data
 - Review sprint events (timeline, completed tasks, blockers)
 - Ask questions like:
 - What went well?
 - What didn't go well?
 - What issues did we face?
 - Use formats: Start / Stop / Continue
3. Generate Insights
 - Group similar issues together
 - Discuss why problems happened (root cause analysis)
 - Identify recurring challenges or bottlenecks
4. Decide What to Do
 - Prioritize the most important issues
 - Define specific action items
 - Assign an owner for each action
 - Ensure actions are realistic and measurable
5. Close the Retrospective
 - Summarize key decisions and action items
 - Ask for quick feedback on the retrospective itself

Below the list, there are two bullet points:

- Start of New Sprint
- Code restructure

On the right side of the Trello board, there is a vertical stack of video call participants:

- Jenine Elaine Dulay
- Tracie Tomon
- Joshua Sta Ana
- Charjoy Cempron

The Windows taskbar at the bottom shows the system clock as 10:14 PM on 4/17/2020.



Sprint 2 Ceremonies

Sprint Planning

The screenshot shows a Trello board titled "BACKLOG" with a list of tasks including "Cloud/Client-side Deployment", "Targeted Dormant Material Export", "Least Viewed Metric tracking", "Text-to-Speech Integration", "Query Duration Logging & 'Upload Date' Tracking", "Feedback Filtering Realignment & Advanced Report Filtering", "CSV Export Enhancement and PDF Report Export", "Deployment diagram", and "Updated ERD". To the right, a video call window displays a "Term 3 Sprint 2 Sprint Planning" meeting with participants Charijoy Compron, Tracie Tomon, Joshua Sta Ana, and Jenine Elaine Dulay. The meeting agenda includes tasks like "Start recording the meeting", "Assign your product owner, scrum master and dev team", "Update your OpenProject Product backlog Sample [Link]", "Create a new board AY 2025-2026 T3 Sprint 02 Sample [Link]", "Create the following lists", "Commit Backlog Items", "Tasks Assigned (Not Started)", "In Progress", "For Review/Testing", "Reviewed/Testing", and "Done".

Daily Standups

Sprint 2 Week 1 Day 01 Daily Scrum Meeting Individual Updated Report (April 24, 2026)

Added by Tracie Tomon about 2 months ago

Answer the following questions:

1. What did you accomplish yesterday?
2. What can I commit today?
3. Anything blocking my progress?

The screenshot shows a daily standup report for April 24, 2026, by Tracie Tomon. It includes a list of questions and answers, such as "What did you accomplish yesterday?" (Started implementing the following tasks: setting the welcome page background blur to 10%, editing the NRI region, and updating the Client Type logic) and "What can I commit today?" (Work on aligning the Sources Information card with the AI Overview section).

Sprint 2 Week 2 Day 03 Daily Scrum Meeting Individual Updated Report (May 5, 2026)

Added by Tracie Tomon about 1 month ago

Answer the following questions:

1. What did you accomplish yesterday?
2. What can I commit today?
3. Anything blocking my progress?

The screenshot shows a daily standup report for May 5, 2026, by Tracie Tomon. It includes a list of questions and answers, such as "What did you accomplish yesterday?" (Updated the PMBOX assigned document specific to configuration management plan "handling of weekly status report" for review by the project manager) and "What can I commit today?" (To review Sprint 2 Week 2 Day 03 Daily Scrum Meeting Individual Updated Report (May 5, 2026) recordings).



Backlog Refinement

Product Backlog Refinement
(April 27, 2026, 9:15 pm)

User Story Points:

1. As an end user, I want the web app to be hosted on the cloud so that I can access it anytime from any browser without installing anything. - 26
2. As an admin, I want to access the admin dashboard via browser so that I can manage the system remotely without needing a local setup. - 26
3. As a Librarian Administrator, I want to export a targeted list of dormant materials (not accessed for 1 month) from the LitPath AI dashboard, so that I can review and manage underperforming thesis and dissertation content efficiently.
4. As a Librarian Administrator, I want to view a Least Viewed Materials metric in the Material Rating tab replacing the Top Least Accessed metric, so that I can better track and identify underperforming thesis and dissertation content in LitPath AI.
5. As an admin, I want the system to automatically log how long each user stays in a session so that I can monitor user engagement and activity.
6. As an admin, I want the upload date of each material to be automatically recorded so that I can track when it was added to the system.

Sprint Review

Term 3 Sprint 2 Sprint Review
(May 11, 2026, 1:30 pm)

Agenda:

1. Progress on the application
2. Documents to be signed
3. About the deployment and AI service
4. Questions about Admin hierarchy
5. User Acceptance Testing
6. Standalone website - Change Request Form

To comply:

1. Feature 14.0 Query duration logging & upload date tracking
2. Dedicated Detail Page per Material Rating Metric with Expanded Content on Click
3. Dedicated page for dormant materials KPI with export button



Sprint Retrospective

Agenda:

1. Set the Stage
2. Gather Data
 - Review sprint events (timeline, completed tasks, blockers)
 - Ask questions like:
 1. What went well?
 2. What didn't go well?
 3. What issues did we face?
 - Use format: Start / Stop / Continue
3. Generate Insights
 - Group similar issues together
 - Discuss why problems happened (root cause analysis)
 - Identify recurring challenges or bottlenecks
4. Decide What to Do
 - Prioritize the most important issues
 - Define specific action items
 - Assign an owner for each action
 - Ensure actions are realistic and measurable
5. Close the Retrospective
 - Summarize key decisions and actions items
 - Ask for quick feedback on the retrospective itself

Specific Action Item:

1. Stop Procrastinating – ALL, whole duration of Sprint 3
2. Continue updating the OpenProject work packages while developing – ALL, whole duration of Sprint 3
3. Dedicate more time on feature implementation and planning design prior to coding – ALL, Sprint Planning (May 8, 9:30 PM)

Other Matter:

Sprint 3 Ceremonies

Sprint Planning

**Term 3 Sprint 3
Sprint Planning**
(May 8, 2026, Friday, 9:52 PM)

Agenda:

- Begin your Sprint Planning meeting in your assigned group channel
- Start recording the meeting
- Assign your product owner, scrum master and dev team
 - Product Owner - char
 - Scrum Master - kaja josh
 - Dev Team - all
- Update your OpenProject Product backlog Sample [link]
- Create a new board AY 2025-2026 T3 Sprint 03 Sample [link]
- Create the following lists:
 1. Committed Backlog Items
 2. Tasks Assigned (Not Started)
 3. In Progress
 4. For Review/Testing
 5. Reviewed/Testing
 6. Done
- Post a link of your recorded Sprint Planning meeting
- Post a screenshot of your recorded Sprint Planning meeting with all participants cameras open
- Post a link and screenshot of your OpenProject Backlogs page
- Post a link and screenshot of your OpenProject AY 2025-2026 T3 Sprint 03 Board
- Post a link and screenshot of the individual sprint backlog item(s) assigned to you.

Other Matters:

1. Sprint Schedule (2-Week Timeline)
2. User Story and Task per backlog
3. Daily Stand up schedule (at 9 or 9:30pm)
 - Sprint 3 Week 1 Day 1 – May 11, Monday 9:30 PM
 - Sprint 3 Week 1 Day 2 – May 12, Tuesday 9:30 PM
4. Backlog Refinement and Story points - tentative, Tuesday, May 12
5. Label the 3 specific actions with the new versions
6. Set up Forum for new sprint

PM Docs

04 Monitoring And Controlling

1. 04 Project Monitoring and Controlling 02 Project Status Report – josh

05 Closing



Daily Standup

Sprint 3 Week 1 Day 01 Daily Scrum Meeting Individual Updated Report (May 11, 2026)

Added by Tracie Tomon about 1 month ago

Answer the following questions:

1. What did you accomplish yesterday?
2. What can I commit today?
3. Anything blocking my progress?

ID RE: Sprint 3 Week 1 Day 01 Daily Scrum Meeting Individual Updated Report (May 11, 2026) - Added by Jenine Elaine Delay about 1 month ago

1. What did you accomplish yesterday?
 - I started working on the OS Closing Document and the RTM, which will still require updates if there are further changes to the test cases.
2. What can I commit today?
 - I will continue progressing on the two PMDOC documents currently in progress.
3. Anything blocking my progress?
 - Academic workload from other subjects.

CS RE: Sprint 3 Week 1 Day 01 Daily Scrum Meeting Individual Updated Report (May 11, 2026) - Added by Charjoy Cempron about 1 month ago

1. What did you accomplish yesterday?
 - I finished revising the risk management plan document and creating the transition out plan document.
2. What can I commit today?
 - I will test the test cases and finish the 3 minutes for 3 meetings.
3. Anything blocking my progress?
 - I don't have my own laptop or desktop at home and other subjects.

TT RE: Sprint 3 Week 1 Day 01 Daily Scrum Meeting Individual Updated Report (May 11, 2026) - Added by Tracie Tomon about 1 month ago

1. What did you accomplish yesterday?
 - Set up message on Forum for Sprint 03 DS report Week 1 Day 1
 - Revised SRS
 - Completed initial test cases
2. What can I commit today?
 - Completed the documents to be signed by the client (GDrive)
 - Complete the remaining PM docs for 04 and 05 phases
 - Set budget and cost in the OpenProject
3. Anything blocking my progress?
 - Other schoolwork, commitments, and work schedule.

JA RE: Sprint 3 Week 1 Day 01 Daily Scrum Meeting Individual Updated Report (May 11, 2026) - Added by Joshua Sta Ana 26 days ago

1. What did you accomplish yesterday?
 - PM docs continuation
2. What can I commit today?
 - Revise assigned open project backlog to align with the client need
 - Continue deployment phase setup
 - PM docs documentation
3. Anything blocking my progress?
 - Limited resources

Sprint 3 Week 2 Day 01 Daily Scrum Meeting Individual Updated Report (May 18, 2026)

Added by Tracie Tomon 26 days ago

Answer the following questions:

1. What did you accomplish yesterday?
2. What can I commit today?
3. Anything blocking my progress?

CS RE: Sprint 3 Week 2 Day 01 Daily Scrum Meeting Individual Updated Report (May 18, 2026) - Added by Charjoy Cempron 24 days ago

1. What did you accomplish yesterday?
 - I finished the two minutes for daily reports and revise the transition out plan and risk register columns.
2. What can I commit today?
 - I will start developing the features that will be assigned to me in the Product Backlog Refinement meeting.
3. Anything blocking my progress?
 - I don't have my own laptop or desktop at home and other subjects.

JA RE: Sprint 3 Week 2 Day 01 Daily Scrum Meeting Individual Updated Report (May 18, 2026) - Added by Joshua Sta Ana 23 days ago

- What did you accomplish yesterday?
 - Arranged documents and prepare revision notes
- What can I commit today?
 - Will re-watch recording of the recent defense and start with assigned PMDOCs revisions
 - Will start continuing project deployment phase according to backlog refinement and pending
- Anything blocking my progress?
 - Work, personal agendas.

TT RE: Sprint 3 Week 2 Day 01 Daily Scrum Meeting Individual Updated Report (May 18, 2026) - Added by Tracie Tomon 23 days ago

1. What did you accomplish yesterday?
 - Set up message on Forum for DS report Week 2 Day 1
 - Revised midterm and final PM docs
 - Final Defense Presentation
 - Prepared for backlog refinement meeting
2. What can I commit today?
 - List all the necessary tasks especially for the final documents and requirements
 - Start coding the remaining features
 - Comments matrix
3. Anything blocking my progress?
 - Other schoolwork, commitments, and work schedule.

ID RE: Sprint 3 Week 2 Day 01 Daily Scrum Meeting Individual Updated Report (May 18, 2026) - Added by Jenine Elaine Delay 2 minutes ago

1. What did you accomplish yesterday?
 - I completed the documents assigned to me and finished my assigned sections of the presentation. I also assisted with other parts of the PPT to help prepare for the presentation.
2. What can I commit today?
 - I will revise my documents and presentation sections based on the panel's comments and recommendations. I will also help update any remaining parts that need improvement.
3. Anything blocking my progress?
 - Academic workload

Backlog Refinement

The screenshot shows a Microsoft Word document titled "Sprint 3 Backlogs" being viewed in a web browser. The document content includes:

- User Story:** As an admin, I want a dedicated logs page so that I can monitor user activities without risk of modification.
- Priority:** High
- Story Points:** 5
- 13. Error Log Filtering for Try-Catch Events**
 - User Story:** As an admin, I want to filter error logs so that I can quickly identify and debug issues.
 - Priority:** High
 - Story Points:** 3
- EPIC 5: UI/UX Consistency**
- 14. Align IT Admin Interface with Library Admin Interface**
 - User Story:** As an admin, I want consistent interfaces so that system navigation is seamless.
 - Priority:** Medium
 - Story Points:** 5
- EPIC 6: Project & Documentation**
- 15. MVP Features Checklist for Acceptance Criteria**
 - User Story:** As a project manager, I want a checklist of completed MVP features so that I can validate project readiness.
 - Priority:** High
 - Story Points:** 3

On the right side of the screen, a video call interface is visible, showing four participants: Tracie Tomon, Charjoy Cempron, Joshua Sta Ana, and Jenine Elaine Delay.



Sprint Review

Test Case ID	Description	Steps	Expected Results	Actual Results	Status
TC_MA_001	Verify login with email in lowercase	1. Enter Email address 2. Enter Password 3. Click 'Login' button	User has registered account with lowercase email	Successful login	Pass
TC_MA_002	Verify login with email in uppercase	1. Enter Email address 2. Enter Password 3. Click 'Login' button	User has registered account with uppercase email	Successful login	Pass
TC_MA_003	Verify login with email in mixed case	1. Enter Email address 2. Enter Password 3. Click 'Login' button	User has registered account with mixed case email	Successful login	Pass
TC_MA_004	Verify login with leading spaces in email	1. Enter Email address 2. Enter Password 3. Click 'Login' button	User has registered account with leading spaces in email	Successful login	Pass
TC_MA_005	Verify login with trailing spaces in email	1. Enter Email address 2. Enter Password 3. Click 'Login' button	User has registered account with trailing spaces in email	Successful login	Pass
TC_MA_006	Verify login preserves user role	1. Enter Email address 2. Enter Password 3. Click 'Login' button	User has 'user' role	Role is correctly set to 'user'	Pass
TC_MA_007	Verify admin login redirects to dashboard	1. Enter Email address 2. Enter Password 3. Click 'Login' button	Admin user logs in	Admin at dashboard	Pass
TC_MA_008	Verify login creates session in database	1. Login with valid credentials 2. Check database for session record	User has registered account	New session record created	Pass

Sprint Retrospective

version 9



04 Monitoring and Controlling Phase

02 Project Status Report

Project Status Summary

Percent Complete: 85%

Scope	Schedule	Cost	Risks	Quality
● On Track	● Slightly Behind	● On Budget	● Low–Moderate	● High

The LitPath AI project is currently 83% complete and remains on track overall, with the majority of core system components fully implemented and integrated. Significant progress has been achieved across frontend and backend development, configuration management processes, and infrastructure deployment. The project has transitioned from the primary development phase into final optimization, validation, and pre-deployment preparation.

Scope remains on track (Green) as all major system features and deliverables defined in the approved requirements have been successfully implemented. Any additional changes are formally controlled and managed through the Configuration Control Board (CCB), ensuring strict adherence to scope boundaries.

The project schedule is slightly behind (Yellow), with an estimated variance of approximately 4–6%, primarily due to additional time allocated for final-stage AI model optimization, embedding dataset validation, and cloud deployment tuning. These activities are critical to ensuring system reliability and performance but are considered part of the system stabilization phase rather than core development.

These schedule variances remain within acceptable control limits and are not expected to impact the overall project delivery timeline, as mitigation measures including parallel workstreams and extended validation efforts have been implemented.

Cost remains on budget (Green), as the project effectively utilizes open-source technologies and existing infrastructure resources. No significant cost overruns or deviations have been observed.

Project risks are currently low to moderate (Yellow), primarily associated with:

- Final optimization of AI model performance
- Completion of dataset validation processes



These risks are actively managed and monitored, with mitigation strategies already in place, including iterative testing, structured validation, and continuous performance evaluation. No high-impact risks currently threaten project success.

Quality remains high (Green), as all completed modules have undergone rigorous functional and regression testing. Additionally, configuration management practices ensure that all system components are properly version-controlled, traceable, and verified prior to release.

Work Planned for Last Month

Feature development (frontend and backend)

Execution of core system development activities, including the implementation of user interface components, backend API services, and system integration consistent with the approved Software Requirements Specification (SRS) and Configuration Management Plan (CMP) baseline.

AI model optimization

Iterative enhancement of the AI recommendation engine through embedding model tuning, parameter adjustment, and performance validation in alignment with configuration item (CI) definitions recorded in the Configuration Management Database (CMDB).

Sprint planning and review

Conduct of structured agile project management activities, including sprint planning, backlog refinement, and sprint review sessions, ensuring alignment of development tasks with approved Configuration Change Requests (CCRs) and project baselines

Adviser consultations

Engagement with academic advisers and stakeholders to validate project direction, review technical decisions, and ensure compliance with institutional and client expectations, thereby supporting governance and quality assurance processes

Work Completed Last Week

Completed advanced search functionality

Successful development and deployment of enhanced search filtering and query processing capabilities, corresponding to CCR-002, with verified improvements in system usability and information retrieval efficiency

Updated Cloudflare configuration

Implementation of infrastructure-level configuration updates to Cloudflare Tunnel, ensuring secure external access and alignment with infrastructure configuration items documented in the CMDB



Improved regression test coverage

Expansion of automated and manual regression testing processes to ensure that newly implemented features and configuration changes do not introduce system defects, supporting configuration verification and quality management objectives

Continued AI model tuning

Ongoing refinement of the AI recommendation engine to improve accuracy, stability, and performance, based on testing results and build validation outputs documented in the Build and Release Summary

Work Planned For Next Week

Finalize AI embedding optimization

Completion of advanced tuning and validation of the embedding model to ensure stable performance and alignment with expected accuracy thresholds prior to User Acceptance Testing (UAT)

Prepare for UAT

Consolidation of all verified configuration items into a stable build baseline, ensuring that all approved CCRs are incorporated and that build readiness criteria (file completeness and CI relationships) are satisfied

Validate dataset

Execution of comprehensive data validation processes to ensure consistency, completeness, and correctness of the AI embedding dataset in accordance with CMDB records and configuration status accounting requirements.

Resolve pending CCRs

Review, evaluation, and disposition of outstanding change requests, including CCR-003, through formal Change Control Board (CCB) processes, ensuring traceability and alignment with project constraints.

Open Issues

AI model performance variability - Under monitoring

The AI recommendation engine is currently exhibiting variability in output accuracy due to ongoing tuning and parameter adjustments; mitigation actions include iterative optimization, controlled testing, and validation against benchmark datasets.

Dataset validation incomplete - Ongoing



The embedding dataset has not yet achieved full validation status, with ongoing checks for data consistency and completeness; interim datasets are sufficient for development and testing but require final verification prior to UAT completion.

Open Risks

AI instability may affect accuracy

There is a risk that incomplete optimization of the AI model may lead to inconsistent or inaccurate recommendations during advanced testing phases or deployment.

Mitigation: Continue tuning

Continuous model tuning, performance benchmarking, and validation through iterative testing cycles aligned with build verification processes.

Dataset quality issues

There is a risk that incomplete or inconsistent dataset preparation may negatively impact AI model performance and overall system reliability.

Mitigation: validation before release

Implementation of structured data validation, cleansing procedures, and cross-verification against defined dataset standards prior to UAT and final release.

Deliverables and Milestones

Milestone	WBS	Planned	Forecasted	Actual	Status
Project Scoping		March	March – April 2025	April 2025	Completed
Information Gathering		April 2025	April 2025	April 2025	Completed
Completion of Project Deliverables and documentation		May 2025	May 2025- June 2025	June 2025	Completed
Deliverable	WBS	Planned	Forecasted	Actual	Status
Design Thinking Process (Stages 1-5)		July	Jul – Aug 2025	Aug 2025	Completed
Core UI development (System UI, Citation Generator)		Sept 2025	Sept – Oct 2025	Oct 2025	Completed
AI Search Engine, Generate Overview feedback mechanism & admin dashboard		October 2025	Oct 2025 – Jan 2026	Jan 2026	Completed
Feature Refinement		May 2026	May 2026	Feb – Mar 2026	Completed
Continued Development & Performance Optimization (incl. Codebase restructuring)		May 2026	May 2026	Mar – May 2026	In Progress



Proxmox Server Deployment and Internal Testing		May 2026	May 2026		In Progress
Bug Fixes & Usability Testing		May 2026	June 2026		Not Started
Final Deployment		Jun – July 2026	Jun – July 2026		Not Started

Open Change Requests

Change Request Name	Change Request Number	Request Date	Current Status
Add xyz Functionality	CCR-001	3/14/20xx	In Review by Change Control Board
Add Redundant Servers	CCR-002	2/17/20xx	Approved and Being Added to the Project Plan

Here's the link for our CCR [LitPath CMP.xlsx](#)



05 Closing Phase

01 Transition Out Plan

Executive Summary

This plan formally documents the transition of the LitPath AI system, deliverables, documentation, and operational responsibilities from team HecTech to the DOST-STII Library. LitPath AI is an AI-powered web application designed to replace the library's existing keyword-based OPAC search with a semantic search system powered by Retrieval-Augmented Generation (RAG) and Gemini AI. Following successful User Acceptance Testing and transition activities, full ownership, hosting, maintenance, and operational responsibility for the system will transfer permanently to DOST-STII.

Transition Approach

For this transition, HecTech will maintain its existing four-member team throughout the transition period to support knowledge transfer, system walkthroughs, and issue resolution. No additional staffing requirements are anticipated. The transition is expected to take 1 day following LitPath AI's successful User Acceptance Testing and will be conducted onsite at DOST-STII Library. HecTech will initiate transition activities upon achieving an average user satisfaction score of at least 3.5 out of 5 from the test group of students and DOST employees. It is assumed that DOST-STII will assign designated representatives to coordinate walkthroughs and documentation acceptance during the transition period.

Transition Team Organization

The following table illustrates the transition team members from HecTech and DOST-STII, along with the roles and responsibilities of each member throughout the transition period.

Organization	Title	Roles/Responsibilities
DOST-STII	Client Representative	Coordinate transition activities with HecTech; receive and review all system documentation and deliverables; provide formal institutional acceptance of the system at transition completion
DOST-STII	End User Representative	Participate in system walkthrough and user acceptance demonstration; provide feedback on usability and functionality during the transition period
HecTech	Project Manager	Chair all transition activities conducted onsite; coordinate and schedule all handover activities; track completion of the transition checklist; provide transition status updates to the Project Sponsor; ensure all deliverables are submitted and accepted
HecTech	Backend Developer	Conduct technical system walkthroughs covering the RAG pipeline, ChromaDB vector search,



		Supabase database, and Gemini AI integration; transfer all backend architecture documentation and production credentials
HecTech	Frontend Developer	Conduct user interface walkthroughs for all system features including search, AI overview display, citation generation, and bookmarking;
HecTech	DevOps Engineer	Confirm completion of all security controls and compliance requirements under RA 10173; conduct final system security review; hand over all QA test reports, test cases, and security audit documentation; support DOST-STII in configuring role-based access for admin accounts

Workforce Transition

For this project transition, all HecTech team members will be present onsite at DOST-STII Library on the day of transition to complete their assigned handover activities. Upon formal acceptance by the Project Sponsor and completion of all training and documentation handover activities, HecTech's involvement with the LitPath AI project will conclude immediately and permanently. As HecTech is a student development team, no workforce continuity or ongoing operational role is assumed after handover. DOST-STII will designate at least one system administrator to receive administrative access, technical documentation, and system training on the day of transition, and this individual will be solely responsible for all system operations from that point forward.

Work Execution During Transition

On the day of transition, the LitPath AI system will be fully deployed and operational on Proxmox server and accessible outside Asia Pacific College campus, having passed User Acceptance Testing with an average satisfaction score of at least 3.5 out of 5. All knowledge transfer sessions, system walkthroughs, and documentation handover will be completed onsite on the transition day. HecTech will deliver the system in a stable, defect-free state and will not be responsible for resolving any issues that arise on DOST-STII's side after the transition is formally accepted and closed. Upon completion of training and documentation handover, HecTech will be fully hands-off, and full operational responsibility including maintenance, troubleshooting, and any future enhancements will rest entirely with DOST-STII.

Subcontracts

The LitPath AI project was developed entirely by the HecTech team with no subcontractors or third-party development engagements. However, the system relies on the following third-party services and platforms that must be acknowledged as part of the transition:

Service / Platform	Provider	Purpose
--------------------	----------	---------



Gemini AI API	Google AI Studio	Semantic query understanding, AI-generated research overviews, result re-ranking
Supabase	Supabase Inc.	PostgreSQL relational database for user accounts, bookmarks, document metadata, etc.
SentenceTransformers	Hugging Face	Embedding generation for RAG pipeline
Production Hosting Environment	DOST-STII (cloud)	Host and serve the LitPath AI application

Property Transition

Institution Furnished Equipment (GFE)

No institution-furnished equipment was provided to HecTech for the development of LitPath AI. All development hardware was sourced and maintained by HecTech. The DOST-STII production hosting environment, managed by DOST-STII, will serve as the operational infrastructure for post-transition. HecTech will provide all necessary deployment documentation to support maintenance of this environment.

HecTech Owned Equipment

All hardware utilized by HecTech during development will remain with the team. These assets are not required for the ongoing operation of the system. The complete LitPath AI codebase, database schemas, and embedding data will be fully migrated to the DOST-STII production environment, ensuring independent access post-transition.

Intellectual Property

All intellectual property resulting from the development of LitPath AI including the codebase, database schema, prompt configurations, and UI/UX designs will be transferred to DOST-STII. DOST-STII will hold full ownership of the deployed system and associated technical documentation upon transition completion. User data handling remains governed by the Data Privacy Act of 2012 (RA 10173).

User Accounts and Passwords

Administrative credentials will be transferred and secured in accordance with system architecture and RA 10173 requirements. HecTech development-environment accounts will be deactivated upon completion.

User Account	Current Holder	Transferred To
Gemini AI API Key	HecTech	DOST-STII System Administrator
Supabase Project	HecTech	DOST-STII System Administrator
GitHub Repository Access	HecTech	DOST-STII System Administrator



System Admin Account	HecTech	DOST-STII System Administrator
----------------------	---------	--------------------------------

Knowledge Transfer

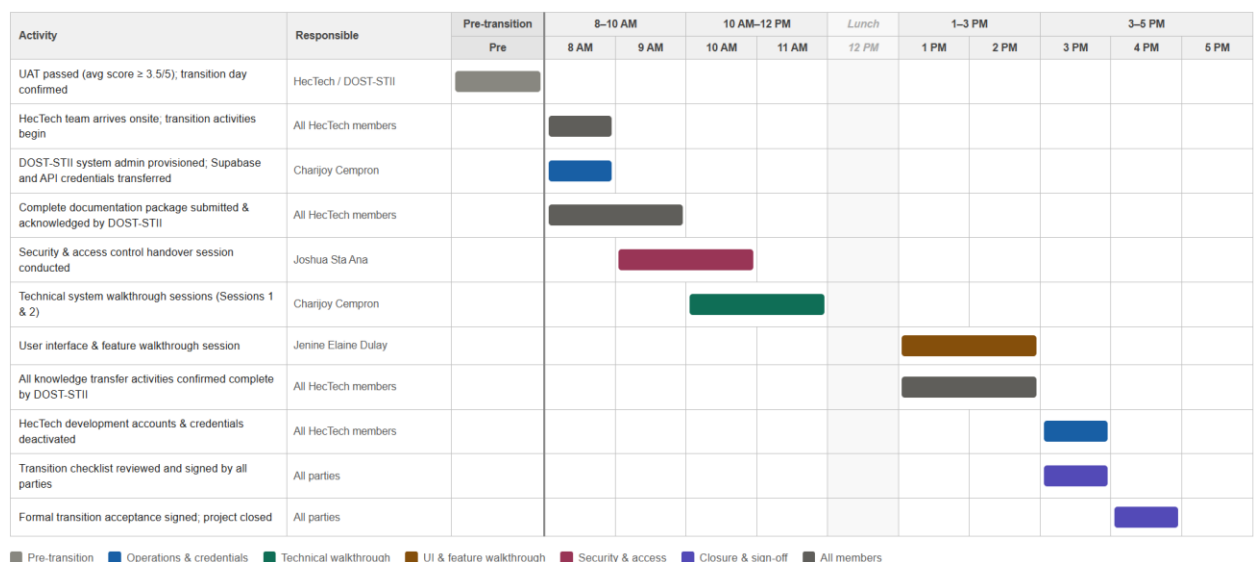
For this transition, knowledge transfer will be conducted onsite at the DOST-STII Library through structured walkthroughs, documentation handover, and support sessions coordinated by the Project Manager.

The Backend Developer will conduct technical walkthroughs covering the system architecture, RAG pipeline, databases, API integrations, deployment, and credential transfer. The Frontend Developer will conduct user-facing walkthroughs of core features such as semantic search, AI-generated overviews, citation generation, and bookmarking. The DevOps Engineer will conduct sessions covering security, role-based access, compliance measures, deployment configuration, and user account management.

All system documentation and test reports will be handed over upon completion of the transition activities, after which full operational responsibility for LitPath AI will transfer to DOST-STII.

Schedule

The following Gantt chart illustrates the schedule for the transition of the LitPath AI system from HecTech to DOST-STII. The transition is a single-day event conducted onsite at the DOST-STII Library on June 1, 2026, initiated upon confirmation that LitPath AI has achieved an average user satisfaction score of at least 3.5 out of 5 during User Acceptance Testing. Any changes to this schedule will require review and approval from the Project Sponsor.





Handover and Acceptance

Formal handover and acceptance of the LitPath AI system will be determined by the completion of all transition checklist activities conducted onsite at DOST-STII Library on the transition day, as validated by the Project Sponsor. The Project Manager will track checklist completion throughout the day and confirm that all items are satisfied before presenting the formal acceptance document for signature.

Transition will be considered complete once all system documentation has been received and acknowledged by DOST-STII, all knowledge transfer and walkthrough sessions have been completed, all user accounts have been properly configured and HecTech development credentials deactivated, and the Proxmox deployment is confirmed stable and operational. The transition checklist must be signed by the DOST-STII Client Representative and the Project Sponsor before the formal acceptance document is presented.

Upon completion, a formal acceptance document will be signed by the HecTech Project Manager and the Project Sponsor. It is only after all signatures are obtained that the transition will be formally closed. The signed document will be archived in the project's shared document repository on SharePoint and submitted as part of the academic closure requirements at Asia Pacific College. From this point forward, HecTech carries no further responsibility for the LitPath AI system, and any issues arising on DOST-STII's side are solely DOST-STII's responsibility to resolve.

02 Project Acceptance

This document establishes formal acceptance of all the deliverables for the [LitPath AI: Smart Pathfinder for Theses and Dissertations](#) project, developed by HecTech for the DOST-Science and Technology Information Institute (DOST-STII). As of the date of this document, the [LitPath AI](#) project is in its final development phase, with all core features substantially complete and final deployment to production scheduled for June to July 2026, in accordance with the approved project schedule.

Upon completion of the final development, bug fixing, usability testing, and deployment phases, the LitPath AI project is expected to have met all acceptance criteria as defined in the Software Requirements Specification (SRS) and the project scope statement. A project audit will be performed prior to final handover to verify that all deliverables meet the defined performance and product requirements. Additionally, a product evaluation will be conducted to confirm that all system components meet the quality and functional requirements established at the outset of the project.

Project Deliverables

The following deliverables are to be formally accepted upon completion and handover of the LitPath AI project:

1. Working Prototype Ready for Deployment: A fully functional standalone web application featuring an AI-powered semantic search engine built on a Retrieval-Augmented



Generation (RAG) pipeline, including a responsive user interface, automated citation generator (APA 7th, MLA 9th, Chicago, and IEEE), bookmarking feature, admin analytics dashboard, and integrated user feedback mechanism.

2. Comprehensive Project Documentation: A complete documentation package including the Technical Manual, User Guide for researchers and library staff, and a Final Report confirming that all project objectives have been successfully met.

Acceptance Criteria

Formal acceptance of the LitPath AI project will be granted upon verification that all of the following criteria have been satisfied:

1. The project delivers a working application with all defined features fully complete and operational.
2. The system achieves an average user satisfaction score of at least 3.5 out of 5 from the test group of students and DOST employees. Feedback must also confirm that the system makes their research process feel faster and easier.
3. The new search system is successfully deployed as a standalone web application via Proxmox and consistently processes queries and displays results faster than the traditional library OPAC system.
4. All key deliverables — the AI engine, the user interface including the citation tool, user feedback system, and all documentation — have been formally signed off on and accepted by the project stakeholders.

Transition to Operations

LitPath AI was developed as a standalone web application. Rather than transitioning to an operations team for manufacturing (as in a traditional product), this section describes the handover of the completed prototype for academic evaluation and any future deployment considerations.

The system is built on a clearly documented technology stack: a Django REST API backend, a React (Vite + Tailwind CSS) frontend, a Supabase (PostgreSQL) cloud database, ChromaDB for vector storage, and the Google Gemini AI API for AI features. This stack was chosen for accessibility and maintainability, allowing future developers or maintainers to onboard with standard documentation.

All source code was committed to a GitHub repository, and the project documentation includes a full appendix of source code structure, business rules, and diagrams, which collectively serve as a technical reference package for any future development team or institutional adopter.

For future deployment, the system would require a production server environment, a stable internet connection for Gemini AI API access, and access to the DOST-STII thesis dataset. The administrative seeding scripts (`seed_admins.py`, `seed_users.py`) and database migration files are included in the repository to facilitate environment setup.



Project Closeout Authorization

Upon formal sign-off of this Project Acceptance document by the Project Sponsor and Project Manager, the Project Manager is authorized to proceed with the formal closeout of the LitPath AI project. The closeout process will include the following activities:

1. Conduct a post-project review to evaluate overall project performance against the original plan, schedule, and objectives.
2. Document and archive lessons learned from all project phases, from project scoping through final deployment.
3. Formally release the HecTech project team from their project responsibilities upon completion of all closeout activities.
4. Close out all procurements and third-party service agreements related to the project.
5. Archive all relevant project documents, including the Business Case, SRS, System Design Document, Change Management Plan, and all the Project Management documents, in a format accessible to DOST-STII for future reference.

Once the closing process is completed, the Project Sponsor will be formally notified of project closure, and the Project Manager will then be released from the project.

By signing below, the undersigned acknowledge and confirm acceptance of all LitPath AI project deliverables as described in this document and authorize the Project Manager to proceed with formal project closeout.

Khasian Eunice M. Romulo Project Sponsor Science Research Specialist II, DOST-STII Date: _____		Tracie Tomon Project Manager HecTech Asia Pacific College Date: _____
Manuel Sebastian S. Sanchez PBL Coordinator Asia Pacific College Date: _____		Gonzalo V. Gumogda Project Adviser Asia Pacific College Date: _____

SPONSOR ACCEPTANCE

Approved by the Project Sponsor:

Khasian Eunice M. Romulo
Science Research Specialist II / Unit Head
(Digital Information and Processing Delivery Unit)

Date: May 14, 2026



03 Post Project Review

Project Summary

The HecTech team is currently in progress with the development of LitPath AI: Smart Pathfinder for Theses and Dissertations, with most core features successfully delivered as planned. The remaining deliverable is the Admin Dashboard for usage analytics and feedback management which is currently being finalized with pending client refinements.

LitPath AI is an AI-powered academic search platform designed to improve the discovery and accessibility of graduate theses and dissertations. The system was built as a standalone web application accessible through standard web browsers, independent of any external library catalog system.

The objective of this project was to design and develop an intelligent search engine that allows students, researchers, and academic professionals to find relevant theses and dissertations through semantic search, rather than simple keyword matching. The system also provides AI-generated overviews of search results, citation generation in multiple formats (APA 7th, MLA 9th, Chicago, and IEEE), bookmarking, search history, a user feedback mechanism, and an administrative analytics dashboard.

Project success was defined as delivering a functional prototype that meets the specified functional and non-functional requirements, including achieving at least 70% search relevance accuracy in the top 10 results which is based on User Feedback during user testing, delivering search results up to the industry standard, and maintaining a target user satisfaction score of at least 3.5 out of 5.

Project Team and Staffing

The LitPath AI project was developed by the HecTech team. The table below provides information about team members and their roles:

Name	Project Role	Contact
Charijoy Cempron	Developer	ctcempron@student.apc.edu.ph
Jenine Elaine Dulay	Developer	jcdulay@student.apc.edu.ph
Joshua Sta. Ana	Developer	Jtstaana2@student.apc.edu.ph
Tracie Tomon	Developer	tctomon@student.apc.edu.ph

The project team operated as a collaborative unit with clearly divided responsibilities across documentation, design, and development. Each member contributed to multiple sprints, as evidenced by the group log. The team utilized Agile/Scrum methodology, conducting regular sprint planning, daily standups, sprint reviews, and retrospectives throughout the project lifecycle. Regular adviser and co-adviser consultations also provided timely guidance and kept the project on track.



Project Deliverables (Planned vs. Actual)

The HecTech LitPath AI project is still to be completed successfully across all core development phases. This section highlights the planned deliverables and compares them to actual deliverables as they occurred.

Planned Deliverable	Actual Deliverable	Summary
AI-powered semantic search using vector embeddings (ChromaDB + SentenceTransformers)	Semantic search implemented using ChromaDB and all-MiniLM-L6-v2 model (built on a Retrieval-Augmented Generation (RAG) pipeline)	Delivered as planned
AI-generated overview summaries via Gemini 3.0 Flash	AI overview generation with inline reference citations implemented via Gemini AI	Delivered as planned
Citation generator in APA, MLA, Chicago, and IEEE formats	Automated citation generation in all four formats implemented	Delivered as planned
Bookmark and search history management	Bookmark feature and research history sidebar with session reload capability implemented	Delivered as planned
User feedback mechanism (ratings and comments)	Item-level thumbs up/down feedback for sources and full CSM form implemented	Delivered as planned
Admin dashboard for usage analytics and feedback management	Administrative dashboard with usage KPIs, charts, and feedback manager implemented	In progress (pending client refinements)

Remaining deliverables covering bug fixes, usability testing, and final deployment are scheduled for completion between June and July 2026.

Transition to Operations

LitPath AI was developed as a standalone web application. Rather than transitioning to an operations team for manufacturing as a traditional product, this section describes the handover of the completed prototype for academic evaluation and any future deployment considerations.

The system is built on a clearly documented technology stack: a Django REST API backend, a React (Vite + Tailwind CSS) frontend, a Supabase (PostgreSQL) cloud database, ChromaDB for vector storage, and the Google Gemini AI API for AI features. This stack was chosen for accessibility and maintainability, allowing future developers or maintainers to onboard with standard documentation.

All source code was committed to a GitHub repository, and the project documentation includes a full appendix of source code structure, business rules, and diagrams, which collectively serve as a technical reference package for any future development team or institutional adopter.



For future deployment, the system would require a production server environment, a stable internet connection for Gemini AI API access, and access to the DOST-STII thesis dataset. The administrative seeding scripts (seed_admins.py, seed_users.py) and database migration files are included in the repository to facilitate environment setup.

Project Costs

The LitPath AI project was completed at zero direct cost to both the HecTech development team and DOST-STII. The entire development phase was carried out using freely available, open-source tools and technologies, including React.js, Django, SentenceTransformers, ChromaDB, and Supabase, all of which are available at no licensing cost. Development and testing were conducted on existing hardware owned by the team, requiring no additional hardware acquisition. The free tiers of the Gemini AI API were used throughout the development and testing phases without incurring any charges.

As a result, no budget overruns occurred, and no additional funding was required at any stage of the project. The absence of direct costs is a significant advantage of this project and demonstrates that a fully functional, AI-powered research platform can be developed and delivered within an academic setting without financial expenditure. The only costs anticipated going forward are post-handover operational expenses, specifically the Gemini AI API subscription and Cloud hosting plan, which will be managed and funded by DOST-STII upon taking ownership of the deployed system.

Cost Item	Dev Phase (HecTech)	Post-Handover (DOST-STII)
DEVELOPMENT COSTS (HecTech)		
Personnel, Honoraria & Allowances	₱0	—
Software Licenses & Tools (React, Django, SentenceTransformers, ChromaDB, Supabase)	₱0	—
Hardware (Existing Proxmox server — no new acquisition)	₱0	—
Testing, QA, & Documentation	₱0	—
Gemini AI API (free tier used during development)	₱0	—
Cloudflare Deployment (free tier used during development)	₱0	—
TOTAL DEVELOPMENT COST (HecTech)	₱0	—
POST-HANDOVER OPERATIONAL COSTS (DOST-STII)		
Gemini AI API (production tier — scaled to usage volume)	—	See recommendation below*
Cloudflare Hosting Plan (production tier — scaled to traffic)	—	See recommendation below*



System Maintenance (existing IT Admin & Librarian Admin roles)

—

₱0 (existing staff)

Note:

* **Gemini AI API** — The team recommends Gemini 2.5 Flash as the most suitable model for an institutional library platform, offering strong AI performance at a cost-efficient rate. The free tier supports up to 1,500 requests per day at no cost, which may be sufficient for initial deployment. Should usage exceed free-tier limits, the paid tier is estimated at approximately \$0.30 per million input tokens and \$2.50 per million output tokens (roughly ₱350 – ₱1,700/month depending on query volume). DOST-STII's IT Administrator may evaluate and select the appropriate tier upon taking ownership of the system.

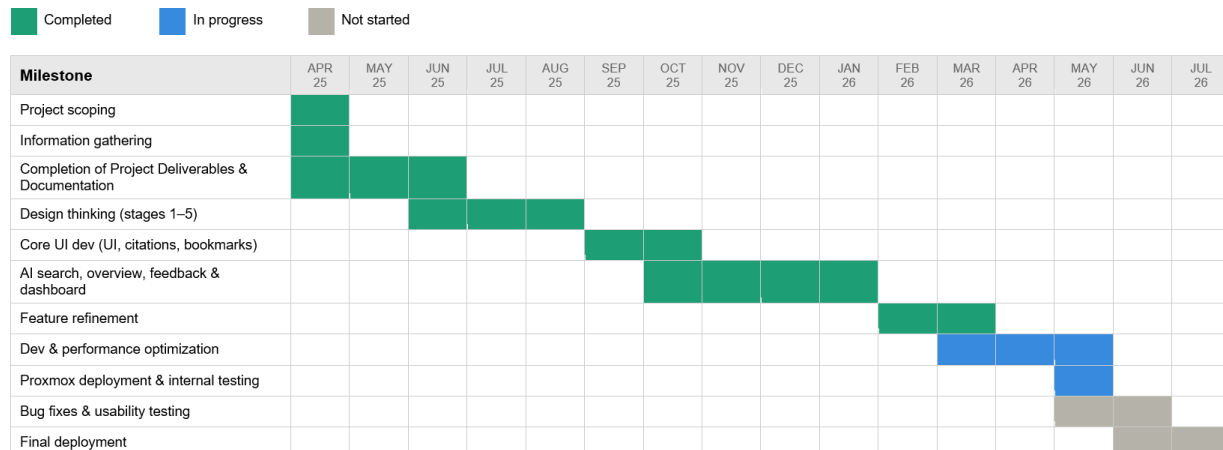
** **Cloudflare Hosting Plan** — A standard cloud hosting plan is estimated at approximately ₱0 – ₱570/month depending on the selected service tier. The free tier may remain sufficient for initial institutional usage. DOST-STII may upgrade to a paid plan as traffic scales.

Project Schedule

The LitPath AI project followed a structured development timeline beginning in April 2025, with final deployment targeted for July 2026. The table below summarizes the key milestones and their completion status.

LitPath AI: Smart Pathfinder for Theses and Dissertations

April 2025 – July 2026



Timeline runs Apr 2025 → Jul 2026

Milestone	Target Date	Status
Project Scoping	April 2025	Completed
Information Gathering	April 2025	Completed
Completion of Project Deliverables & Documentation	June 2025	Completed
Design Thinking Process (Stages 1–5)	Aug 2025	Completed



Core UI Development (System UI, Citation Generator, Bookmark)	Sep – Oct 2025	Completed
AI Search Engine, Generate Overview, Feedback Mechanism & Admin Dashboard	Oct 2025 – Jan 2026	Completed
Feature Refinement	Feb – Mar 2026	Completed
Continued Development & Performance Optimization (incl. codebase restructuring)	Mar – May 2026	In Progress
Proxmox Server Deployment & Internal Testing	May 2026	In Progress
Bug Fixes & Usability Testing	May – Jun 2026	Not Started
Final Deployment	Jun – Jul 2026	Not Started

Early phases covering Project Scoping, Information Gathering, and initial Documentation were all completed on schedule by June 2025. The Design Thinking Process was finalized in August 2025, followed by Core UI Development from September to October 2025, which delivered the main interface, citation generator, and bookmark feature as planned.

Development of the more complex features including the AI Search Engine, Generate Overview, Feedback Mechanism, and Admin Dashboard was completed between October 2025 and January 2026. Feature Refinement and Client Feedback Integration followed from February to March 2026, incorporating evaluator and stakeholder input into the system.

As of May 2026, the team is actively working through Continued Development and Performance Optimization alongside Proxmox Server Deployment and Internal Testing. Bug Fixes, Usability Testing, and Final Deployment remain on schedule for May through July 2026.

Recommendations

Recommendation #1:

Expand LitPath AI's Document Scope beyond Theses and Dissertations. Future development teams should consider expanding the system to support additional document types, particularly Filipiniana articles and locally produced academic journals. Currently, LitPath AI is limited to theses and dissertations, which restricts the range of sources available to researchers. Broadening the document scope would require updates to the parsing pipeline and metadata schema but would greatly increase the platform's value as a comprehensive research tool for Filipino academic institutions.

Recommendation #2:

Set Up Error Logging and Monitoring Using Sentry Before Final Deployment. As recommended by the client, the team could integrate Sentry for Django into the production environment prior to final deployment. Sentry provides real-time error tracking, detailed stack traces, and alerting capabilities that will allow the team to quickly identify and resolve issues after the system goes live on the Proxmox server. Configuring Sentry before deployment



rather than after ensures that any errors encountered during the initial rollout are captured immediately and do not go unnoticed.

Sponsor Acceptance

Approved by the Project Sponsor:

Date: May 14, 2026

Khasian Eunice M. Romulo
Science Research Specialist II / Unit Head
(Digital Information and Processing Delivery Unit)

04 Lessons Learned

Introduction

Capturing lessons learned is an integral part of every project and serves several important purposes. While the finalization of a formal lessons learned document is completed during the project closeout process, capturing lessons learned should occur throughout the project lifecycle to ensure all information is documented in a timely and accurate manner. The lessons learned document serves as a valuable tool for future project managers assigned to similar projects, describing not only what went wrong and how to avoid similar occurrences, but also what went well and how future projects may benefit from these practices.

The purpose of this Lessons Learned document for the LitPath AI: Smart Pathfinder for Theses and Dissertations project is to capture the project's lessons in a formal document for use by other project managers on similar future academic technology projects. This document may be used during new project planning to understand what problems occurred, how they were handled, and how they may be avoided. It also details what went well and why, so that other project teams may apply these practices. This document will be formally communicated to the Project Sponsor and stored in the organizational assets and archives via OpenProject and SharePoint.

Lessons Learned Approach

The lessons learned approach describes how this document was created, what it consists of, and how lessons are categorized. For the LitPath AI project, lessons learned were captured continuously throughout the project lifecycle rather than only at close out, ensuring that details and context were preserved accurately as events occurred.

Lessons were gathered through the following methods:



1. **Sprint Retrospectives in OpenProject:** At the end of each two-week sprint, the HecTech team conducted retrospectives to reflect on what went well, what did not, and what could be improved. These retrospective notes served as the primary source of lessons learned throughout the project.
2. **Team Meetings and Daily Standups:** Issues, blockers, and successes raised during daily standups and weekly team meetings were logged and tracked, providing real-time insights into team dynamics, technical challenges, and process effectiveness.
3. **Group Journal and Log Entries:** The team maintained a group log documenting significant events, decisions, and observations throughout the project, serving as a chronological reference for compiling lessons learned during the closeout phase.

The lessons learned in this document are categorized by project knowledge area, consistent with standard project management practice. Knowledge areas represented include cost management, risk management, time management, scope management, human resource management, integration management, and technology and infrastructure management. Knowledge areas for which no lessons were documented are omitted.

Lessons Learned from this Project

The following table presents the lessons learned for the LitPath AI project, categorized by project knowledge area. Each lesson includes a description of the issue or success, its impact on the project, and a recommendation for consideration on similar future projects. Both challenges and successes are documented to provide a balanced and complete reference.

Category	Issue Name	Problem/Success	Impact	Recommendation
Scope Management	Limited Document Scope — Theses and Dissertations Only	PROBLEM — LitPath AI was developed to support only theses and dissertations, which restricts the range of academic sources available to researchers using the platform.	Researchers looking for Filipiniana articles or locally produced academic journals cannot find them through LitPath AI, limiting the platform's value as a comprehensive research discovery tool for Filipino academic institutions.	Future development teams should consider expanding the system to support additional document types, particularly Filipiniana articles and locally produced academic journals. This would require updates to the parsing pipeline and metadata schema but would significantly increase the platform's research utility.



Technology & Infrastructure Management	Absence of Error Logging and Monitoring Before Deployment	PROBLEM — The system was deployed without integrating a real-time error tracking tool such as Sentry for Django. Error monitoring was not configured prior to production deployment, meaning any issues encountered during initial rollout risked going undetected.	Without error logging in place at deployment, the team would have limited visibility into production errors, stack traces, and system failures, increasing the time needed to identify and resolve issues after go-live.	Integrate Sentry for Django into the production environment prior to final deployment. Configuring error logging before deployment rather than after ensures that any errors encountered during the initial rollout are captured immediately and addressed without delay.
--	---	---	--	---

Lessons Learned Knowledge Base / Database

The Lessons Learned Knowledge Base contains historical information from completed projects and serves as a centralized repository for organizational project assets. All project lessons learned, risk outcomes, issue logs, and techniques that worked well should be stored here to ensure ease of access and reference for future project teams.

The lessons learned for the LitPath AI project are stored across two platforms that serve as the team's organizational knowledge repository:

1. OpenProject: Sprint retrospective notes, issue logs, and change request records generated throughout the project lifecycle are archived within the LitPath AI project space in OpenProject, cataloged by sprint and knowledge area for structured retrieval.
2. SharePoint: The formal Lessons Learned document, along with all other project management documents — including the Business Case, Feasibility Study, SRS, System Design Document, Change Management Plan, Post-Project Review, and Project Acceptance — are archived in the team's SharePoint repository under the LitPath AI project folder, accessible to DOST-STII, APC faculty, and future project teams.

This information will be valuable for any future team assigned to develop or extend an AI-powered academic discovery system, particularly within a DOST or APC institutional context.

Lessons Learned Applied from Previous Projects

The LitPath AI project is the first project of its kind developed by the HecTech team. As such, no direct organizational lessons learned database from prior similar projects was available at the outset. However, the team drew on general best practices and publicly available knowledge from comparable academic and open-source AI development projects to inform key planning decisions:



1. Agile/Scrum methodology practices from APC's project management curriculum were applied in structuring the sprint cadence, retrospective process, and OpenProject setup, informing the team's approach to managing scope, tracking progress, and handling change requests.
2. Published guidance on RAG pipeline architecture and vector database integration informed the team's technology selection decisions, particularly the choice of ChromaDB and SentenceTransformers. Referencing community documentation and case studies helped the team avoid known pitfalls in semantic search implementation.
3. General software engineering best practices around API dependency risk — specifically the recommendation to plan fallback mechanisms for third-party services — were applied when integrating the Gemini AI API, leading to the early implementation of the ChromaDB-based fallback search.

Process Improvement Recommendations

Based on the lessons learned documented in this report, the following process improvement recommendations are proposed for the HecTech team and for future project teams undertaking similar academic software development projects. It is important that these recommendations are reviewed, approved, and incorporated into future project planning processes to drive continuous improvement.

1. **Introduce a Feature Freeze Milestone Before Deployment:** Future projects should include a formal feature freeze checkpoint in the schedule, after which no new development is merged into the main codebase. This ensures the deployment and testing environment remains stable and that the team's attention is fully directed toward quality assurance and deployment preparation.
2. **Conduct Client Requirements Elicitation for Admin Features Early:** Administrative and client-facing features should be prototyped and reviewed with the client before full development begins. Wireframes or mockups presented during planning can surface client expectations early and prevent late-stage scope changes.
3. **Address Technical Debt Within Each Sprint:** Rather than deferring performance optimization to a dedicated phase, future teams should allocate approximately 20% of each sprint's capacity to refactoring and technical debt reduction. This prevents the accumulation of issues that require disruptive mid-project restructuring.
4. **Formalize a Change Control Process from Day One:** The Change Management Plan should be introduced and communicated to all stakeholders at the project kickoff. All scope or technical change requests should flow through the defined CCB process from the first sprint, ensuring changes are evaluated and tracked consistently throughout the project lifecycle.



06 Agile

03 Agile Team Charter

Project Vision

To provide DOST-STII Library users (students, researchers, and DOST employees) with a fast, intelligent, and accessible way to discover relevant theses and dissertations through a standalone AI-powered web application, replacing the limitations of traditional keyword-based search with semantic understanding and intelligent research support tools.

Mission and Objectives

To design, develop, and deploy LitPath AI, a web application that leverages vector embeddings and large language models to deliver semantic search results, intelligent summaries, citation generation, bookmarking, research history, and follow-up question capabilities and deploy it on a Proxmox server instance on dedicated hardware at DOST-STII by June–July 2026.

Success Criteria

- The system achieves a minimum of 70% relevance accuracy in its top search results, validated through user feedback and expert review.
- The system achieves an average user satisfaction score of at least 3.5 out of 5 from usability testing participants (student researchers and DOST employees).
- Search queries are processed and results displayed faster than the traditional library OPAC system.
- All key deliverables: AI search engine, citation generator, bookmark feature, research history, follow-up questions, admin dashboard, and documentation are formally accepted by the Project Sponsor.
- The system is successfully deployed on the Proxmox server instance at DOST-STII.

Team Members and Roles

Name	Role	Responsibilities
Tracie Tomon	Project Manager / Scrum master	Ensures the project meets its goals while supporting the team's productivity and adherence to Agile practices.
Joshua Sta Ana	Product Owner	Responsible for maximizing a product's value by setting its vision and roadmap, aligning it with business goals and customer needs, and making key decisions throughout its lifecycle.



Charijoy Cempron	Developer	Design, development, and testing of assigned features and work packages
Jenine Dulay	Developer	Design, development, and testing of assigned features and work packages

Team Communication

The team communicates primarily through Facebook Messenger and Microsoft Teams for day-to-day updates, questions, and coordination. Video calls and virtual meetings are conducted through Microsoft Teams when needed.

The team meets during class hours and whenever coordination is needed outside of class. During active sprints, the team conducts daily stand-ups throughout the sprint period up until the sprint review. Stand-ups are held to discuss what each member did, what they plan to do next, and any blockers they are facing.

All project documents and files are stored and shared through Microsoft SharePoint to ensure all team members always have access to the latest versions.

Tools

Category	Tools
Communication	Facebook Messenger, Microsoft Teams,
Development	GitHub, Visual Studio Code
Project Tracking	OpenProject, Trello
Document Management	Microsoft SharePoint
UI/UX Prototyping	Figma
Deployment Infrastructure	Proxmox server on dedicated local hardware

Rules of Behavior

- Attend all scheduled team meetings. Notify the group at least 2 hours in advance via Microsoft Teams if unable to attend.
- Update assigned tasks in OpenProject within 24 hours of completing them.
- All formal documents must be uploaded to SharePoint using the agreed file naming convention before submission.
- Raise blockers or concerns with the Project Manager immediately and do not wait until the next meeting.
- Scope of change requests must go through the Project Manager before being brought to the client or instructor.
- Feedback from the client or instructor must be shared with the full team within 24 hours of receipt.
- Code and design output must be peer-reviewed by at least one other team member before being marked complete.



SPONSOR ACCEPTANCE

Approved by the Project Sponsor:

Date: April 1, 2026

Khasian Eunice Romulo
Science Research Specialist II / Unit Head
(Digital Information and Processing Delivery Unit)

03B Agile Team Values

Group Name: HecTech

Date: March 27, 2026

PROJMAN MI231

- Copy this document to the Shared tab of your group channel.
- Only one document for the whole group. Each group member, add your name, signature and date signed below.
- Add a link to the video recording of the group discussion for this task

Part 1 |[Link](#)|

Part 2 |[Link](#)|

Questions for Group Discussion:

Using the values below as a starting point, decide your own values as a team.

You can update, add or remove from the list below

- Edit the values and rubrics listed below
- Include evaluation criteria
- Each team member commits to the values that you have all decided to follow for your group.
- Print and sign your name below.

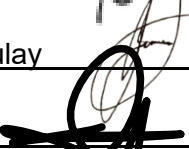
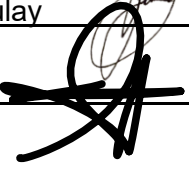
I, as a member of **HecTech**, promise to do my very best to abide by the values of our group listed below. We have discussed and agreed to follow the values listed below.



I understand that I will evaluate myself and my peers based on the agreed upon values in an objective manner during the midterms and finals of the term.

Group Leader: Tracie Tomon  March 18, 2026

Members:

- Charijoy Cempron  March 18, 2026
- Jenine Elaine Dulay  March 18, 2026
- Joshua Sta Ana  March 18, 2026

The Base Layer

- **Motivation (Why am I doing this?)**

Our team is motivated by both our studies and personal growth. We want to graduate successfully, gain real project experience, improve our knowledge and skills for our future careers, and also help our stakeholders by providing useful and reliable solutions.

- **Leadership (Who is steering the ship?)**

We trust our assigned leader to guide the team, set our direction, and make sure we stay on track with our goals.

- **Negotiation (Do I discuss, negotiate and bargain for my work tasks?) (Sprint Planning)**

During sprint planning, we actively negotiate the timeframes of our work packages to ensure we have realistic deadlines. We also bargain and discuss task assignments to ensure that the quality and deliverables of our outputs are never compromised.

- **Team Building (What activities have we done/ can we do to foster Teamwork and Camaraderie?)**

We hold regular meetings to discuss tasks and align our goals. We also meet in person, like in the library, to work together and strengthen our teamwork. Sometimes, we also go on food trips or hang out while talking about our recent performance and how we can improve as a team.

- **Communication (How can I communicate with my team members in a more consistent, transparent, and timely manner?)**



We value initiative and practice a "tap system" to keep each other accountable. Furthermore, we make it a habit to inform the team of our availability as a sign of respect, striving to provide regular updates so no one is left in the dark.

Values – What are the values of our team? (Discuss and define your Team Values) All team members must commit to following the values agreed on.

- **Initiative** – We take things first on our own resolve and capability and seek assistance only if necessary.
- **Commitment** – Giving your best to your assigned task to make sure the outcome is high-quality, rather than just doing the bare minimum to pass.
- **Transparency** – Being honest about what you can and can't do.
- **Flexibility** – Willing to adjust plans, take on different roles, and adapt to challenges to help the team meet our goals.
- **Effective Communication** - Clearly conveying ideas, updates, and concerns while actively listening to teammates.
- **Accountability** - Taking ownership of your responsibilities and being answerable to the team for your actions and results.

Beliefs – What beliefs do we bring into the team?

- Past experiences that help us re-shape and grow.
- Learning starts when we start doing it.
- Mistakes are inevitable and are part of learning that helps us improve as a team.
- Good communication leads to better results and fewer problems.
- Supporting each other leads to better outcomes for the whole team.
- Respecting each other's time, ideas, and differences.

Traditions – What shared activities, ceremonies, can we have as a team?

- A short update from each member about their progress and any hindrances.
- Setting time to help each other when someone is stuck on a task.



- Weekly meetings sharing what went well, what can be improved for the next week, assigning tasks and reminding the team about upcoming submissions.
- Keeping notes or summaries after meetings so everyone stays informed.
- Setting goals before starting new tasks.
- A simple way of recognizing finished tasks or milestones like giving simple thank you message to members for their contributions.

Talk by colonist.io Startup Co-Founder Demi Yilmaz - Co-founder, Hiring & Marketing

<https://colonist.io/values>

Questions for Group Discussion:

Using the values above as a starting point, decide on your own values as a team. You can add or remove from the list above.

Write down the values you have decided on.

1. **Initiative** – we should act on our own, using our skills and judgment.
2. **Commitment** – we give our best effort on assigned tasks to ensure high-quality results which reflects our dedication and responsibility.
3. **Transparency** – to build trust within the team and prevents misunderstandings, we should be honest on what we can and cannot do.
4. **Flexibility** – to keep the team's resiliency and efficiency, we should be adaptable to the changes in order to support the team and achieve our goals.
5. **Effective Communication** – clearly sharing our ideas, updates, and concerns while actively listening to teammates to make sure everyone stays informed and aligned.
6. **Accountability** – we take ownership of our responsibilities to strengthen reliability and trust within the team.

Include evaluation criteria similar to the example above.

1. Initiative

- a. We act without waiting to be told
 - -1: Waits for instructions before starting any task



- 0: Starts tasks when given direction
- +1: Acts independently and looks for what needs to be done next
- b. We solve problems on our own before asking for help
 - -1: Immediately asks others without attempting to figure it out first
 - 0: Makes some effort before seeking help
 - +1: Exhausts all available resources and only asks when truly necessary
- c. We look for ways to contribute beyond what is assigned
 - -1: Only does the minimum of what was assigned
 - 0: Completes assigned tasks and nothing more
 - +1: Looks for additional ways to add value to the team

2. Commitment

- a. We manage our workload responsibly
 - -1: Overcommitting or underdelivering
 - 0: Managing tasks effectively
 - +1: Planning and avoiding delays
- b. We communicate potential risks to commitments
 - -1: Informing other team members only after the issues have already caused delays.
 - 0: Reporting issues when they become noticeable
 - +1: Identifying risks early and suggest possible solutions
- c. We follow through deadlines and agreements
 - -1: Frequently misses deadlines
 - 0: Regularly meet agreed timelines and deliverables
 - 1: Consistently delivering on time and can each be relied upon

3. Transparency

- a. We always communicate honestly
 - -1: Hiding or lying to avoid problems
 - 0: Telling the truth when asked
 - +1: Openly sharing information even if it is not convenient.
- b. We follow through on what we say
 - -1: Often failing to deliver or overlooking commitments.



- 0: Completing tasks as expected.
- +1: Consistently delivering and adding or suggesting improvements.
- c. We take initiative to address problems and improve situations
 - -1: Ignoring problems or waiting for others to fix them.
 - 0: Acting when it was told what to do.
 - +1: Identifying issues and taking actions to resolve them.

4. Flexibility

- a. We adapt to changes in plans or roles
 - -1: Resists changes and struggles to adjust
 - 0: Accepts changes when asked
 - +1: Embraces changes quickly and helps others adjust too
- b. We take on different responsibilities when the team needs it
 - -1: Refuses tasks outside their usual role
 - 0: Takes on different tasks when assigned
 - +1: Willingly steps up to fill gaps without being asked
- c. We stay composed and solution-focused under pressure
 - -1: Gets overwhelmed and shuts down when challenges arise
 - 0: Manages pressure but needs some guidance
 - +1: Stays calm, adapts quickly, and keeps the team moving forward

5. Effective Communication

- a. We communicate clearly
 - -1: Messages are unclear or confusing.
 - 0: Communication is understandable.
 - +1: Messages are clear and easy to follow.
- b. We keep the team members informed
 - -1: Failing to update others.
 - 0: Provide updates when asked.
 - +1: Consistently share updates without being asked.
- c. We raise concerns
 - -1: Avoid discussing problems.
 - 0: Mention issues when necessary.



- +1: Addresses concerns early to prevent bigger problems

6. Accountability

- a. We take responsibilities for our actions and outcomes
 - -1: Always blaming each other for the workaround that hasn't gone well and ignore the consequences.
 - 0: Responsible to all the actions done and commitments said.
 - +1: We admit our mistakes and accept it thus opening ourselves to correction for improvement.
- b. We admit mistakes and learn from them
 - -1: We hide mistakes or deny responsibility
 - 0: We admit mistakes when pointed out
 - +1: We openly admit mistakes, accept feedback, and use them as opportunities to improve
- c. We are keeping ourselves informed of the consequences
 - -1: Ignoring the outcome of our actions and all that may affect it.
 - 0: Observing compassion and avoiding harming others.
 - +1: Choosing battles over my ego.



07 Project Documents

04 System Design Document

Introduction

This System Design Document provides a comprehensive description of the system architecture and design for LitPath AI: Smart Pathfinder for Theses and Dissertations. The system is designed to enhance research discovery within the DOST-STII Library by addressing the limitations of the existing Online Public Access Catalog (OPAC), particularly in terms of search relevance, performance, and usability.

LitPath AI is a standalone, AI-powered web application that complements the existing OPAC system. It introduces semantic search, automated citation generation, bookmarking, and analytics to improve the accessibility and usability of theses and dissertations.

Purpose

The purpose of this document is to define how LitPath AI will be constructed from a technical and architectural perspective. It ensures that the system design aligns with the requirements specified in the Software Requirements Specification (SRS) and the goals outlined in the Business Case.

This document describes the system's architecture, components, database structure, hardware and software design, and security mechanisms. It serves as a reference for developers, stakeholders, and evaluators to understand how the system will be implemented.

System Overview

LitPath AI is a web-based application designed to improve the discovery of academic research materials, specifically theses and dissertations stored in the DOST-STII Library.

The system leverages AI-powered semantic search using Retrieval-Augmented Generation (RAG) and Gemini AI to interpret user queries and return contextually relevant results. Unlike traditional keyword-based systems, LitPath AI understands user intent, enabling more accurate and efficient research.

Key functionalities include:

- AI-powered semantic search with ranked results
- AI-generated overview summaries with citations
- Advanced filtering by subject and publication date
- Automated citation generation (APA, MLA, Chicago, IEEE)
- Bookmarking and research history tracking
- User feedback collection



- Administrative analytics dashboard

Design Constraints

The design and implementation of LitPath AI are subject to several constraints:

- The system depends on external AI services (Gemini AI), which introduces limitations such as API latency and rate limits.
- Development is limited to a four-member team with no external funding for development resources.
- Full-text access to theses and dissertations is out of scope for the public; only metadata and abstracts are used.
- The system must be deployed within the project timeline ending June 2026.
- The system must comply with institutional security and data privacy policies.

Roles and Responsibilities

The development and design of LitPath AI involve multiple roles:

Name	Role	Phone	Email
Tracie Tomon	Project Manager	09922492939	tctomon@student.apc.edu.ph
Jenine Elaine Dulay	UI/UX Designer	09458471401	jcdulay@student.apc.edu.ph
Charijoy Cempron	System Architect	09764935474	ctcempron@student.apc.edu.ph
Joshua Sta Ana	Security Lead	09684787041	jtstaana2@student.apc.edu.ph

Project References

The system design is based on the following references:

- Software Requirements Specification (SRS) for LitPath AI v2.0
- LitPath AI Business Case Document

System Architecture

LitPath AI follows a three-tier architecture:

1. Presentation Layer (Frontend)
2. Application Layer (Backend & AI Processing)
3. Data Layer (Databases and External Systems)

Hardware:

The system is currently deployed in a local Proxmox-based server environment with the following specifications:

Processor:

Intel Core i5-6500 @ 3.20GHz (4 cores, x86_64 architecture)

Memory:

32 GB DDR4 RAM



Storage:

- 512 GB SSD (primary system and application storage)
- 500 GB HDD (backup storage)
- Total: 1 TB

Performance Benchmarks:

- Buffered Reads: 515.55 MB/sec
- Average Seek Time: 0.06 ms
- FSYNC/sec: 583.03
- Regex Processing: 4,154,772 ops/sec

Network:

Ethernet (Intel I219-LM)

System Unit:

HP ProDesk 600 G2 MT

This hardware supports development, testing, and temporary deployment. Production deployment will transition to a cloud-based environment managed by DOST-STII.

Software:

The system is composed of modular components:

Frontend

- React.js
- Tailwind CSS
- Vite (build tool)

Backend

- Django (Python-based web framework)
- RESTful API architecture

AI and Search Components

- Gemini AI API (for summarization and query understanding)
- SentenceTransformers (embedding generation)
- ChromaDB (vector database for semantic search)
- Retrieval-Augmented Generation (RAG) pipeline

Database Systems

- Supabase (PostgreSQL) for relational data
- ChromaDB for vector embeddings

Key Modules

- User Authentication Module
- Search and Retrieval Module
- AI Analysis Module
- Bookmarking Module
- Citation Generator Module
- Feedback Module



- Admin Analytics Dashboard

Database Design

The system uses hybrid database architecture:

Relational Database (Supabase PostgreSQL) stores:

- User accounts
- Sessions
- Bookmarks
- Search history
- Feedback data
- Logs

Vector Database (ChromaDB) stores:

- Document embeddings
- Semantic representations of theses

Data Structure Characteristics

- Metadata includes title, author, abstract, year, subject
- Indexed for fast retrieval and filtering
- Supports semantic similarity search

The system ensures consistency between frontend modules and database fields to support reporting, filtering, and analytics.

Hardware and Software Detailed Design

Hardware:

The system leverages existing hardware infrastructure for development and testing. No additional hardware is required. The system is optimized for SSD-based performance and sufficient RAM allocation to support AI processing workloads.

Software:

Search Module

- Accepts natural language queries
- Converts queries into embeddings
- Retrieves relevant documents via vector similarity
- Reranks results using AI

AI Overview Module

- Uses Gemini AI to generate summaries
- Produces contextual explanations with references

User Interaction Modules

- Bookmarking system for saving research



- Citation generator supporting multiple formats
- Search history tracking

Admin Module

- Displays analytics on search trends
- Provides insights into user behavior

System Security and Integrity Controls

The system incorporates multiple layers of security:

Authentication and Authorization

- Secure login with email and password
- Role-based access control (User, Admin)
- Session management with secure tokens

Data Protection

- Secure API communication (HTTPS)
- Password hashing and encryption

System Controls

- Firewalls and network restrictions
- Controlled API access to AI services

Audit and Monitoring

- Logging of user activities (searches, bookmarks, feedback)
- Analytics for system monitoring

Backup and Recovery

- Backup storage on HDD
- Capability to restore system in case of failure

SPONSOR ACCEPTANCE

Approved by the Project Sponsor:

Khasian Eunice Romulo
Science Research Specialist II / Unit Head
(Digital Information and Processing Delivery Unit)

Date: April 1, 2026